FORCED NONINJECTIVITY AND INFINITE SHEET RESOLUTION IN CANONICAL LOCALIZATION

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ABSTRACT. Kahn's localization construction for ψ-polynomial-like maps produces, at every renormalization stage, a localized inclusion between marked covers. The construction is functorial and preserves immersions; the sources never assert that it preserves embeddings. We prove that at every primitively renormalizable ordinary quadratic-like restriction of a quadratic polynomial (period $p > 1$), the canonically localized inclusion is *forced* to be noninjective: the degree budget 2^p against central degree 2 produces an excess component of the pulled central continuum, and a peripheral loop around it lies outside the source localization subgroup while dying in the target. The failure is resolved canonically: the localized inclusion factors uniquely (within the based commuting intermediate-cover class) as a covering onto an embedded intermediate cover, the fiber of the covering is a pointed coset set, and this sheet defect is *infinite*, so no finite sheet enumeration is complete. In fact the defect has a canonical group model — the kernel of the source-to-target hole-filling homomorphism — and this kernel surjects, by peripheral winding numbers, onto $\mathbb{Z}^{2^p-2}$, where $2^p - 2$ is the elementary degree budget of $F = f^p$ against central degree 2: canonical localization forgets a hole-filling kernel with exponentially many independent peripheral directions. The novelty claimed here is the winding surjection and the kernel identification, not the count. The resolution propagates through Kahn's exact pullback towers on the full, unselected complementary-cover locus, and — under an explicit strong stage-alignment hypothesis — to the Kahn–Lyubich direct-decoration depths, where it forces noninjectivity of the induced tile lifts and of the glued pseudo-piece immersion. Finally, the infinite canonical resolution is incompatible with any whole boundary-extended Kahn–Lyubich tile factorization satisfying the stated annular-tower typing: an end-compatible comparison of that type forces a trivial deck kernel. Two abstract counterexamples (an exactly certified Blaschke quotient for Kahn's annulus hypotheses, and a complete abstract ψ-quadratic-like pair with a remote self-collision) show that no argument from the stated hypotheses alone can restore injectivity. These results correct an embedding-preservation route proposed in the author's research program, not in the sources: they contradict nothing stated in the published sources, and they do not by themselves decide any global rigidity statement. The corrective conclusion is structural: the canonical localized object at an ordinary primitive canonical seed is not an embedded pseudo-piece but an embedded cover *together with* an infinite coset-valued sheet fibration and its canonical transport.

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1. Introduction

1.1. **Setting.** Renormalization of quadratic-like maps [DH85], in the generality needed for a priori bounds, takes place in Kahn's category of ψ-polynomial-like maps [Kah23]: pairs (i, f) of a holomorphic *immersion* and a proper branched covering between disks, with a common connected filled preimage. A central device of [Kah23], used again in the Kahn–Lyubich decoration papers [KL08, KL09], is *localization*: passing to marked covers determined by the fundamental group of a neighborhood of a marked full continuum, so that the dynamics near one small filled Julia set can be studied with the rest of the cycle removed. Canonical renormalization is the dynamical localization of the p-th iterate at the central small filled Julia set $A = K_p$, with the cycle $B = \mathcal{K}_p$ removed.

The sources are careful about what this construction preserves. The localization functor preserves immersions and branched coverings; the localized inclusion leg is asserted to be an immersion, never an embedding; and Kahn notes explicitly that the map which was an embedding in the quadratic-like setting has become an immersion in the ψ-quadratic-like setting. We are not aware of any statement in the literature determining whether this caution is necessary at the canonical stages themselves — whether the canonically localized inclusion might in fact be injective, so that localization could be treated as embedding-preserving in arguments that need a globally embedded local model. Section 1.3 records what we did examine and what the closest published anticipation is.

1.2. **Main results.** We show that the caution is necessary in the strongest possible sense: noninjectivity is not merely possible but *forced*, at every ordinary primitive canonical seed, and the failure carries a canonical infinite structure.

Throughout, $f\colon U^1 \to U^0$ is an ordinary quadratic-like restriction of a quadratic polynomial, primitively renormalizable of period $p > 1$; $F = f^p\colon U^p \to U^0$; $A = K_p$ is the central small filled Julia set and $B = \mathcal{K}_p$ its cycle; $T' = (U^p \setminus F^{-1}B) \cup A$ and $T = (U^0 \setminus B) \cup A$ are the marked surfaces of Kahn's canonical localization, with localization subgroups $H_s \leq \pi_1(T' \setminus A)$ and $H_t \leq \pi_1(T \setminus A)$, localization covers $\widehat{\Omega}_s$ and $\widehat{\Omega}_t$ of the complementary surfaces $\Omega_s = T' \setminus A$ and $\Omega_t = T \setminus A$, complement-cover lift $\widehat{a}\colon \widehat{\Omega}_s \to \widehat{\Omega}_t$, and localized inclusion $\widehat{\imath}\colon T'_A \to T_A$.

Theorem 1.1 (Forced noninjectivity; Theorem 4.8)**.** *At every ordinary primitive canonical seed of period $p > 1$, the canonically localized inclusion $\widehat{\imath}$ is not injective. Explicitly, there are an excess component $X \neq A$ of*

$F^{-1}(A)$ and a based loop $\gamma \subset U^p \setminus F^{-1}B$ with $[\gamma] \notin H_s$ and $(i^p)_*[\gamma] = 1 \in H_t$, and the lift of γ has distinct endpoints identified by $\widehat{\imath}$.

Theorem 1.2 (Canonical resolution; Theorems 5.8 and 5.13)**.** *The complement-cover lift $\widehat{a}$ of i^p factors exactly as $\widehat{a} = e \circ q$, where $q\colon \widehat{\Omega}_s \to P_K$ is a surjective covering onto the basepoint component P_K of the corresponding pullback cover and $e\colon P_K \hookrightarrow \widehat{\Omega}_t$ is a topological embedding. Every point collision of $\widehat{\imath}$ lies in the complement-cover part, where $\widehat{\imath}$ agrees with $\widehat{a}$, and is a sheet identification of q; the based fiber of q is the pointed coset set K/H_s with $K = (i^p)_*^{-1}(H_t)$ the reflected subgroup; and P_K is the unique connected intermediate cover, in the based commuting comparison class of Theorem 5.13, on which the remaining leg embeds. No extension of the factorization across the filled completion is asserted.*

Remark 1.3 (The structural shape of Theorem 1.2 is not new). The shape asserted by Theorem 1.2 — an immersion that is a covering onto an embedded image away from a proper compactum — is already an axiomatised notion in the recent literature. Dudko–Luo–Lyubich [DLL25, §2.1] call an immersion $\iota\colon A \to B$ a *covering embedding rel $S \subset B$* when S is ι-proper and ι restricted to $A \setminus \iota^{-1}(S)$ is a covering onto its image, and they impose this as Items (V)–(VI) of their ψ-ql Siegel definition. Their setting is different (Siegel rather than primitive) and, decisively, the property is *imposed* there as part of a definition. The contribution of Theorem 1.2 is not the notion but its canonical derivation at the seed: the factorization is produced, its intermediate factor is shown to be unique in a stated comparison class, and its fiber is identified with an explicit dynamical coset set.

Theorem 1.4 (Infinite sheet defect; Corollary 5.11)**.** *At every ordinary primitive canonical seed, $[K : H_s] = \infty$: the cosets $[\gamma]^n H_s$ are pairwise distinct. Consequently every non-singleton fiber of $\widehat{\imath}$ (equivalently, every fiber over the embedded resolved complement component) is infinite and no finite raw sheet enumeration of the canonical localization is complete.*

Theorem 1.5 (Kernel normal form; Theorem 7.1)**.** *At every ordinary primitive canonical seed, with the shared local neighborhood and based subgroup representatives of Definition 4.2, the hole-filling homomorphism $a_* = (i^p)_*$ restricts to an isomorphism $H_s \to H_t$, and the reflected subgroup splits as $K = N_{\mathrm{fill}} \rtimes H_s$ with $N_{\mathrm{fill}} = \ker a_*$. Consequently $n \mapsto nH_s$ is a canonical pointed bijection $N_{\mathrm{fill}} \to K/H_s$: the sheet fiber carries a canonical group model, and canonical localization forgets exactly the kernel of the source-to-target hole-filling map.*

Theorem 1.6 (Exact excess count and exponential defect; Theorem 7.3, Corollary 7.4)**.** *At every ordinary primitive canonical seed of period $p > 1$ the central preimage decomposes as $F^{-1}(A) = A \sqcup X_1 \sqcup \cdots \sqcup X_{2^p-2}$, where each excess component X_j is a full continuum disjoint from B mapped homeomorphically onto A by F. The peripheral winding vector about the*

excess components is a surjective homomorphism $N_{\mathrm{fill}} \twoheadrightarrow \mathbb{Z}^{2^p-2}$*; hence* $\operatorname{rank} H_1(N_{\mathrm{fill}};\mathbb{Z}) \geq 2^p-2$*, and the forgotten sheet fiber carries at least* 2^p-2 *independent peripheral winding coordinates.*

Two calibrations of Theorem 1.6. The count 2^p-2 is an elementary degree budget — $\deg F = 2^p$, the only critical point of F whose critical value lies in A is the origin, so every other component of $F^{-1}(A)$ is univalent — and we claim no novelty for it; the existence of non-central components is in any case structurally built into the hypothesis of the Kahn–Lyubich Covering Lemma ([KL08, §4]: A is a component of $f^{-1}(B)$ with big degree D and local degree d, and $D > d$ forces further components). What is claimed is the peripheral-winding surjection $N_{\mathrm{fill}} \twoheadrightarrow \mathbb{Z}^{2^p-2}$ and the resulting rank bound. Likewise, the abstract covering-space factorization is standard; the contribution is the identification of the actual dynamical subgroup defect and its canonical kernel model, not the invention of intermediate covers.

Theorem 1.7 (Exact propagation; Proposition 8.6, Corollary 8.8, Theorem 9.6)**.** *The factorization and its fiber data are preserved by every full, unselected exact Kahn equalizer on the complementary-cover locus, hence by every depth of Kahn's pullback iteration of the same seed. Under the strong stage-alignment hypothesis (Definition 9.1), the collision transports to the exact decoration depths* $m = \nu q + 1$*,* $n = (\nu-1)q + 1$*, where it forces non-injectivity of the Kahn–Lyubich ambient annulus arrow, of every compatible full universal-cover tile lift, and of the full glued pseudo-piece immersion.*

Theorem 1.8 (No aligned annular realization of the canonical infinite resolution is a whole KL tile; Theorem 10.5, Corollary 10.6)**.** *Whenever the canonical resolved factorization restricts, at an aligned stage, to a covering/embedding factorization of a degree-one immersion of finite-modulus annuli, the full annular-tower typing of Definition 8.1 is automatic (Theorem 10.1), and no end-compatible whole-tile comparison square can identify the canonical resolved factorization with a complete boundary-extended Kahn–Lyubich tile factorization carrying the two source-locked infinitely-wrapping vertices: any such comparison forces the deck kernel to be trivial, while the canonical resolution has infinite sheet defect. The statement is conditional only on the aligned-stage annular factorization; the typing is derived, not assumed.*

Two abstract counterexamples calibrate the main theorems. A degree-one annular immersion satisfying all stated hypotheses of Kahn's annulus lemma can fail to be injective (Theorem B.10; an exactly certified Blaschke quotient), and the complete abstract ψ-quadratic-like axioms admit a pair whose immersion has an explicit remote self-collision (Theorem B.13). Thus no proof from the stated hypotheses alone — degree, immersion, properness of the companion branch, common connected inverse — can restore global injectivity; the forced collision of Theorem 1.1 is a fact about the

actual dynamical construction, established by the actual excess-component geometry.

1.3. **Relation to the literature.** Because the results above are stated against two specific sources, we record what was examined, what the closest published anticipation is, and how much of the argument is short once one knows to look for it.

What the primary sources assert. Kahn's localization functor is asserted to preserve immersions and branched coverings and nothing more ([Kah23, §§7.4.1–7.4.2, pp. 43–45]); at the quadratic-like stage the arrow i is literally the inclusion of nested domains, and a footnote announces that in the ψ-quadratic-like sections it will be replaced by an immersion ([Kah23, §6, p. 31 and footnote 15]); and the setup ingredient of the present argument is already visible in the source, since [Kah23, §6.1, pp. 31–32] exhibits the hole-filling leg $U^{n+1} \setminus f^{-1}\mathcal{K} \subset U^{n+1} \setminus \mathcal{K}$ as one side of a triangle the text explicitly calls non-commutative, and controls passage along it quantitatively via Property D. On the Kahn–Lyubich side the pseudo-piece inclusion arrow is never upgraded beyond an immersion [KL08, §3.2.2, Lemma 3.4].

The closest published anticipation. It is by the source authors themselves, and it deserves to be in the introduction rather than in an appendix. In the proof of [KL08, §3.2.2, Lemma 3.4(ii), p. 68 of the journal pagination], for puzzle pieces $Y \subset Z$ the components Y_i are embedded into components $Z_{j(i)}$, and the component index map $j = j(i)$ is described as "surjective but not necessarily injective". That is the collapsing-of-components mechanism, in print, since 2008. Theorem 1.1 differs from it in three specific ways.

(1) It is a *forcing* statement, not a possibility statement: at every ordinary primitive canonical seed the collision must occur, whereas "not necessarily injective" asserts only that injectivity is not guaranteed.
(2) It concerns the *glued immersion*, not the component index map j. Noninjectivity of j does not imply noninjectivity of the glued map: distinct universal-cover tiles can occupy disjoint sheets over one target tile, so a two-to-one j is compatible with an injective glued arrow.
(3) It concerns Kahn's localization functor at a canonical *seed*, not Kahn–Lyubich pseudo-puzzle tiles at an arbitrary puzzle depth.

On difficulty. We do not present Theorem 1.1 as hard. In Kahn's own path model of the canonical localization ([Kah23, §7.4.5, pp. 46–47]), where the source class space is formed rel $f^{-p}(K_p)$ and the target rel $\mathcal{K}_p$ and the source proves only the forward direction of path equivalence, the failure of injectivity is a short argument once one thinks to look for it: a loop around an excess component of $f^{-p}(K_p) \setminus \mathcal{K}_p$ is nontrivial in the source class space and trivial in the target. We are not aware of anyone having recorded it, and the substance of this paper lies in what follows from it — the kernel normal form, the infinitude $[K : H_s] = \infty$, the rank-$(2^p - 2)$ winding surjection, and the propagation and no-go results.

Post-2008 work on this machinery. We examined the papers that actually use the ψ-ql/localization apparatus and found no statement of, and nothing contradicting, Theorem 1.1. The most recent restatement of the construction is by two of the three source authors: Kahn–Kapiamba–Lyubich [KKL26, §§2.2–2.3] recall localization via subsurface covers and renormalization, and still assert only that the pair (f^n, ι^n) *lifts* to a ψ-ql map — no embedding, injectivity, or sheet statement. The same holds in [KL26], where ι is used on complements with no injectivity claim; in the Dudko–Lyubich renormalization papers [DL22, DL23]; in the Siegel variation [DLL25], whose §2.1 axiomatises the covering-embedding shape discussed in Remark 1.3; in [CS15]; and in [BLOT22]. Dudko's survey [Dud25] does not treat the localization functor at this level of detail. We make no claim of an exhaustive search: the negative result rests on the citation graphs of the two sources together with targeted full-text checks, not on a complete bibliographic sweep.

1.4. **What these results do and do not assert.** Three boundaries govern everything below and are restated where they apply.

First, the results contradict nothing stated in the published sources [Kah23] and [KL08]; the exact identification points on which this interface rests are listed in Section 13. Those sources assert immersion preservation, a homeomorphism on filled and marked sets, an embedded inner collar, and the forward direction of path equivalence — never a global embedding of the localized inclusion. In particular the embedded inner collar of [Kah23, Theorem 7.3, p. 37] and [KL08, Lemma 3.1, p. 10] is *not* in tension with Theorem 1.1: the forced collision is located outside the collar, on a peripheral loop around an excess component in the outer part of the source, as spelled out in Remark 4.10. What is refuted is a proposed *extension* of the sources: any argument treating canonical localization as embedding-preserving. Appendix A documents where the sources themselves demote the inclusion to an immersion.

Second, the direct-decoration consequences in Theorem 1.7 are conditional on the strong stage-alignment hypothesis (Definition 9.1) for the same canonical seed; the alignment and realization quantifiers are stated exactly in Proposition 4.11, and no source-locked parameter realization is claimed. As Remark 9.2 explains, that hypothesis is not a comparison of two constructions — the Kahn–Lyubich ambient arrow is literally an iterate of Kahn's equalizer tower — but an identification of two *seeds*: on the reading this paper intends, in which the Kahn–Lyubich puzzle machinery is run over the canonical seed itself, the identifications are the identity and Theorem 1.7 is unconditional; it is the distinct-seed reading that remains open.

Third, the results are corrective, not constructive, with respect to any global program: they show that the full embedded pseudo-piece cannot be the universal carrier invariant, and that a valid replacement must retain the coset-valued sheet fibration and its transport, or synchronize the removed obstacle packet. They do not by themselves prove or refute any rigidity,

a priori bounds, or local connectivity statement, and we make no claim in that direction.

1.5. **Organization.** Section 2 fixes the ψ-polynomial-like background and the localization construction. All citations into the two primary sources are given by published section, statement number and page; the SHA-256–locked TEX archives on which the paraphrases were originally prepared are retained only as a provenance concordance in Appendix D. Section 3 develops the criterion layer: the subgroup-reflection criterion and peripheral isolation of pulled full continua. Section 4 proves Theorem 1.1; Section 5 proves Theorems 1.2 and 1.4 and constructs the sheet-transport groupoid; Section 6 gives the exact coset tests for enlargement. Section 7 proves Theorems 1.5 and 1.6: the kernel normal form, the exact excess count, and the exponential winding defect of the hole-filling kernel. Section 8 proves the base-change half of Theorem 1.7; Section 9 transports the collision to tiles and pseudo-pieces; Section 10 proves Theorem 1.8. Section 11 separates the ordinary canonical seed from a general bold Kahn–Lyubich stage. Section 12 states what these results change in localization arguments. Section 13 records the outcome of the source check on the identifications on which the interface with the sources rests, together with the exact scope of the machine-checked certificates. Appendix A collects the annulus immersion lemma and the sources' immersion disclosures; Appendix B gives the two abstract counterexamples. Appendix C records the certificate conventions; Appendix D records the source-lock data.

Provenance and verification disclosure. This paper was prepared with extensive machine assistance, both in the research phase and in drafting, within a source-locked, machine-audited research framework; the supporting dependency ledger and validation scripts are available with the manuscript sources. The machine-verified content is strictly delimited (exact rational certificates for Appendix B; source-lock hashes; arithmetic and routing checks) and is described in Section 13. We stand behind the unconditional core of the paper — the forced collision, the canonical resolution, the kernel normal form, and the exact excess count. The Kahn-side identifications on which that core rests have now been checked against the published sources and are recorded as confirmed, with their published loci, in Section 13; specialist confirmation is invited but the checks are no longer unperformed. The Kahn–Lyubich consequences are stated as precise implications under the strong stage-alignment hypothesis (Definition 9.1), whose real content — identifying two seeds, not two constructions — is analysed in Remark 9.2. On the reading this paper intends, in which the Kahn–Lyubich puzzle machinery is run over the canonical seed itself, those consequences are unconditional; supplying the identifications for a *different* seed is the one source-side proposition that remains open.

2. Background: ψ-polynomial-like maps and localization

This section fixes definitions and records, in paraphrase, exactly what the published sources provide. Citations read "Kahn I, §X / Theorem N, p. P" for [Kah23] in the numbering of arXiv:math/0609045v3 (28 December 2023, 63 pp.), and "KL II, Lemma N, p. P" for [KL08] in the numbering of Ann. Sci. Éc. Norm. Supér. (4) **41** (2008) 57–84 and of the identically sectioned arXiv:math/0609046v2. The SHA-256–locked TEX archives from which the paraphrases were prepared, and their concordance to these published loci, are recorded in Appendix D.

Path convention. For paths α, β with $\alpha(1) = \beta(0)$, the concatenation $\alpha * \beta$ traverses α first. All displayed loop products below use this convention.

Definition 2.1 (FJ-set; Kahn I, §1.5)**.** A set $K \subset \mathbb{C}$ (or in a Riemann-surface disk) is an *FJ-set* ("filled-Julia-like set") if it is compact, connected, full, and has more than one point.

Definition 2.2 (ψ-polynomial-like map; Kahn I, Definition 7.1, p. 36)**.** A *ψ-polynomial-like map* (with connected Julia set) of degree d is a pair of holomorphic maps $(i, f)\colon \mathbf{U}' \to \mathbf{U}$ together with an FJ-set $K \subset \mathbf{U}$ such that:

(1) $\mathbf{U}'$ and $\mathbf{U}$ are disks;
(2) i is an immersion;
(3) f is a proper branched covering of degree d;
(4) $i^{-1}(K) = f^{-1}(K) =: K'$; and
(5) K' is connected.

K is the *filled Julia set* of (i, f). The source records (Definition 7.1 and the remark following it, p. 36; §7.3.2, p. 42): K' is again an FJ-set; connectivity of K' is equivalent to all branch values of f lying in K; the set K is determined by the pair (i, f); and $i\colon K' \to K$ is a bijection (used later as a homeomorphism). A degree-two ψ-polynomial-like map is *ψ-quadratic-like*.

Definition 2.3 (marked surfaces, local pairs, local covers; Kahn I, §7.4.1, p. 44)**.** A *marked Riemann surface* is a pair (S, A) with S a Riemann surface and $A \subset S$ compact, connected, and full (no component of $S \setminus A$ has compact closure in S). A *morphism* $g\colon (S, A) \to (T, B)$ is a holomorphic map with $A = g^{-1}(B)$. The pair (S, A) is *local* if every component of $S \setminus A$ is an annulus; in that case S deformation-retracts into any neighborhood of A. For every (S, A) and every neighborhood $V \supset A$ there is a subsurface $U \subset V$ containing A with (U, A) local; such U is a *local neighborhood* of A. A morphism $g\colon (S, A) \to (T, B)$ is a *local cover* if $g\colon A \to B$ is a bijection and $g\colon S \setminus A \to T \setminus B$ is a covering map.

Definition 2.4 (localization; Kahn I, §7.4.1, Theorem 7.12, p. 44)**.** The *localization* of a marked Riemann surface (S, A) is the (unique up to isomorphism) local marked Riemann surface $(S_A, \widehat{A})$ equipped with a local

cover $\pi_{S,A}\colon (S_A, \widehat{A}) \to (S, A)$ produced by the localization theorem below; for a morphism g, its unique commuting lift $\widehat{g}$ is the *localization of* g.

Definition 2.5 (canonical seed and localization subgroups; local notes, forced-collision Sec. 5; Kahn I, §§7.4.1–7.4.3, pp. 44–46)**.** Let $f\colon U^1 \to U^0$ be a quadratic-like restriction of a quadratic polynomial, primitively renormalizable of period $p > 1$; put $F = f^p\colon U^p \to U^0$, $a = i^p\colon U^p \hookrightarrow U^0$ (the physical inclusion, so $a^{-1}(A) = A$), $A = K_p$, $B = \mathcal{K}_p$, and

$$T = (U^0 \setminus B) \cup A, \qquad T' = (U^p \setminus F^{-1}B) \cup A,$$
$$\Omega_s = T' \setminus A = U^p \setminus F^{-1}B, \qquad \Omega_t = T \setminus A = U^0 \setminus B.$$

Fix a Jordan local neighborhood $D_A \Subset T'$ of A disjoint from every other component of $F^{-1}B$, and a basepoint $z_* \in D_A \setminus A$. The *source localization subgroup* is

$$H_s = \operatorname{im}\bigl[\pi_1(D_A \setminus A, z_*) \longrightarrow \pi_1(\Omega_s, z_*)\bigr],$$

and $H_t \leq \pi_1(\Omega_t, z_*)$ is defined analogously from a local neighborhood of A in T. These are based representatives of the (unbased, conjugacy-well-defined) covering subgroups by which Kahn's localization covers of Ω_s and Ω_t are constructed. Kahn's own requirement on a local neighborhood is weaker than what we take: he asks only that every component of $U \setminus A$ be an annulus (Kahn I, §7.4.1, p. 44). Restricting to *Jordan* D_A is our own specialization; it is legitimate by Lemma 4.1, which shows that a Jordan neighborhood of a full continuum is a local pair in Kahn's sense, and it is convenient because it makes the winding computations of Section 4 immediate.

Definition 2.6 (KL direct-decoration stage; KL II, §2.1, p. 6)**.** In the decoration combinatorics of KL II (satellite period q, escape depth ν), the principal piece is $P = Y^{(\nu-1)q+1}$; since $f^q(P) \supset P$ and the critical value $f^q(0)$ is not in P, the piece P has two univalent f^q-pullbacks $Q_L, Q_R \subset P$ of depth $\nu q + 1$, attached to α and α' respectively. The *direct-decoration depths* are

$$m = \nu q + 1, \qquad n = (\nu - 1)q + 1.$$

Beyond the definitions, the body of the paper uses five statements of the sources, each a paraphrase with published citations: the polynomial-like restriction theorem, the exact pullback tower, the localization theorem, dynamical localization, and canonical renormalization. Kahn's annulus immersion lemma and the four source passages demoting the localized inclusion from an embedding to an immersion are needed only for calibration and scope; they are collected, under the same citation conventions, in Appendix A.

Theorem 2.7 (polynomial-like restriction; Kahn I, Theorem 7.3, p. 37)**.** *Let $F = (i, f)\colon \mathbf{U}' \to \mathbf{U}$ be a ψ-polynomial-like map of degree $d \geq 2$ with filled Julia set K and $K' = i^{-1}(K)$. Then there is an open set $U \subset \mathbf{U}$ with $K \subset U$ such that, writing $U' = f^{-1}(U) \subset \mathbf{U}'$:*

(1) $f\colon U' \to U$ *is a proper degree-d branched covering;*
(2) $i|_{U'}$ *is an embedding;*
(3) $i(U') \subset U$*; and*
(4) $\operatorname{mod}\big(U \setminus i(U')\big) \geq \mu > 0$*, where μ can be taken uniform over all such maps with d bounded and* $\operatorname{mod}(\mathbf{U} \setminus K)$ *bounded below.*

In particular $f \circ (i|_{U'})^{-1}\colon i(U') \to U$ is a polynomial-like map of degree d: every ψ-polynomial-like map carries an embedded *polynomial-like germ, with modulus controlled by the modulus of the ambient pair.*

Proof (faithful paraphrase of Kahn I, proof of Theorem 7.3, pp. 38–39). Step 1: local injectivity near K'. If i were injective on no neighborhood of K', there would be sequences $z_k \to K'$, $w_k \to K'$ with $z_k \neq w_k$ and $i(z_k) = i(w_k)$. Passing to subsequences with $z_k \to z \in K'$ and $w_k \to w \in K'$ gives $i(z) = i(w)$; since $i\colon K' \to K$ is a bijection, $z = w$, and then the failure of injectivity in every neighborhood of z forces z to be a critical point of i, contradicting that i is an immersion.

Step 2: transport to round annuli. Uniformize $\phi\colon \mathbb{A}(1, r) \to \mathbf{U} \setminus K$ and $\phi'\colon \mathbb{A}(1, r') \to \mathbf{U}' \setminus K'$, where $r' = r^{1/d}$ because f restricts to a proper degree-d map between the annular ends. Set $I = \phi^{-1} \circ i \circ \phi'\colon \mathbb{A}(1, r') \to \mathbb{A}(1, r)$, a holomorphic immersion. Since $|I(z)| \to 1$ as $|z| \to 1$, the reflection principle for harmonic functions extends I continuously to $\mathbb{T} \to \mathbb{T}$ (Kahn I, remark following Lemma 7.4, p. 38); by Step 1 the extension is injective near $\mathbb{T}$, and a doubling/open-mapping argument of that remark shows the boundary map is injective, hence of degree one.

Step 3: apply the annulus immersion lemma. By Lemma A.1 and the combined Remark (Kahn I, p. 38) there are uniform $1 < s < \rho < r$ with $I|_{\mathbb{A}(1,\rho')}$ an embedding, $\rho' = \rho^{1/d}$, and $I(\mathbb{A}(1, \rho')) \subset \mathbb{A}(1, s)$. Define $U' = K' \cup \phi'(\mathbb{A}(1, \rho'))$ and $U = K \cup \phi(\mathbb{A}(1, \rho))$. Then $f\colon U' \to U$ is a proper degree-d branched covering, $i|_{U'}$ is an embedding, $i(U') \subset K \cup \phi(\mathbb{A}(1, s)) \subset U$, and $\operatorname{mod}(U \setminus i(U'))$ is bounded below by a positive quantity determined by $s < \rho$, uniform for d bounded and $\operatorname{mod}(\mathbf{U} \setminus K) = \frac{1}{2\pi} \log r$ bounded below. □

Proposition 2.8 (pullback tower; Kahn I, §§7.3.1–7.3.2, pp. 38–43)**.** (1) *(Existence; Lemma 7.8, p. 40.) In the category of complex 1-manifolds, the pullback of a diagram $g\colon G \to V$, $h\colon H \to V$ exists whenever g (or, symmetrically, h) is an immersion; its underlying topological space is the topological fiber product $E = \{(x, y) \in G \times H : g(x) = h(y)\}$ with the coordinate projections π_G, π_H (Kahn describes E as the* equalizer*; §7.3.1, p. 39).*

(2) *(Inheritance; Lemma 7.7, p. 40 and Lemma 7.9, p. 40.) Each projection inherits the properties of the opposite leg: if g is proper (resp. a local homeomorphism, a homeomorphism, injective, surjective, bijective), then so is π_H; and for $y \in H$ the fiber $\pi_H^{-1}(y)$ is in natural bijection with $g^{-1}(h(y))$. If g is an immersion and h is proper of degree d, then π_H is an immersion and π_G is proper of degree d, and the local degree of π_G at $z \in E$ equals the local degree of h at $\pi_H(z)$.*

(3) *(One ψ-polynomial-like pullback; §7.3.2, p. 42 and Lemma 7.11, p. 42.) For a degree-d ψ-polynomial-like map $(i,f)\colon (\mathbf{U}',K')\to(\mathbf{U},K)$, the pullback of $i\colon \mathbf{U}'\to\mathbf{U}$ along $f\colon \mathbf{U}'\to\mathbf{U}$ is*

$$\mathbf{U}''=\{(x,y)\in \mathbf{U}'\times\mathbf{U}' :\ i(x)=f(y)\},\qquad i_1(x,y)=y,\quad f_1(x,y)=x,$$

and $(i_1,f_1)\colon(\mathbf{U}'',K'')\to(\mathbf{U}',K')$ is again a degree-d ψ-polynomial-like map, where $K''=i_1^{-1}(K')=f_1^{-1}(K')$.

(4) *(Tower; §7.3.2, p. 43.) Relabeling (i,f) as $(i_0,f_0)\colon(\mathbf{U}^1,K^1)\to(\mathbf{U}^0,K^0)$ and iterating produces degree-d ψ-polynomial-like maps $(i_n,f_n)\colon(\mathbf{U}^{n+1},K^{n+1})\to(\mathbf{U}^n,K^n)$ with the commuting squares*

$$i_n\circ f_{n+1}=f_n\circ i_{n+1}\qquad(n\in\mathbb{N}),$$

each square being a genuine categorical pullback (fiber product) of the previous stage.

(5) *(Germ preservation; §7.3.3, p. 43.) If $U'\subset\mathbf{U}'$, $U\subset\mathbf{U}$ are such that $(i,f)\colon U'\to U$ is polynomial-like ($i(U')\Subset U$ and $i|_{U'}$ an embedding), then the pullback U'' of $(i,f)\colon U'\to U$ sits inside $\mathbf{U}''$ and $(i_1,f_1)\colon U''\to U'$ is polynomial-like; consequently the polynomial-like germ of (i_1,f_1) equals that of (i,f).*

Proof (faithful paraphrase of Kahn I, §7.3.1, pp. 40–41 and §7.3.2, p. 42). (1) By the inheritance statement for local homeomorphisms, $\pi_H\colon E\to H$ is a local homeomorphism when g is an immersion; pulling the charts of H back through π_H endows E with a complex structure for which π_H is holomorphic, and locally $\pi_G=g^{-1}\circ h\circ\pi_H$, hence π_G is holomorphic. The universal property is checked by factoring any competitor $F\to E$ through the topological pullback and verifying holomorphy locally via $(\pi_H|_{\widehat{U}})^{-1}\circ f_H$ (proof of Lemma 7.8, pp. 40–41). Kahn notes the pullback may be disconnected, which is why he writes "complex 1-manifold" rather than "Riemann surface" (§7.3.1, p. 41).

(2) Properness: for compact $B\subset H$, the set $\pi_H^{-1}(B)$ is a closed subset of $g^{-1}(h(B))\times B$, hence compact (proof of Lemma 7.7, p. 40). Local homeomorphism: over a neighborhood on which g is invertible, π_H has the explicit inverse $y\mapsto((g|_U)^{-1}(h(y)),y)$ (ibid.). The set-theoretic statements and the fiber correspondence are elementary (ibid.). When h is proper of degree d, evaluating the fiber correspondence at a regular value gives $|\pi_G^{-1}(x)|=d$, and the local-degree statement follows by comparing small neighborhoods (proof of Lemma 7.9, pp. 40–41).

(3) By (1)–(2), i_1 is an immersion and f_1 is proper of degree d. For $(x,y)\in\mathbf{U}''$, the chain $x\in K'\iff i(x)\in K\iff f(y)\in K\iff y\in K'$ shows $i_1^{-1}(K')=f_1^{-1}(K')=:K''$ (§7.3.2, p. 42). Since $i\colon K'\to K$ is a homeomorphism, so is $i_1\colon K''\to K'$ (inheritance), whence K'' is connected (§7.3.2, p. 42). Every component of $\mathbf{U}''$ meets K'' because f_1 is proper onto $\mathbf{U}'$; hence $\mathbf{U}''$ is connected (§7.3.2, p. 42). A point of $\mathbf{U}''$ is critical for f_1 exactly when its i_1-image is critical for f, with the same multiplicity; therefore all critical points of f_1 lie in K'' and their total multiplicity is $d-1$,

so Riemann–Hurwitz gives $\chi(\mathbf{U}'') = 1$ and $\mathbf{U}''$ is a disk (§7.3.2, p. 42). This verifies all clauses of the ψ-polynomial-like definition (Kahn I, Lemma 7.11, p. 42).

(4) Apply (3) inductively; the commuting relation is the defining property of the pullback square at each stage (§7.3.2, p. 43).

(5) The pullback of the polynomial-like restriction embeds in the pullback of the ambient pair; the restricted pair $(i_1, f_1)\colon U'' \to U'$ satisfies the polynomial-like conditions, and its germ at the filled Julia set coincides with that of (i, f) (§7.3.3, p. 43). □

Theorem 2.9 (localization; Kahn I, Theorem 7.12, p. 44)**.** *For every marked Riemann surface (S, A) there exists a pair consisting of a local marked Riemann surface $(S_A, \widehat{A})$ and a local cover $\pi_{S,A}\colon (S_A, \widehat{A}) \to (S, A)$, and this pair is unique up to isomorphism. For every morphism $g\colon (S, A) \to (T, B)$ there is a unique continuous map $\widehat{g}\colon S_A \to T_B$ with $\pi_{T,B} \circ \widehat{g} = g \circ \pi_{S,A}$, and this $\widehat{g}$ is a morphism $(S_A, \widehat{A}) \to (T_B, \widehat{B})$. These assignments define a functor from marked Riemann surfaces to local marked Riemann surfaces, and the maps $\pi_{S,A}$ form a natural transformation from localization to the identity functor (§7.4.1, pp. 44–45). Moreover (§7.4.1, p. 45): if g is an immersion then $\widehat{g}$ is an immersion, and if $g\colon S \setminus A \to T \setminus B$ is a covering map then $\widehat{g}$ is a branched covering.*

Remark 2.10 (what the theorem does *not* assert). The source asserts functoriality, uniqueness of the commuting lift, and preservation of immersions and of coverings off the marked set. It does *not* assert that $\widehat{g}$ is an embedding when g is an embedding; it does not assert that the naturality square $\pi_{T,B} \circ \widehat{g} = g \circ \pi_{S,A}$ is Cartesian; and it does not assert any subgroup-reflection property of the covering subgroups involved. In the path model of the canonical renormalization (Kahn I, §7.4.5, pp. 46–47) the source proves only the forward direction: source-equivalent paths are mapped to target-equivalent paths.

Proof (faithful paraphrase of Kahn I, proof of Theorem 7.12, p. 44). Construction when $S \setminus A$ is connected (§7.4.1, p. 44; the only case used for the canonical renormalization). Choose a local neighborhood U of A and let $\pi'_{S,A}\colon S'_A \to S \setminus A$ be the covering of $S \setminus A$ corresponding to (the image of) the fundamental group of $U \setminus A$. The subsurface $U \setminus A$ lifts canonically to S'_A. Define S_A as the quotient of the disjoint union $U \sqcup S'_A$ obtained by identifying each point of $U \setminus A$ with its image under this lift; let $\widehat{A} \subset S_A$ be the image of A (and $\widehat{U}$ the image of U). Then $S_A \setminus \widehat{A}$ is isomorphic to S'_A, whose fundamental group is $\mathbb{Z}$, so $(S_A, \widehat{A})$ is local. Define $\pi_{S,A} = \pi'_{S,A}$ on S'_A and as the inverse of the quotient map on $\widehat{U}$; the two definitions agree on the overlap and make $\pi_{S,A}$ a local cover.

General case (§7.4.1, p. 44). With U a local neighborhood of A, take S'_A to be the disjoint union, over the components E of $S \setminus A$, of the annular covers of E induced by $E \cap U$, and proceed as before.

Lift of a morphism (§7.4.1, p. 44). Given $g\colon (S,A)\to(T,B)$, define $\widehat{g}$ on $\widehat{A}$ as $(\pi_{T,B}|_{\widehat{B}})^{-1}\circ g\circ\pi_{S,A}$, and on each component of $S_A\setminus\widehat{A}$ as the lift of $g\circ\pi_{S,A}$ through the covering $\pi_{T,B}$; such a lift exists because g carries the local-neighborhood subgroup of the source into that of the target (implicit in the source's "let $\widehat{g}$ be the lift"), and it is the unique continuous map commuting with the projections. One checks directly that $\widehat{g}$ is a morphism.

Functoriality and uniqueness (§7.4.1, p. 44). Uniqueness of commuting lifts implies that the lift of a composition is the composition of the lifts. Uniqueness of the pair $((S_A,\widehat{A}),\pi_{S,A})$ follows by lifting the identity morphism between two candidate localizations.

Preservation statements (§7.4.1, p. 45). $\widehat{g}$ is locally a lift of g through local homeomorphisms, hence an immersion when g is; when g restricts to a covering $S\setminus A\to T\setminus B$, the lifted map on complements is a covering and the completed map is a branched covering. □

Proposition 2.11 (dynamical localization; Kahn I, §7.4.2, p. 45)**.** *Let $(i,f)\colon(\mathbf{U}',K')\to(\mathbf{U},K)$ be a ψ-polynomial-like map of degree d, and suppose:*

1. *$A\subset K$ is an FJ-set, and $A'=i^{-1}(A)$;*
2. *$B\subset K$ is compact and $\mathbf{U}\setminus B$ is connected;*
3. *A is a component of B;*
4. *A' is a component of $f^{-1}(A)$;*
5. *$i^{-1}(B)\subset f^{-1}(B)$ (see Appendix D for the caveat on the source's literal wording, which reads B' for B in Kahn I, §7.4.2, p. 45); and*
6. *all branch values of f lie in B.*

Set $T=(\mathbf{U}\setminus B)\cup A$ and $T'=(\mathbf{U}'\setminus f^{-1}B)\cup A'$. Then (T,A) and (T',A') are marked Riemann surfaces and the restrictions of i and f are morphisms. Writing $\mathbf{V}=T_A$ and $\mathbf{V}'=T'_{A'}$ for the localizations:

(a) *$\mathbf{V}$ and $\mathbf{V}'$ are disks;*

(b) *the localized pair $(\widehat{i},\widehat{f})\colon(\mathbf{V}',\widehat{A'})\to(\mathbf{V},\widehat{A})$ is a ψ-polynomial-like map whose degree is the local degree $d_{A'}$ of f at A', with $\widehat{i}$ an immersion and $\widehat{f}$ a branched covering;*

(c) *since $\pi_{T,A}\colon\mathbf{V}\setminus\widehat{A}\to\mathbf{U}\setminus B$ is a covering map, the Poincaré length of the core curve of $\mathbf{V}\setminus\widehat{A}$ equals the Poincaré length of the closed geodesic of $\mathbf{U}\setminus B$ that goes around A.*

The pair $(\widehat{i},\widehat{f})$ is the dynamical localization of f at A with the removal of B.

Proof (faithful paraphrase of Kahn I, §7.4.2, p. 45). Hypotheses (1)–(6) make (T,A) and (T',A') marked Riemann surfaces and make the restrictions of i and f morphisms: the marked sets pull back correctly by (1), (4) and (5), and the removal of $f^{-1}B$ on the source side is compatible with the branch values by (6). Apply the localization theorem to obtain $(\widehat{i},\widehat{f})\colon(T'_{A'},\widehat{A'})\to(T_A,\widehat{A})$ together with the local covers $\pi_{T,A}$ and $\pi_{T',A'}$. Since $T\setminus A=\mathbf{U}\setminus B$

is connected by (2) (and correspondingly on the source side), the localized complements are connected covers; since A and A' are FJ-sets and the localized pairs are local marked surfaces with connected complement, the localized surfaces $\mathbf{V} = T_A$ and $\mathbf{V}' = T'_{A'}$ are disks (using the source's characterization at Kahn I, §7.4.1, p. 44: a local pair with connected complement is exactly a disk marked by an FJ-set). By the preservation statements of the localization theorem, $\widehat{i}$ is an immersion and $\widehat{f}$ is a branched covering; the source records (§7.4.2, p. 45) that its degree equals the local degree $d_{A'}$ of f at A' — near the marked set $\widehat{f}$ is conjugate by the local covers to f near A'. Hence $(\widehat{i}, \widehat{f})$ satisfies all clauses of the ψ-polynomial-like definition with filled Julia set completed from $\widehat{A}$; it is ψ-polynomial-like of degree $d_{A'}$ (§7.4.2, p. 45). Finally, $\pi_{T,A}$ restricts to a covering $\mathbf{V} \setminus \widehat{A} \to \mathbf{U} \setminus B$, and a holomorphic covering is a local isometry for the Poincaré metrics; the core curve of $\mathbf{V} \setminus \widehat{A}$ covers, with degree one, the closed geodesic of $\mathbf{U} \setminus B$ in the free homotopy class going around A, so the two lengths are equal (§7.4.2, p. 45). □

Theorem 2.12 (canonical renormalization; Kahn I, §7.4.3, p. 45)**.** *Let $(i, f)\colon (\mathbf{U}^1, K') \to (\mathbf{U}^0, K)$ be a ψ-quadratic-like map admitting a quadratic-like restriction that is primitively renormalizable of period p; let K_p be the central small filled Julia set, $\mathcal{K}_p$ the union of the small filled Julia sets, and $K'_p = i^{-1}(K_p)$ (iterate notation). Then $A = K_p$ and $B = \mathcal{K}_p$ satisfy hypotheses (1)–(6) of the dynamical-localization proposition for the iterate $(i^p, f^p)\colon \mathbf{U}^p \to \mathbf{U}^0$ of the pullback tower, and the resulting dynamical localization $(\widehat{i}, \widehat{f})\colon (\mathbf{V}', \widehat{K}'_p) \to (\mathbf{V}, \widehat{K}_p)$ — the* canonical renormalization of (i, f) of period p *— is a ψ-quadratic-like map.*

Theorem 2.13 (crucial length properties; Kahn I, §7.4.3, p. 46)**.** *Let $(i, f)\colon \mathbf{U}' \to \mathbf{U}$ be a ψ-quadratic-like map that is primitively renormalizable of periods p and pq $(p, q > 1)$, and let $(j, g)\colon \mathbf{V}' \to \mathbf{V}$ be its canonical renormalization of period p. Then*

$$|\gamma(g)| = |\gamma_p(f)|, \qquad |\gamma_q(g)| \;\geq\; |\gamma_{pq}(f)|.$$

Proof (faithful paraphrase of Kahn I, §7.4.3, p. 45 and 578–591). Canonical renormalization. Form the iterate $(i^p, f^p)\colon \mathbf{U}^p \to \mathbf{U}^0$ via the pullback tower. The data $A = K_p$, $B = \mathcal{K}_p$ satisfy the hypotheses of dynamical localization for this iterate: K_p is an FJ-set and a component of the compact set $\mathcal{K}_p$ (by primitivity the small filled Julia sets are pairwise disjoint); $\mathbf{U}^0 \setminus \mathcal{K}_p$ is connected; K'_p is a component of $(f^p)^{-1}(K_p)$; and the branch values of f^p lie in $\mathcal{K}_p$. The dynamical localization of f^p at K_p with removal of $\mathcal{K}_p$ is then ψ-polynomial-like of degree equal to the local degree of f^p at K'_p, namely 2; this is the canonical renormalization (§7.4.3, pp. 45–46).

Length properties. The equality $|\gamma(g)| = |\gamma_p(f)|$ is exactly conclusion (c) of the dynamical-localization proposition applied to this localization (§7.4.3, p. 46). For the inequality, write $K(g), \mathcal{K}_q(g)$ for the objects of g

and $K, \mathcal{K}_p, \mathcal{K}_{pq}$ for those of f, and let

$$\pi\colon (\mathbf{V}, K(g)) \longrightarrow \big((\mathbf{U} \setminus \mathcal{K}_p) \cup K_p,\ K_p\big)$$

be the local cover constructed in the localization (§7.4.3, p. 46). Then $\pi(\mathbf{V} \setminus \mathcal{K}_q(g)) \subseteq \mathbf{U} \setminus \mathcal{K}_{pq}$: indeed $\pi(\mathbf{V} \setminus K(g)) \subseteq \mathbf{U} \setminus \mathcal{K}_p \subseteq \mathbf{U} \setminus \mathcal{K}_{pq}$, while $\pi(K(g) \setminus \mathcal{K}_q(g)) = K_p \setminus \mathcal{K}_{pq}$ (§7.4.3, p. 46). Thus π maps the hyperbolic surface $\mathbf{V} \setminus \mathcal{K}_q(g)$ holomorphically into $\mathbf{U} \setminus \mathcal{K}_{pq}$, carrying $\gamma_q(g)$ into the free homotopy class of $\gamma_{pq}(f)$; the Schwarz–Pick lemma (Poincaré-metric contraction under holomorphic maps) gives $|\gamma_q(g)| \geq |\gamma_{pq}(f)|$ (§7.4.3, p. 46). □

Calibration by two abstract counterexamples. Two abstract counterexamples calibrate everything that follows: an exactly certified Blaschke quotient shows Kahn's annulus hypotheses alone do not imply global injectivity, and a complete abstract ψ-quadratic-like pair exhibits a remote self-collision (Appendix B). After those examples, any injectivity assertion for an actual localized inclusion must use its dynamical construction history; the body of this paper shows the canonical answer is negative for dynamical reasons.

3. The criterion layer

The forced collision of Section 4 is proved against two pieces of infrastructure: an exact covering-space criterion measuring when a localized lift is injective, and a finite-topology toolkit isolating pulled full continua. Both are elementary; both are stated and proved in full because the direction of every subgroup inclusion is load-bearing later.

3.1. **The subgroup-reflection criterion.**

Notation and standing conventions.

- $K \Subset U$ means K is compactly contained in U. A *full continuum* is a nonempty compact connected set $K \subset \mathbb{C}$ with connected complement $\mathbb{C} \setminus K$.
- $f : U^1 \to U^0$ is an ordinary quadratic-like restriction of a quadratic polynomial; $U^r = f^{-r}(U^0)$, so $U^{r+1} \subset U^r$ and each $f : U^{r+1} \to U^r$ is quadratic-like (Kahn I, §6, p. 31). For a primitively renormalizable stage of period $p > 1$ we write $F = f^p : U^p \to U^0$, proper of degree 2^p, and $i^p : U^p \hookrightarrow U^0$ for the physical inclusion.
- K_p is the little filled Julia set, $K_p(j) = f^j(K_p)$ for $0 \leq j < p$, and $\mathcal{K}_p = \bigcup_{j=0}^{p-1} K_p(j)$; primitivity means the $K_p(j)$ are pairwise disjoint (Kahn I, §1.1). We abbreviate $A = K_p$, $B = \mathcal{K}_p$.
- The canonical-localization marked surfaces are $T' = (U^p \setminus F^{-1}B) \cup A$ and $T = (U^0 \setminus B) \cup A$, with complementary surfaces $\Omega_s = U^p \setminus F^{-1}B$ and $\Omega_t = U^0 \setminus B$ (Kahn I, §§7.4.2–7.4.3, pp. 45–46).
- For a subgroup $H \leq \pi_1(X, x)$ of a connected, locally path-connected, semilocally simply connected space, $X_H \to X$ denotes the connected based cover corresponding to H. The *localization subgroups* $H_s \leq$

$\pi_1(\Omega_s, z_*)$, $H_t \le \pi_1(\Omega_t, z_*)$ are the images of the fundamental groups of deleted local neighborhoods of the marked continuum A (Kahn I, §7.4.1, Theorem 7.12, p. 44). Kahn requires of a local neighborhood only that every component of its complement of A be an annulus; the specialization to *Jordan* neighborhoods made in this paper is our own, and is justified by Lemma 4.1.

- Boldface domains $\mathbf{U}^m$ are the stages of the exact equalizer tower of Section 4 over the canonical seed, with complementary annuli $A_m = \mathbf{U}^m \setminus K^m$. Boldface domains $\mathbf{U}^m_{\mathrm{KL}}$, with $A^{\mathrm{KL}}_m = \mathbf{U}^m_{\mathrm{KL}} \setminus K(f)$, are the ψ-quadratic-like (bold) domains on which the Kahn–Lyubich pseudo-puzzle of KL II, §3.2 is run. These two families of symbols are kept strictly distinct: comparing them is the entire content of the strong stage-alignment hypothesis (Definition 9.1), and they may be identified only where that hypothesis has been declared to be in force. $\boldsymbol{\iota}$ denotes the inclusion-induced pseudo-piece *immersion* and $\mathbf{g}$ the dynamical branched-covering arrow; these are distinct arrows (KL II, Lemma 3.4, pp. 12–13). We follow KL in reserving the word "inclusion" for the puzzle-piece relation $Y \subset Z$: the ambient arrow is an immersion, and is never proved to be an embedding.
- (q, ν) are what KL II call the *decoration parameters* (KL II, Corollary 5.14 and Theorem 5.15, pp. 23–24), q being the denominator of the combinatorial rotation number at the α-fixed point (KL II, §2.1, p. 5). Where this paper says "decoration period q and depth index ν" it means exactly KL's decoration parameters.
- For a connected annulus, $\pi_\bullet : \widetilde{A}_\bullet \to A_\bullet$ is a universal cover with positively oriented deck generator $T_\bullet$; the (annular) *degree* of a map of annuli is the integer it induces on $\pi_1 \cong \mathbb{Z}$ with respect to the chosen orientations.
- $\mathrm{Ind}_x(\gamma)$ is the winding number of a loop γ about a point $x \in \mathbb{C} \setminus |\gamma|$.
- Citations of the form "Kahn I, §X / Theorem N, p. P" refer to [Kah23] in the numbering of arXiv:math/0609045v3 (28 December 2023); citations of the form "KL II, Lemma N, p. P" refer to [KL08] in the numbering of Ann. Sci. Éc. Norm. Supér. (4) **41** (2008) 57–84 and of the identically sectioned arXiv:math/0609046v2. The SHA-256–locked local TEX archives and their line-number concordance are recorded in Appendix D only.

Definition 3.1 (Localized lift data)**.** Let X_s, X_t be connected, locally path-connected, semilocally simply connected spaces, and let

$$p_s : \widetilde{X}_s \to X_s, \qquad p_t : \widetilde{X}_t \to X_t$$

be connected based covers corresponding to subgroups $H_s \le \pi_1(X_s, x_s)$ and $H_t \le \pi_1(X_t, x_t)$. Given a based embedding $a : (X_s, x_s) \hookrightarrow (X_t, x_t)$, a *based lift* of a is a map $A : \widetilde{X}_s \to \widetilde{X}_t$ with

$$p_t \circ A = a \circ p_s.$$

By the standard lifting criterion [Hat02], such an A exists if and only if $a_*(H_s) \subseteq H_t$.

Definition 3.2 (Canonical localization subgroups at a primitive stage)**.** Let f, $F = f^p$, $A = K_p$, $B = \mathcal{K}_p$, Ω_s, Ω_t be as in the Notation. Fix a Jordan local neighborhood $D_A \Subset T'$ of A meeting no other component of $F^{-1}B$, and a basepoint $z_* \in D_A \setminus A$. The *source localization subgroup* is

$$H_s = \operatorname{im}\big[\pi_1(D_A \setminus A, z_*) \longrightarrow \pi_1(\Omega_s, z_*)\big],$$

and the *target localization subgroup* $H_t \leq \pi_1(\Omega_t, z_*)$ is defined analogously from a local neighborhood of A in T. The *localization covers* $\widehat{\Omega}_s \to \Omega_s$, $\widehat{\Omega}_t \to \Omega_t$ are the connected covers corresponding to H_s, H_t, and the *localized inclusion* $\widehat{i} : T'_A \to T_A$ is Kahn's unique commuting lift of the physical inclusion $a = i^p$, completed by adjoining the marked continuum A.

Definition 3.3 (Degree-one annular lift datum)**.** Let A_s, A_t be connected annuli with universal covers $\pi_s : \widetilde{A}_s \to A_s$, $\pi_t : \widetilde{A}_t \to A_t$ and positively oriented deck generators T_s, T_t. A *degree-one annular lift datum* is a holomorphic immersion $I : A_s \to A_t$ whose induced homomorphism on fundamental groups is multiplication by $+1$, together with a lift $L : \widetilde{A}_s \to \widetilde{A}_t$ satisfying $\pi_t \circ L = I \circ \pi_s$.

Theorem 3.4 (Subgroup-reflection criterion)**.** *Let X_s, X_t be connected, locally path-connected, semilocally simply connected spaces, let*

$$p_s : \widetilde{X}_s \to X_s, \qquad p_t : \widetilde{X}_t \to X_t$$

be connected based covers corresponding to subgroups $H_s \leq \pi_1(X_s, x_s)$, $H_t \leq \pi_1(X_t, x_t)$, let $a : (X_s, x_s) \hookrightarrow (X_t, x_t)$ be a based embedding, and let $A : \widetilde{X}_s \to \widetilde{X}_t$ be a based lift with $p_t \circ A = a \circ p_s$. The existence of A forces

$$a_*(H_s) \subseteq H_t.$$

Then

$$A \text{ is injective} \quad \Longleftrightarrow \quad a_*^{-1}(H_t) = H_s.$$

If the equality fails, then every class

$$[\gamma] \in a_*^{-1}(H_t) \setminus H_s$$

is an exact collision certificate: the lift of γ to $\widetilde{X}_s$ starting at the base lift has distinct endpoints, and A identifies those endpoints.

Proof. Choose base lifts $\widetilde{x}_s \in p_s^{-1}(x_s)$ and $\widetilde{x}_t = A(\widetilde{x}_s)$. The standard lifting criterion states that the based lift A exists if and only if

$$a_*\big(p_{s*}\pi_1(\widetilde{X}_s, \widetilde{x}_s)\big) \subseteq p_{t*}\pi_1(\widetilde{X}_t, \widetilde{x}_t),$$

which is exactly the inclusion $a_*(H_s) \subseteq H_t$.

Step 1: reflection equality implies injectivity. Assume $a_*^{-1}(H_t) = H_s$, and let $\widetilde{u}, \widetilde{v} \in \widetilde{X}_s$ satisfy $A(\widetilde{u}) = A(\widetilde{v})$. Projecting and using $p_t \circ A = a \circ p_s$,

$$a(p_s(\widetilde{u})) = p_t(A(\widetilde{u})) = p_t(A(\widetilde{v})) = a(p_s(\widetilde{v})).$$

Since a is injective,

$$p_s(\widetilde{u}) = p_s(\widetilde{v}) =: u.$$

Choose paths $\widetilde{\alpha}, \widetilde{\beta}$ in $\widetilde{X}_s$ from $\widetilde{x}_s$ to $\widetilde{u}$ and $\widetilde{v}$ respectively, and put $\alpha = p_s \circ \widetilde{\alpha}$, $\beta = p_s \circ \widetilde{\beta}$; both run from x_s to u. By the covering endpoint criterion, the source endpoints agree, $\widetilde{u} = \widetilde{v}$, if and only if the loop $\gamma = \alpha * \beta^{-1}$ has class in H_s.

On the target side, $A \circ \widetilde{\alpha}$ and $A \circ \widetilde{\beta}$ are the lifts, from the common base lift $\widetilde{x}_t$, of $a \circ \alpha$ and $a \circ \beta$. The hypothesis $A(\widetilde{u}) = A(\widetilde{v})$ says that these lifts have the same endpoint; by the endpoint criterion applied in the target,

$$a_*[\gamma] = [(a\alpha) * (a\beta)^{-1}] \in H_t.$$

The assumed reflection equality gives $[\gamma] \in H_s$, hence $\widetilde{u} = \widetilde{v}$. Thus A is injective.

Step 2: injectivity implies reflection equality. Assume A is injective. The lift-existence inclusion gives

$$H_s \subseteq a_*^{-1}(H_t).$$

For the reverse inclusion, let $[\gamma] \in a_*^{-1}(H_t)$. Lift γ from $\widetilde{x}_s$ and denote its endpoint by $\widetilde{y}_s$. The path $A \circ \widetilde{\gamma}$ is the lift of $a \circ \gamma$ from $\widetilde{x}_t$. Since $a_*[\gamma] \in H_t$, this target lift is closed. Consequently

$$A(\widetilde{y}_s) = A(\widetilde{x}_s).$$

Injectivity of A gives $\widetilde{y}_s = \widetilde{x}_s$, i.e. the source lift of γ is closed, so $[\gamma] \in H_s$. Hence $a_*^{-1}(H_t) \subseteq H_s$, and with the previous inclusion this proves the equivalence.

Step 3: exact failure certificate. If $[\gamma] \in a_*^{-1}(H_t) \setminus H_s$, then the source lift of γ begins at $\widetilde{x}_s$ and ends at a distinct point $\widetilde{y}_s$ (because $[\gamma] \notin H_s$), while the target lift of $a \circ \gamma$ is closed (because $a_*[\gamma] \in H_t$). Therefore

$$\widetilde{y}_s \neq \widetilde{x}_s, \qquad A(\widetilde{y}_s) = A(\widetilde{x}_s).$$

This is a point-level collision of A, not merely a failure of a label map; the loop class records the exact sheet displacement that becomes invisible after target localization. □

Corollary 3.5 (Path-reflection formulation)**.** *In the setting of Theorem 3.4, call two source paths α, β with common endpoints* source equivalent *if $[\alpha\beta^{-1}] \in H_s$, and* target equivalent *if $a_*[\alpha\beta^{-1}] \in H_t$. Then A is injective if and only if*

target equivalence of two source paths $\Longrightarrow$ source equivalence of those paths.

The opposite implication (source equivalence implies target equivalence) holds automatically by functoriality.

Proof. Let α, β be source paths with the same endpoints. By the covering endpoint criterion, their lifts from one common source lift have the same endpoint exactly when $[\alpha\beta^{-1}] \in H_s$, and the lifts of their target images from one common target lift have the same endpoint exactly when $a_*[\alpha\beta^{-1}] \in H_t$.

Hence the displayed implication holds for all pairs of source paths if and only if $a_*^{-1}(H_t) \subseteq H_s$; together with the automatic inclusion $H_s \subseteq a_*^{-1}(H_t)$ this is the reflection equality, which by Theorem 3.4 is equivalent to injectivity of A. The opposite implication is functoriality of a_* together with $a_*(H_s) \subseteq H_t$. □

Corollary 3.6 (Conditional revised-or-restricted embedding route)**.** *Consider a KL localization stage whose underlying map is a physical embedding a, with source and target localization subgroups H_s, H_t. If*

$$a_*^{-1}(H_t) = H_s,$$

then the localized inclusion is an embedding on the covered complementary component. If this equality holds at every stage through the complete direct-decoration horizon, then all induced degree-one universal-cover tile maps are injective; together with the separately proved component-bijection and tile-gluing criterion, the corresponding fixed pseudo-piece immersion is an interior holomorphic embedding.

Proof. The first assertion is Theorem 3.4 applied to the stage's localization covers: the reflection equality forces the localized lift to be injective, and it is an embedding on the covered complementary component (an injective holomorphic immersion is an open injective map, hence a homeomorphism onto its image). For the second assertion, stagewise injectivity of the localized base maps upgrades, via the tile/base equivalence (Theorem 9.8, proved in Section 9 below), to injectivity of every induced degree-one universal-cover tile lift. The final assertion combines tile injectivity with the fixed-pair component bijection through the tile-gluing criterion (Proposition 9.11, also proved in Section 9 below) to conclude that the glued pseudo-piece immersion is an interior holomorphic embedding. No circularity arises from the forward references: nothing in Sections 3–8 uses this corollary. □

3.2. **Peripheral isolation of pulled full continua.**

Standing conventions.

- A *full continuum* is a nonempty compact connected set $K \subset \mathbb{C}$ whose complement $\mathbb{C} \setminus K$ is connected. We write $K \Subset U$ for compact containment.
- $f\colon U^1 \to U^0$ is an *ordinary* quadratic-like restriction of a quadratic polynomial, primitively renormalizable of period $p > 1$. We set $U^r = f^{-r}(U^0)$; then $U^{r+1} \subset U^r$ and every restriction $f\colon U^{r+1} \to U^r$ is quadratic-like (Kahn I, §6, p. 31, arXiv:math/0609045v3).
- $F = f^p\colon U^p \to U^0$, a proper holomorphic map of degree 2^p; $i^p\colon U^p \hookrightarrow U^0$ is the *physical inclusion*.
- K_p is the filled Julia set of the p-renormalization; $K_p(j) = f^j(K_p)$ for $0 \le j < p$; $\mathcal{K}_p = \bigcup_{j=0}^{p-1} K_p(j)$. Primitivity means the $K_p(j)$ are pairwise disjoint (Kahn I, §1.1).

- $A = K_p$, $B = \mathcal{K}_p$. The marked source and target surfaces of the canonical localization are
$$T' = (U^p \setminus F^{-1}B) \cup A, \qquad T = (U^0 \setminus B) \cup A, \tag{1.1}$$
and $a = i^p \colon T' \to T$ is the restriction of the physical inclusion. For the physical inclusion $A' = (i^p)^{-1}(A) = A$: the marked continuum is literally the same set on both sides.
- $\Omega_s = T' \setminus A = U^p \setminus F^{-1}B$ and $\Omega_t = T \setminus A = U^0 \setminus B$.
- $D_A \Subset T'$ is a fixed Jordan local neighborhood of A, disjoint from every other component of $F^{-1}B$; $z_* \in D_A \setminus A$ is the localization basepoint. The *localization subgroups* are
$$H_s = \operatorname{im}\big[\pi_1(D_A \setminus A, z_*) \longrightarrow \pi_1(\Omega_s, z_*)\big], \tag{5.2}$$
and $H_t \leq \pi_1(\Omega_t, a(z_*))$, defined analogously from a Jordan local neighborhood of A in T; these are the subgroups to which Kahn's connected-complement localization covers correspond (Kahn I, §7.4.1, Theorem 7.12, p. 44). Kahn requires of a local neighborhood only that every component of its complement of A be an annulus; the Jordan choice is our own specialization, justified by Lemma 4.1.
- $\pi_s \colon \widehat{\Omega}_s \to \Omega_s$, $\pi_t \colon \widehat{\Omega}_t \to \Omega_t$ are the localization covers; $\widehat{a} \colon \widehat{\Omega}_s \to \widehat{\Omega}_t$ is the unique commuting lift of a; $\widehat{i} \colon T'_A \to T_A$ is Kahn's localized inclusion between the completed localized marked surfaces.
- For $x \in \mathbb{C} \setminus \Omega_s$ (in practice $x \in X \subset F^{-1}B$), $\mathrm{Ind}_x \colon \pi_1(\Omega_s, z_*) \to \mathbb{Z}$ is the winding-number homomorphism about x.
- *Pullback dynasty:* $(i_r, f_r) \colon (\mathbf{U}^{r+1}, K^{r+1}) \to (\mathbf{U}^r, K^r)$, $r \geq 0$, is Kahn's exact pullback iteration of the canonical seed $(i_0, f_0) = (\widehat{i}, \widehat{f})$, realized by the equalizers
$$\mathbf{U}^{r+2} = \{(x, y) : i_r(x) = f_r(y)\}, \quad i_{r+1}(x, y) = y, \quad f_{r+1}(x, y) = x \tag{2.1}$$
(Kahn I, §7.3.2, pp. 42–43). $A_r = \mathbf{U}^r \setminus K^r$ is the complementary annulus at depth r. For $m > n \geq 0$, $I_{m,n} = i_n \circ i_{n+1} \circ \cdots \circ i_{m-1} \colon A_m \to A_n$.
- *KL direct-decoration data:* $P = Y^{(\nu-1)q+1}$ with the two univalent f^q-pullbacks $Q_L, Q_R \subset P$ of depth $\nu q + 1$ (KL II, §2.1, p. 6, and Figure 2.1, p. 7). The *exact direct-decoration depths* are $m = \nu q + 1$ and $n = (\nu - 1)q + 1$.
- *Tile conventions:* T_s, T_t generate the deck groups of the universal covers of the source and target annuli; a *compatible universal-cover tile lift* L of $I_{m,n}$ is a lift to those universal covers satisfying the degree-one equivariance $L \circ T_s = T_t \circ L$. KL's pseudo-pieces are glued from universal-cover tiles of complementary components, and each source tile is embedded in its source pseudo-piece, with pairwise disjoint images (KL II, proof of Lemma 3.3, p. 12).

Definition 3.7 (Marked surfaces, local covers, localization; after Kahn)**.** A *marked Riemann surface* is a pair (S, A) with S a Riemann surface and

$A \subset S$ compact, connected, and full in the sense that no component of $S \setminus A$ has compact closure in S; a morphism $g\colon (S, A) \to (T, B)$ is a holomorphic map with $A = g^{-1}(B)$. The pair is *local* if every component of $S \setminus A$ is an annulus; a *local neighborhood* of A is a subsurface $U \supset A$ with (U, A) local. A morphism is a *local cover* if it restricts to a bijection of the marked continua and to a covering map of their complements. Kahn proves that every (S, A) admits a *localization*: a local pair $(S_A, \widehat{A})$ with a local cover $\pi_{S,A}\colon (S_A, \widehat{A}) \to (S, A)$, unique up to isomorphism. When $S\setminus A$ is connected, $S_A \setminus \widehat{A}$ is the cover of $S \setminus A$ corresponding to the image of $\pi_1(U \setminus A)$ for a local neighborhood U of A, completed by regluing U along its natural lift. Every morphism g has a unique commuting localized lift $\widehat{g}$, and $\widehat{g}$ is an immersion whenever g is (Kahn I, §7.4.1, Theorem 7.12, pp. 43–45, arXiv:math/0609045v3). The source asserts preservation of *immersion*; it never asserts preservation of embedding.

Definition 3.8 (Dynamical localization and canonical renormalization)**.** Given a ψ-polynomial-like $(i, f)\colon (\mathbf{U}', K') \to (\mathbf{U}, K)$ of degree d, an FJ-set $A \subset K$ with $A' = i^{-1}(A)$,[1] and a compact $B \subset K$ satisfying Kahn's hypotheses (in particular A is a component of B, B is compact with $\mathbf{U} \setminus B$ connected, A' is a component of $f^{-1}(A)$, and all branch values of f lie in B), set

$$T = (\mathbf{U} \setminus B) \cup A, \qquad T' = (\mathbf{U}' \setminus f^{-1}B) \cup A'.$$

Localizing the restricted morphisms yields the *dynamical localization* $(\widehat{i}, \widehat{f})\colon (T'_{A'}, \widehat{A'}) \to (T_A, \widehat{A})$, a ψ-polynomial-like map in which $\widehat{i}$ is an immersion and $\widehat{f}$ is a branched cover of degree equal to the local degree of f at A' (Kahn I, §7.4.2, p. 45). The *canonical renormalization* of a primitively p-renormalizable ψ-quadratic-like map is the dynamical localization of (i^p, f^p) at $A = K_p$ with removal of $B = \mathcal{K}_p$; its central local degree is two (Kahn I, §7.4.3, p. 45). For an ordinary primitive seed the marked surfaces are exactly T', T of (1.1), with $A' = A$.

This choice of B is not the only one available in the source, and the distinction is load-bearing here. Kahn I, §7.4.4, p. 46 describes two *further* natural renormalizations obtained as dynamical localizations of the *same* iterate (i^p, f^p) at the *same* $A = K_p$, but with a different removal set: one takes $B := A \cup \mathcal{K}_{pq}$, and a second variant is built from the postcritical set and the geodesic γ_p (it is for that variant that the pq-renormalization is the q-renormalization of the p-renormalization). Throughout this paper "canonical renormalization" always means the §7.4.3 construction with $B =$

[1] At Kahn I, §7.4.2, p. 45 the hypothesis list for dynamical localization reads $A' = i^{-1}(A)$, and §7.4.3, p. 45 then writes $K'_p = i^{-1}(K_p)$ while simultaneously instructing the reader to replace (i, f) by the iterate $(i^p, f^p)\colon \mathbf{U}^p \to \mathbf{U}^0$. Read with the un-iterated i, the two prescriptions name different maps. We use throughout the intended reading, in which i denotes the inclusion leg of the map actually being localized: $A' = (i^p)^{-1}(A) = A$ at the ordinary primitive seed, since $K_p \subset U^p$. This matters because Definition 3.9 turns on distinguishing A' from the other components X of $F^{-1}(A)$.

$\mathcal{K}_p$. The restriction is essential rather than decorative: the excess-component and removal-asymmetry argument of Section 4 turns on the source removing all of $F^{-1}B$ while the target removes only B, and enlarging B to $A \cup \mathcal{K}_{pq}$ changes that asymmetry. We make no claim about the §7.4.4 variants.

Definition 3.9 (Excess component)**.** An *excess component* is a component $X \neq A$ of $F^{-1}A$. (For the physical inclusion $A' = (i^p)^{-1}A = A$; the selected X is a *different* component of $F^{-1}A$ and must not be confused with A'.)

Definition 3.10 (Direct-decoration alignment)**.** A canonical seed *carries KL direct-decoration data* with *decoration parameters* (q, ν) — KL's own term (KL II, Corollary 5.14 and Theorem 5.15, pp. 23–24) — if the Yoccoz-puzzle apparatus of KL II applies to it, producing $P = Y^{(\nu-1)q+1}$ and the two univalent f^q-pullbacks $Q_L, Q_R \subset P$ of depth $\nu q + 1$ (KL II, §2.1, p. 6, and Figure 2.1, p. 7). The *exact direct-decoration depths* are $m = \nu q + 1$ and $n = (\nu - 1)q + 1$.

Lemma 3.11 (Jordan-neighborhood basis)**.** *Let $K \Subset U \subset \mathbb{C}$ be a nonempty full continuum and let $\mathcal{O}$ be any neighborhood of K compactly contained in U. Then there is a Jordan domain V with*

$$K \subset V \subset \overline{V} \subset \mathcal{O}. \tag{3.1}$$

Moreover there is a nested sequence of such Jordan domains with

$$K \subset V_{r+1}, \qquad \overline{V}_{r+1} \subset V_r, \qquad \bigcap_{r \geq 1} \overline{V}_r = K. \tag{3.2}$$

Proof. If K is a singleton, round disks centered at that point satisfy (3.1) and (3.2). Otherwise, fullness and connectedness of K make $\widehat{\mathbb{C}} \setminus K$ a simply connected domain of the sphere containing ∞. Uniformize it by the exterior of the closed unit disk, normalized at infinity: let $\Phi\colon \{|w| > 1\} \cup \{\infty\} \to \widehat{\mathbb{C}} \setminus K$ be the conformal isomorphism with $\Phi(\infty) = \infty$. For each radius $r > 1$ the image $\Phi(\{|w| = r\})$ is a Jordan curve in the complement of K, and its bounded complementary side $V(r)$ contains K. Every point z of the complement of K has uniformizing modulus $|\Phi^{-1}(z)|$ strictly greater than 1, so $z \notin \overline{V(r)}$ as soon as $r < |\Phi^{-1}(z)|$; hence $\bigcap_{r>1} \overline{V(r)} = K$. The closed sets $\overline{V(r)} \setminus \mathcal{O}$ are compact, decrease as $r \downarrow 1$, and have empty intersection, so some $V = V(r)$ with r sufficiently close to 1 satisfies $\overline{V} \subset \mathcal{O}$; this gives (3.1). Choosing the radii $r_k \downarrow 1$ strictly monotonically gives the nested sequence of (3.2), and the intersection equals K because every point of the complement has uniformizing modulus strictly greater than 1. □

Lemma 3.12 (One quadratic pullback)**.** *Let $q\colon U' \to U$ be a proper holomorphic map of degree two between plane topological disks, and let $K \Subset U$ be a nonempty full continuum. Let c be the unique critical point of q and $v = q(c)$.*

(1) *If $v \in K$, then $q^{-1}K$ is a single full continuum, and the local degree of q on it is two.*

(2) *If $v \notin K$, then $q^{-1}K$ is the disjoint union of two full continua, each mapped homeomorphically onto K by q.*

Proof. Choose the nested Jordan basis V_r of Lemma 3.11, with each ∂V_r avoiding v; if $v \notin K$, choose $\overline{V}_1$ disjoint from v.

Case $v \notin K$. Then $v \notin V_r$ for every r, so $q\colon q^{-1}(V_r) \to V_r$ is an unbranched proper covering of total degree two. Since V_r is simply connected, its preimage has exactly two components W_r^0, W_r^1, and each restriction $q\colon W_r^k \to V_r$ is a biholomorphism. The two inverse branches of q on V_1 label the components coherently for all r, and the nesting $\overline{V}_{r+1} \subset V_r$ gives

$$\overline{W}_{r+1}^k \subset W_r^k, \qquad k = 0, 1.$$

Set $L_k = \bigcap_{r\geq 1} \overline{W}_r^k$. As a nested intersection of nonempty compact connected sets, each L_k is nonempty, compact, and connected. It is full because

$$\mathbb{C} \setminus L_k = \bigcup_{r\geq 1} \big(\mathbb{C} \setminus \overline{W}_r^k\big)$$

is an increasing union of connected sets: each $\overline{W}_r^k$ is a closed Jordan disk, since ∂V_r avoids the critical value and the inverse branch carries V_r and its regular boundary biholomorphically. The inverse branches identify each L_k homeomorphically with $K = \bigcap_r \overline{V}_r$, and $q^{-1}K = L_0 \sqcup L_1$.

Case $v \in K$. Every V_r contains v. The component W_r of $q^{-1}(V_r)$ containing the critical point c maps with degree at least two; since the total degree is two, its degree is exactly two and W_r exhausts $q^{-1}(V_r)$. Riemann–Hurwitz for the proper degree-two map $W_r \to V_r$ with its single simple critical point gives

$$\chi(W_r) = 2\chi(V_r) - 1 = 1,$$

so W_r is a disk. Because ∂V_r contains no critical value, ∂W_r is a regular Jordan curve. The compact Jordan disks $\overline{W}_r$ are nested, and

$$q^{-1}K = \bigcap_{r\geq 1} \overline{W}_r.$$

The same increasing-complement argument as in the first case shows that this intersection is connected and full.

For completeness, the bordered Riemann–Hurwitz step can be obtained by doubling W_r and V_r across their regular boundaries. The doubled degree-two map of closed surfaces has exactly the critical point and its mirror, so

$$2 - 2g = 2 \cdot 2 - 2 = 2,$$

whence $g = 0$; and since the double of a planar domain with b boundary components has genus $b - 1$, we get $b = 1$, i.e. W_r is a disk with Jordan boundary. □

Corollary 3.13 (Finite full p-fold pullback)**.** *Under a chain of p quadratic-like pullbacks, the inverse image of one full continuum is a union of at most 2^p pairwise disjoint full continua. Every resulting component maps onto its*

parent continuum with positive integer degree. The inverse image of a finite pairwise-disjoint family of full continua is again a finite pairwise-disjoint family of full continua.

Proof. Apply Lemma 3.12 successively along the chain. At each stage every parent continuum has either one degree-two child or two degree-one children, and every child maps onto its parent. Children of different parents are disjoint because their images lie in the disjoint parent continua. Induction over the p stages gives the bound 2^p, the stated surjectivity, and a positive product degree for every terminal component. Applying this separately to the members of a finite pairwise-disjoint family proves the last statement. □

Corollary 3.14 (Isolating Jordan disks)**.** *Let $C_1, \dots, C_N \Subset U$ be pairwise disjoint full continua. For each i there is a Jordan domain $D_i \Subset U$ such that*

$$C_i \subset D_i, \qquad \overline{D}_i \cap C_j = \varnothing \quad (j \neq i). \tag{3.3}$$

The D_i may be chosen pairwise disjoint.

Proof. A finite pairwise-disjoint family of compacta in U has positive minimum pairwise distance and positive distance from ∂U. Apply Lemma 3.11 inside pairwise-disjoint metric neighborhoods of the C_i. □

Lemma 3.15 (Connectivity after finitely many full deletions)**.** *Let U be a plane topological disk and let $C_1, \dots, C_N \Subset U$ be pairwise disjoint full continua. Then*

$$U \setminus \bigcup_{i=1}^{N} C_i$$

is path connected.

Proof. Applying Lemma 3.11 twice inside the disjoint neighborhoods used in Corollary 3.14, choose pairwise-disjoint Jordan domains

$$C_i \subset D_i^-, \qquad \overline{D_i^-} \subset D_i^+, \qquad \overline{D_i^+} \subset U.$$

The finitely connected plane domain $U_0 = U \setminus \bigcup_{i=1}^{N} \overline{D_i^-}$ is connected: after a homeomorphism $U \cong \mathbb{D}$, removing finitely many pairwise-disjoint closed Jordan disks compactly contained in $\mathbb{D}$ leaves a disk with finitely many holes. As an open connected subset of the plane, U_0 is path connected.

For each i, the open set $D_i^+ \setminus C_i$ is connected. Indeed $\mathbb{C} \setminus C_i$ is connected by fullness, and its intersection with the Jordan disk D_i^+ cannot have a component separated from the outer collar of ∂D_i^+: such a component would be a bounded complementary component of C_i, contradicting fullness. Hence $D_i^+ \setminus C_i$ is path connected. It contains the nonempty annulus $D_i^+ \setminus \overline{D_i^-}$, which is also contained in U_0; thus every $D_i^+ \setminus C_i$ has nonempty connected

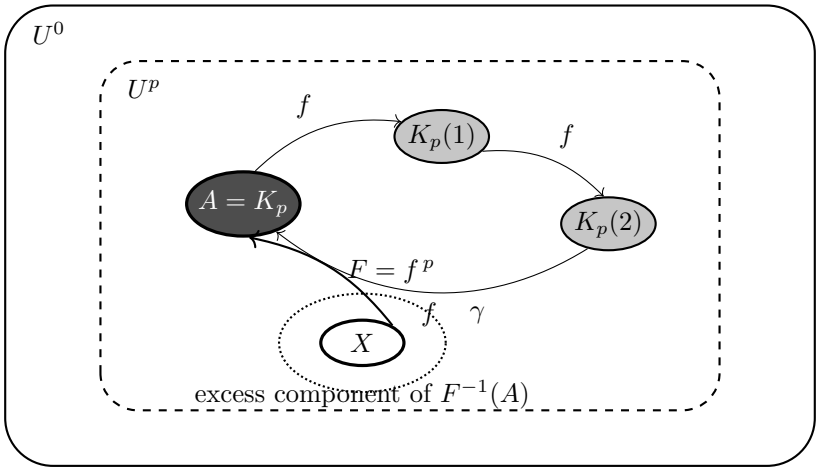


FIGURE 1. The canonical seed of a primitive renormalization of period p (drawn for $p = 3$). The cycle $B = \mathcal{K}_p = \bigcup_j K_p(j)$ is removed and the central continuum $A = K_p$ is retained on both sides. Because $\deg F = 2^p$ exceeds the central degree 2, the preimage $F^{-1}(A)$ has a component $X \neq A$. A based loop γ encircling X lies outside the source localization subgroup but is contracted by the physical inclusion in the target: its localized lift has distinct endpoints that the localized inclusion identifies.

overlap with U_0. Their union is path connected and equals

$$U_0 \cup \bigcup_{i=1}^{N} \big(D_i^+ \setminus C_i\big) = U \setminus \bigcup_{i=1}^{N} C_i. \qquad \square$$

4. Forced noninjectivity at ordinary primitive canonical seeds

This section proves Theorem 1.1. The mechanism is summarized by Figures 1 and 2: the degree budget produces an excess component X of $F^{-1}(A)$; peripheral isolation puts an isolating Jordan disk around X; the winding number about X defines a homomorphism killing the source localization subgroup but not the peripheral class; and the target contracts the peripheral loop across the isolating disk, since the target removes only B while the source removes all of $F^{-1}B$.

Lemma 4.1 (Jordan neighborhoods are local pairs)**.** *Let $A \subset \mathbb{C}$ be a nondegenerate full continuum compactly contained in a Jordan disk D_A. Then $D_A \setminus A$ is a topological annulus; in particular (D_A, A) is a local pair in the sense of the localization construction (Definition 2.4).*

Proof. A connected component of $D_A \setminus A$ whose closure does not meet ∂D_A would be a bounded component of $\mathbb{C} \setminus A$, contradicting fullness of A; hence

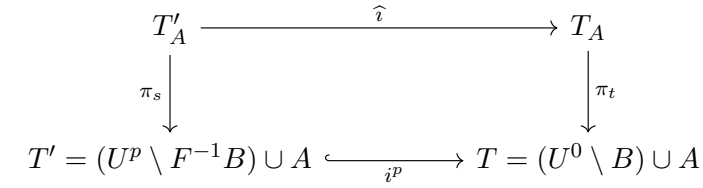


FIGURE 2. The localization covers. The covers π_s, π_t are the based covers classified by the localization subgroups $H_s \leq \pi_1(T' \setminus A)$ and $H_t \leq \pi_1(T \setminus A)$; the localized inclusion $\widehat{\imath}$ is the unique based lift of the physical inclusion i^p. The forced collision is the statement that $\widehat{\imath}$ is never injective at an ordinary primitive canonical seed of period $p > 1$: the class $[\gamma]$ of Figure 1 lies outside H_s while $(i^p)_*[\gamma]$ is trivial in H_t.

$D_A \setminus A$ is connected. In the sphere $\widehat{\mathbb{C}}$, the complement of $D_A \setminus A$ has exactly the two components A and $\widehat{\mathbb{C}} \setminus D_A$, so $D_A \setminus A$ is a doubly connected plane domain, i.e. an annulus. Its fundamental group is infinite cyclic, generated by a core curve, which is the defining property of a local pair. □

Definition 4.2 (Shared local neighborhood and based subgroup representatives)**.** Fix once and for all a Jordan disk $D_A \supset A$ with $\overline{D_A} \subset U^p$ and $\overline{D_A} \cap \big(F^{-1}B \setminus A\big) = \varnothing$; such a disk exists because $F^{-1}B \setminus A$ is a compact set disjoint from the component A of $F^{-1}B$. Since $\Omega_s = T' \setminus A \subset T \setminus A = \Omega_t$ and $D_A \setminus A \subset \Omega_s$, the same disk is a valid local neighborhood on both sides. With a common basepoint and connecting path fixed in $D_A \setminus A$, define

$$H_s = \operatorname{im}\big[\pi_1(D_A \setminus A) \to \pi_1(\Omega_s)\big], \qquad H_t = \operatorname{im}\big[\pi_1(D_A \setminus A) \to \pi_1(\Omega_t)\big].$$

These are the based representatives of the localization subgroups used throughout; the collision certificate is representative-independent (the winding homomorphism kills every conjugate of H_s).

Lemma 4.3 (Injectivity of the completion)**.** *In the connected-complement localization construction (Theorem 2.9), the natural map from the complement covering surface $\widehat{\Omega}$ into the completed localized surface S_A is injective.*

Proof. The completed surface is the quotient of the disjoint union $U \sqcup \widehat{\Omega}$ in which each point $u \in U \setminus A$ is identified with its image $\ell(u) \in \widehat{\Omega}$ under the canonical lift ℓ of $U \setminus A$. Every identification therefore pairs one point of U with one point of $\widehat{\Omega}$; no two points of $\widehat{\Omega}$ are ever identified with one another, since ℓ is single-valued. Hence the composition $\widehat{\Omega} \hookrightarrow U \sqcup \widehat{\Omega} \to S_A$ is injective. □

Remark 4.4. Lemma 4.3, applied on both sides, shows that the filled completion identifies no pair of distinct complement-cover points. Consequently every point collision of the completed localized inclusion $\widehat{\imath}$ arises from a

collision of the complement-cover lift $\widehat{a}$, and conversely; this is the bridge invoked in Theorem 4.8, now proved rather than delegated to the verification checklist of Section 13.

Proposition 4.5 (Excess central pullback)**.** *There is a component*

$$X \subset F^{-1}A, \qquad X \neq A. \tag{4.1}$$

Every such component satisfies

$$X \cap B = \varnothing. \tag{4.2}$$

Remark 4.6 (Complete component certificate). By Corollary 3.13 the family of excess components is finite. Fix any deterministic order on the finite pulled-component inventory and choose the first X; no theorem depends on this choice. Its complete finite itinerary is

$$\tau(X) = \big(X,\ f(X), \dots, f^p(X) = A;\ \varsigma_0, \dots, \varsigma_{p-1}\big), \tag{4.4}$$

where ς_r records the actual component (sheet) of $f^{-1}(f^{r+1}X)$ containing $f^r X$. If a stage already carries external or puzzle markings, those markings order the ς_r; otherwise (4.4) is a geometric component itinerary, not an invented symbolic word. By Corollaries 3.13 and 3.14 there is a Jordan domain $D_X \Subset U^p$ with

$$X \subset D_X, \qquad \overline{D}_X \cap \big((F^{-1}B \setminus X) \cup B\big) = \varnothing. \tag{4.5}$$

The peripheral access used in the collision is the based homotopy class of ∂D_X: a genuine access in the complement of the actual pulled obstacle family, not a ribbon-tree label.

Proof. The global map $F = f^p \colon U^p \to U^0$ is proper of degree 2^p (Kahn I, §6, p. 31). By the definition of the primitive quadratic-like renormalization, the central component A has local degree two under F (Kahn I, §7.4.3, p. 45).

Choose a sufficiently small Jordan neighborhood V of A inside the range of the quadratic-like renormalization. The central component of $F^{-1}(V)$ maps onto V with degree two. The degrees of all components of $F^{-1}(V)$ sum to 2^p, and since $p > 1$,

$$2^p - 2 > 0,$$

so there is a component W of $F^{-1}(V)$ other than the central one. The proper map $F \colon W \to V$ is onto, so W contains a point of $F^{-1}(A)$. The component X of $F^{-1}(A)$ through that point lies in W, hence is not the central component A. This proves (4.1).

For $0 \le j < p$ write $K_j = K_p(j) = f^j(A)$. Since $F = f^p$ commutes with f^j on these sets and $F(A) = A$,

$$F(K_j) = f^j(F(A)) = f^j(A) = K_j. \tag{4.3}$$

Suppose $X \cap K_j \neq \varnothing$. If $j \neq 0$, a point of the intersection is mapped by F into both A (because $X \subset F^{-1}A$) and K_j (by (4.3) and invariance), contradicting the primitive disjointness $A \cap K_j = \varnothing$. If $j = 0$, then $X \cap A \neq \varnothing$, so the two

connected subsets X and A of $F^{-1}A$ lie in one component, forcing $X = A$, again a contradiction. Hence (4.2). □

Theorem 4.7 (Localization subgroup defect)**.** *Let X and D_X be as in Proposition 4.5 and (4.5), and let H_s, H_t be the localization subgroups of (5.2). Let $\eta \subset \Omega_s$ be any path from z_* to ∂D_X and set*

$$\gamma = \eta * \partial D_X * \eta^{-1}. \tag{5.3}$$

Then

$$[\gamma] \notin H_s, \qquad a_*[\gamma] = 1 \in H_t,$$

and consequently

$$a_*^{-1}(H_t) \supsetneq H_s : \tag{5.7}$$

the exact failure of the subgroup-reflection equality on the actual canonical-localization data.

Proof. First, $a = i^p$ maps Ω_s into Ω_t: if $z \in B$, then $z \in K_j$ for some j, so $F(z) \in F(K_j) = K_j \subset B$ by (4.3); hence $B \subset F^{-1}B$, and therefore $U^p \setminus F^{-1}B \subset U^0 \setminus B$.

The removed family $F^{-1}B$ is a finite union of pairwise disjoint full continua by Corollary 3.13, so $\Omega_s = U^p \setminus F^{-1}B$ is path connected by Lemma 3.15; the path η exists, and γ of (5.3) is a well-defined based loop at z_*.

Fix $x \in X$. Winding number about x defines a homomorphism Ind_x on $\pi_1(\Omega_s, z_*)$, since $x \in X \subset F^{-1}B$ is disjoint from Ω_s. Every element of H_s is represented, before inclusion, by a loop in $D_A \setminus A$; the Jordan disk D_A is disjoint from every component of $F^{-1}B$ other than A, so $x \notin D_A$, and any loop inside the simply connected D_A is null-homotopic in $\mathbb{C} \setminus \{x\}$. Consequently

$$\mathrm{Ind}_x(H_s) = \{0\}. \tag{5.4}$$

On the other hand, the path and reverse path in (5.3) cancel in winding number, while ∂D_X (positively oriented) has winding number one about $x \in X \subset D_X$. Thus

$$\mathrm{Ind}_x(\gamma) = 1, \qquad [\gamma] \notin H_s. \tag{5.5}$$

For the target: (4.5) gives $\overline{D}_X \cap B = \varnothing$, and $D_X \subset U^p \subset U^0$; hence $D_X \subset \Omega_t$, and ∂D_X contracts across the disk D_X inside Ω_t. Therefore

$$a_*[\gamma] = 1 \in H_t. \tag{5.6}$$

Combining (5.5) and (5.6), $[\gamma] \in a_*^{-1}(H_t) \setminus H_s$, which is (5.7). □

Theorem 4.8 (Forced collision at ordinary primitive canonical seeds)**.** *Let $f\colon U^1 \to U^0$ be an ordinary quadratic-like restriction of a quadratic polynomial, primitively renormalizable of period $p > 1$, and let*

$$\widehat{i}\colon T'_A \longrightarrow T_A$$

be Kahn's localized inclusion of the canonical renormalization (the dynamical localization of (i^p, F) at $A = K_p$ with removal of $B = \mathcal{K}_p$). Then $\widehat{i}$ is not

injective. *More precisely, there are a component $X \neq A$ of $F^{-1}A$, a based loop $\gamma \subset U^p \setminus F^{-1}B$, and localization subgroups H_s, H_t such that*

$$[\gamma] \notin H_s, \qquad (i^p)_*[\gamma] = 1 \in H_t, \tag{1.2}$$

and the lift of γ to the source localization cover has distinct endpoints which $\widehat{i}$ identifies.

Remark 4.9 (Collision certificate). The proof yields the explicit certificate

$$\mathfrak{C}_{\mathrm{can}} = \big(X,\ D_X,\ [\gamma],\ \tau(X),\ H_s,\ H_t,\ H_s,\ H_s[\gamma],\ \widehat{z}_*,\ \widehat{z}'_*,\ \widehat{a}(\widehat{z}_*)\big), \tag{6.3}$$

recording the first excess component, the peripheral access, the complete p-step component itinerary, the source and target subgroups, the two distinguished source cosets, the distinct source sheets, and the common target point. No critical-value crossing and no numerical approximation is used to create the collision.

Proof. Take X, D_X, γ, H_s, H_t from Proposition 4.5, Remark 4.6, and Theorem 4.7; then (1.2) is (5.5)–(5.6).

Let $\pi_s\colon \widehat{\Omega}_s \to \Omega_s$ and $\pi_t\colon \widehat{\Omega}_t \to \Omega_t$ be the localization covers, corresponding to the subgroups H_s and H_t (Kahn I, §7.4.1, Theorem 7.12, p. 44), and let $\widehat{a}\colon \widehat{\Omega}_s \to \widehat{\Omega}_t$ be the unique commuting lift of $a = i^p$. Choose a source lift $\widehat{z}_* \in \pi_s^{-1}(z_*)$ and lift γ from $\widehat{z}_*$; write $\widehat{z}'_*$ for the terminal endpoint of the lifted path. By the covering-subgroup endpoint criterion — a loop lifts closed at $\widehat{z}_*$ exactly when its class lies in H_s — together with (5.5),

$$\widehat{z}'_* \neq \widehat{z}_*. \tag{6.1}$$

The $\widehat{a}$-image of the lifted path is the target lift, from $\widehat{a}(\widehat{z}_*)$, of the loop $a \circ \gamma$. By (5.6) the class of $a \circ \gamma$ is trivial, in particular lies in H_t, so this target lift is closed. Therefore

$$\widehat{a}(\widehat{z}'_*) = \widehat{a}(\widehat{z}_*). \tag{6.2}$$

Both points lie in the complementary annulus part of the localized source. Completing the localized marked surfaces by adjoining the marked continuum A glues A in along a local neighborhood and does not identify distinct points of the complement cover. Thus (6.1)–(6.2) exhibit two distinct points of T'_A with equal $\widehat{i}$-image in T_A: an actual point collision of Kahn's localized inclusion.

In path language: if α is any source path from the marked base access to z_*, the two classes $[\alpha]$ and $[\alpha * \gamma]$ are distinct in the source localization and equal after target localization. Kahn's forward path-equivalence statement (Kahn I, §7.4.5, pp. 46–47) — source-equivalent paths become target-equivalent — is respected; it is the (unstated) converse that fails. □

Remark 4.10 (Compatibility with the embedded inner collar). The published sources state that i *is* an embedding near the filled set: Kahn I, Theorem 7.3, p. 37 produces $U \supset K$ with $U' = f^{-1}U$ such that $i|_{U'}$ is an embedding and $\mathrm{mod}(U \setminus i(U')) \geq \mu > 0$; Kahn I, Lemma 7.4(2), p. 37 gives the uniformized form (an embedding on a round sub-annulus adjacent to the inner end); and

KL II, Lemma 3.1, p. 10 repeats that i is an embedding in a neighborhood of K'. Read quickly, this looks like a contradiction of Theorem 4.8. It is not: the two statements concern disjoint parts of the source, and it is worth locating the collision precisely.

Theorem 4.8 places the colliding points on the lift of the peripheral loop γ around the excess component X. By Proposition 4.5, X is a component of $F^{-1}(A)$ with $X \neq A$ and $X \cap B = \varnothing$, and its isolating disk D_X is disjoint from the isolating disk D_A of the central continuum. Thus γ lies in the *outer* part of the source $\Omega_s = U^p \setminus F^{-1}B$, at positive distance from A, whereas Theorem 7.3's embedding statement concerns a quadratic-like restriction shrunk toward K': the collar on which injectivity is proved separates K' from the outer end and, being disjoint from D_X, contains no point of $|\gamma|$. Injectivity on an inner collar and noninjectivity on the complement of that collar are simply different assertions; both hold.

We emphasize that nothing here weakens the sources. On the contrary, the same inner-collar embedding is what supplies the annular degree $+1$ in Lemma 9.5. What the sources do not supply, and what the counterexamples of Appendix B exist to forestall, is any extension of the inner-collar statement to a global one: Theorem B.10 exhibits a degree-one annular immersion satisfying all the hypotheses of Kahn's annulus lemma that is nevertheless noninjective, and Theorem B.13 exhibits a complete abstract ψ-quadratic-like pair with a remote self-collision compatible with the embedded inner collar. Those examples show that the gap between "embedded near K'" and "embedded" cannot be closed from the stated hypotheses; the present theorem shows that at the canonical seed it is not merely open but actually open in the negative direction.

Proposition 4.11 (Alignment and realization quantifiers)**.** *Items (1)–(2) below are proved without further hypotheses.*

(1) *Every first canonical localization of an ordinary primitively p-renormalizable quadratic polynomial, $p > 1$, has the collision with certificate (6.3).*
(2) *That collision propagates through every pullback depth of that same seed.*
(3) *If, in addition, the same seed carries KL direct-decoration data and is strongly aligned at the relevant depths in the sense of Definition 9.1, then the corresponding Kahn–Lyubich ambient ψ-ql arrow, its full tile lifts, and the glued pseudo-piece immersion are noninjective.*

Item (3) is an implication about an aligned realized stage. It does not infer that an arbitrary abstract canonical seed has a particular decoration.

Remark 4.12 (Separate realization routes; not used here). A separate realization statement may be supplied either by

- a named quadratic parameter whose primitive renormalization straightens to a non-satellite KL decoration germ, or
- the standard primitive-tuning theorem, applied to a primitive post-critically finite quadratic base and a non-satellite quadratic germ.

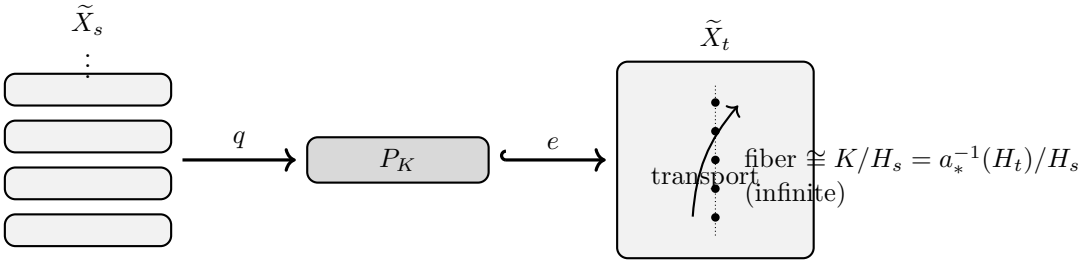


FIGURE 3. The canonical intermediate-cover resolution. The unresolved immersed pseudo-piece is replaced by the factorization $\widetilde{X}_s \xrightarrow{q} P_K \xhookrightarrow{e} \widetilde{X}_t$: a covering onto the embedded intermediate factor (unique in the based commuting comparison class) followed by an embedding. The sheet fiber over the embedded factor is a coset space detected by the winding homomorphism; it is infinite, and canonical transport moves sheets along paths in the base. Exact Kahn base change preserves the entire structure.

Until such a realization is source-locked, the present results are not used to change an existential statement about a narrower parameter subclass. What is settled unconditionally is the behavior of the actual canonical-localization map at ordinary primitive canonical seeds: any proof that treats that canonical localization as embedding-preserving is invalid.

Proof. Item (1) restates Theorem 4.8 together with the certificate of Remark 4.9. Item (2) restates Theorem 9.6(1)–(2), proved in Section 9 below. Item (3) restates Theorem 9.6(3)–(4), also proved in Section 9 below; no circularity arises from the forward reference, since nothing before Section 9 uses items (2)–(3). Its hypotheses — that the seed carries KL direct-decoration data and is strongly aligned at the relevant depths in the sense of Definition 9.1 — are carried verbatim into the conclusion and are not discharged here. No realization claim is made, and no empty-class or stage-alignment inference is hidden in any of the three items. □

Remark 4.13 (KL-facing item and strong alignment). Item (3) is the Kahn–Lyubich-facing item: it restates the KL-facing items (3)–(4) of Theorem 9.6, and it carries the strong stage-alignment hypothesis (Definition 9.1 of Section 9) explicitly in its statement. This is as it must be: existence of the KL direct-decoration data alone does not supply the identifications Θ_m, Θ_n required by that definition, which is exactly why item (3) assumes strong alignment in addition to the decoration data.

5. The canonical sheet resolution and the infinite defect

This section proves Theorems 1.2 and 1.4. The results are stated for an abstract based embedding a between bases of connected covers, with based lift A. The factorization is summarized by Figure 3.

Notational scope: within Sections 5–10 several symbols are reused in the abstract covering and tile settings, and each is to be resolved at its local declaration. Specifically: A is the localized lift of the base embedding a (here and again in Section 8), not the marked continuum $A = K_p$ of Sections 2–4; $K = a_*^{-1}(H_t)$ is the reflected subgroup, not a filled Julia set; T names the target of the resolved factorization in Definition 8.1, not the marked surface $(U^0 \setminus B) \cup A$; $H = \langle h \rangle$ is the wrapping group of the annular-tower typing in Section 10, distinct from the localization subgroups H_s, H_t; q and e denote the resolved covering and embedding legs (with tower iterates q_r, e_r), while in the direct-decoration depth formulas $m = \nu q + 1$, $n = (\nu - 1)q + 1$ of Section 9 the letter q is the Kahn–Lyubich decoration period; L denotes the intermediate subgroup in the proof of Theorem 5.13 and, in Sections 9–10, a universal-cover tile lift; P denotes the pullback $X_s \times_{X_t} \widehat{X}_t$ (with basepoint component P_K) in this section and the Kahn–Lyubich decoration piece $P = Y^{(\nu-1)q+1}$ in Section 9; and $\mathbf{U}^r$ denotes the complementary-cover equalizer stage of Sections 4 and 8, whose two usages are reconciled by the identity $i_r^{-1}(K^r) = K^{r+1} = f_r^{-1}(K^r)$: the complement-locus equalizer is the complement of the equalizer's filled set, so the stagewise complementary annuli $A_r = \mathbf{U}^r \setminus K^r$ are well defined at every depth.

The canonical localization of Section 4 is the instantiation $a = i^p$, $X_s = \Omega_s$, $X_t = \Omega_t$, $A = \widehat{a}$ (the complement-cover lift of Theorem 4.8); Kahn's localized inclusion $\widehat{\imath}$ is obtained from $\widehat{a}$ by the filled completion, which identifies no complement-cover points. The pointed coset set K/H_s is *not* asserted to be a group unless $H_s \trianglelefteq K$ is separately proved.

Notation and standing conventions. All spaces in this part are Hausdorff, connected, locally path-connected, and semilocally simply connected; all covering spaces are connected and based. These hypotheses are satisfied by the planar complementary surfaces used in the canonical KL localization. Path products and induced fundamental-group maps are based; every statement involving an unbased cover or subgroup is to be read after the displayed basepoints have been fixed.

For a based covering $p\colon (\widehat{X}, \widehat{x}) \to (X, x)$ we write

$$H \;=\; p_*\pi_1(\widehat{X}, \widehat{x}) \;\leq\; \pi_1(X, x)$$

for its characteristic subgroup.

For subgroups $H \leq K$ of a group,

$$K/H = \{\, kH : k \in K \,\} \tag{1}$$

denotes only the *pointed set* of cosets with distinguished point H. No multiplication on (1) is asserted unless $H \triangleleft K$. With the usual path-endpoint

convention a cover fiber is naturally the right-coset set $H\backslash K$; inversion gives the canonical pointed bijection $Hk \longmapsto k^{-1}H$ with (1). Thus the notation K/H does not silently assume normality.

For composable paths we use the paper-wide convention of Section 2: $\delta_1 * \delta_2$ means first traverse δ_1, then traverse δ_2, and in fundamental groups $[\delta_1][\delta_2] = [\delta_1 * \delta_2]$ (the first-traversed factor is written leftmost). In the endpoint description, continuing the fiber point labelled $H_s k$ by a based loop g appends g after k and gives $H_s kg$; under the inversion identification $H_s k \mapsto k^{-1}H_s$, that transport is

$$k^{-1}H_s \;\longmapsto\; g^{-1}k^{-1}H_s. \tag{2}$$

Formula (2) replaces any naive multiplication on K/H_s and remains valid when H_s is not normal in K.

Fixed symbols of this part: $a\colon (X_s, x_s) \hookrightarrow (X_t, x_t)$ is a based topological embedding (at the canonical KL primitive seed the spaces are the complementary surfaces, $X_s = \Omega_s$, $X_t = \Omega_t$, and $a = i^p$); $p_s\colon (\widehat{X}_s, \widehat{x}_s) \to (X_s, x_s)$ and $p_t\colon (\widehat{X}_t, \widehat{x}_t) \to (X_t, x_t)$ are connected based covers with subgroups H_s, H_t; $A\colon (\widehat{X}_s, \widehat{x}_s) \to (\widehat{X}_t, \widehat{x}_t)$ is a based lift satisfying $p_t A = a p_s$; $K := a_*^{-1}(H_t) \le \pi_1(X_s, x_s)$; $P = X_s \times_{X_t} \widehat{X}_t$ is the pullback and P_K the path component of the basepoint $(x_s, \widehat{x}_t)$, with first and second projections r and e; $q\colon \widehat{X}_s \to P_K$ is the canonical comparison map of Theorem 5.8; $\mathcal{S}_{\mathrm{loc}} = K/H_s$ is the based localization-sheet fiber; M_δ denotes sheet transport along a path δ in P_K; $\Pi_1(P_K)$ is the fundamental groupoid of P_K; and $\mathcal{G}_q$ is the canonical sheet-transport groupoid of Proposition 5.15. $\mathbf{Set}_{\mathrm{bij}}$ denotes the category of sets and bijections. A map defined on an open interval is *topologically proper* if preimages of compact sets are compact.

Definition 5.1 (Canonical localization covering data)**.** Let $a\colon (X_s, x_s) \hookrightarrow (X_t, x_t)$ be a based topological embedding, and let $p_s\colon (\widehat{X}_s, \widehat{x}_s) \to (X_s, x_s)$, $p_t\colon (\widehat{X}_t, \widehat{x}_t) \to (X_t, x_t)$ be connected based covers with characteristic subgroups H_s, H_t. Suppose a based lift $A\colon (\widehat{X}_s, \widehat{x}_s) \to (\widehat{X}_t, \widehat{x}_t)$ exists with

$$p_t A = a p_s. \tag{3}$$

The lifting criterion then gives $a_*(H_s) \le H_t$. Define

$$K := a_*^{-1}(H_t) \;\le\; \pi_1(X_s, x_s), \tag{4}$$

so that $H_s \le K$.

Definition 5.2 (Canonical pullback component)**.** With the data of Definition 5.1, form the pullback

$$P = X_s \times_{X_t} \widehat{X}_t = \big\{(x, \widehat{y}) : a(x) = p_t(\widehat{y})\big\}, \tag{5}$$

and let P_K be the path component of the basepoint $(x_s, \widehat{x}_t)$. Write

$$r\colon P_K \to X_s, \quad r(x, \widehat{y}) = x, \qquad e\colon P_K \to \widehat{X}_t, \quad e(x, \widehat{y}) = \widehat{y}. \tag{6}$$

Definition 5.3 (Sheet transport)**.** Let $q\colon \widehat{X}_s \to P_K$ be the covering of Theorem 5.8. For every path $\delta\colon [0,1] \to P_K$ and every $\widehat{u} \in q^{-1}(\delta(0))$, let $\widetilde{\delta}_{\widehat{u}}$ be the unique q-lift of δ beginning at $\widehat{u}$, and define

$$M_\delta(\widehat{u}) = \widetilde{\delta}_{\widehat{u}}(1) \in q^{-1}(\delta(1)). \tag{7}$$

Definition 5.4 (Enlargement data and the based coset map)**.** Let $H \leq K \leq G$ and $H^+ \leq K^+ \leq G^+$ be subgroup chains, and let $\varphi\colon G \to G^+$ be a homomorphism such that

$$\varphi(H) \leq H^+, \qquad \varphi(K) \leq K^+. \tag{8}$$

Then

$$\overline{\varphi}\colon K/H \longrightarrow K^+/H^+, \qquad kH \longmapsto \varphi(k)H^+ \tag{9}$$

is well defined: if $k_1H = k_2H$ then $k_2^{-1}k_1 \in H$, so $\varphi(k_2)^{-1}\varphi(k_1) \in \varphi(H) \leq H^+$, whence $\varphi(k_1)H^+ = \varphi(k_2)H^+$.

Lemma 5.5 (Subgroup of the pullback component)**.** *With the data of Definitions 5.1 and 5.2, the first projection $r\colon P_K \to X_s$ is the connected based cover of X_s corresponding to the subgroup $K = a_*^{-1}(H_t)$.*

Proof. The pullback of a covering is a covering. The pullback P is locally path-connected, so its path components are open. Moreover, the selected component P_K maps onto X_s: since a connected, locally path-connected space is path connected, for any $x \in X_s$ choose a path from x_s to x; its lift to the pullback from $(x_s, \widehat{x}_t)$ stays in P_K and ends over x. Hence the restriction $r\colon P_K \to X_s$ is itself a connected covering.

It remains to compute its subgroup. Let γ be a based loop in X_s. Its lift to P_K from $(x_s, \widehat{x}_t)$ is

$$t \longmapsto \big(\gamma(t), \widehat{a\gamma}(t)\big), \tag{10}$$

where $\widehat{a\gamma}$ is the lift of $a\gamma$ to $\widehat{X}_t$ from $\widehat{x}_t$. The lift (10) is closed if and only if $\widehat{a\gamma}$ is closed. By the cover–subgroup criterion, this is equivalent to $a_*[\gamma] \in H_t$, which, by (4), is equivalent to $[\gamma] \in K$. Therefore

$$r_*\pi_1\big(P_K, (x_s, \widehat{x}_t)\big) = K.$$

This also proves that the selected path component is precisely the based connected cover classified by K. □

Lemma 5.6 (The resolved leg is embedded)**.** *The second projection*

$$e\colon P_K \hookrightarrow \widehat{X}_t \tag{11}$$

is a topological embedding.

Proof. First consider the second projection on the entire pullback P. Its image is $p_t^{-1}(a(X_s))$. Because a is an embedding, the formula

$$\widehat{y} \longmapsto \big(a^{-1}(p_t(\widehat{y})),\, \widehat{y}\big) \tag{12}$$

is a continuous inverse on this image. Thus the second projection identifies P homeomorphically with $p_t^{-1}(a(X_s))$. Restricting this homeomorphism to P_K proves (11). □

Remark 5.7. The embedding hypothesis on a is essential: if a has a base-space self-intersection, equality of second coordinates in (5) need not determine the first coordinate, and Lemma 5.6 can fail before any subgroup collision is considered; compare Section 11.

Theorem 5.8 (Canonical sheet resolution)**.** *With the data of Definitions 5.1 and 5.2, define*

$$q\colon \widehat{X}_s \longrightarrow P_K, \qquad q(\widehat{u}) = \big(p_s(\widehat{u}),\, A(\widehat{u})\big). \tag{13}$$

Then:

(1) *q is the unique based map over X_s from the H_s-cover to the K-cover;*
(2) *q is a surjective covering map;*
(3) *the original lift factors exactly as*
$$A = e \circ q; \tag{14}$$
(4) *for all $\widehat{u}, \widehat{v} \in \widehat{X}_s$,*
$$A(\widehat{u}) = A(\widehat{v}) \quad \Longleftrightarrow \quad q(\widehat{u}) = q(\widehat{v}); \tag{15}$$
(5) *equivalently, the kernel pairs agree:*
$$\widehat{X}_s \times_{P_K} \widehat{X}_s = \widehat{X}_s \times_{\widehat{X}_t} \widehat{X}_s; \tag{16}$$
(6) *the fiber of q over the based point is the pointed coset set*
$$\mathcal{S}_{\mathrm{loc}} = K/H_s; \tag{17}$$
(7) *q is injective if and only if $K = H_s$; and*
(8) *if $[K : H_s] < \infty$, then q has exactly $[K : H_s]$ sheets. No finiteness is asserted when the index is infinite.*

Proof. The formula (13) lands in the pullback (5) because (3) gives

$$a(p_s(\widehat{u})) = p_t(A(\widehat{u})).$$

It lands in P_K because $\widehat{X}_s$ is path connected and q sends the basepoint to $(x_s, \widehat{x}_t)$. Moreover,

$$rq = p_s. \tag{18}$$

Since $H_s \leq K$, the lifting criterion for covers over the common base X_s gives a unique based map from the H_s-cover to the K-cover. Thus (13) is that map, proving (1).

We verify surjectivity directly rather than hiding it in cover classification. Let $z \in P_K$, and choose a path δ in P_K from the based point to z. Lift $r\delta$ to $\widehat{X}_s$ from $\widehat{x}_s$, obtaining a path $\widehat{\delta}$. Both δ and $q\widehat{\delta}$ are lifts of $r\delta$ to P_K from the same initial point. Uniqueness of path lifting gives

$$q(\widehat{\delta}(1)) = \delta(1) = z.$$

Hence q is onto. Choose a path-connected neighborhood $U \subset X_s$ evenly covered by both p_s and r. Each component of $p_s^{-1}(U)$ is mapped by q into one component of $r^{-1}(U)$; because both projections to U are homeomorphisms and $rq = p_s$, this restriction of q is a homeomorphism onto that target

sheet. Surjectivity of q ensures that every target sheet is met. Therefore q is a covering map, proving (2).

The second coordinate in (13) is $A(\widehat{u})$, so (14) is immediate, proving (3). Since e is injective by Lemma 5.6,

$$A(\widehat{u}) = A(\widehat{v}) \iff e(q(\widehat{u})) = e(q(\widehat{v})) \iff q(\widehat{u}) = q(\widehat{v}),$$

which proves (4) and (5).

The standard fiber classification for the map from an H_s-cover to a K-cover identifies its based fiber with the right-coset set $H_s\backslash K$. The inversion convention ($H_s k \mapsto k^{-1}H_s$) identifies that pointed set with K/H_s, proving (6). The map is one-sheeted exactly when the two subgroups agree, proving (7); in finite index its sheet number is $[K : H_s]$, proving (8). □

Corollary 5.9 (Exact meaning of the forced KL collision)**.** *For an ordinary primitive KL canonical localization, Theorem 4.8 (via display (5.7) of Theorem 4.7) gives*

$$H_s \subsetneq K = (i^p)_*^{-1}(H_t). \tag{19}$$

Consequently q in (13) *is a nontrivial covering and e is an embedded resolved pullback component. The peripheral loop around the proved excess component gives a non-base point of the based fiber* (17)*. All point collisions of the full localized inclusion A occur in fibers of this same q; none is left unexplained by the factorization.*

Proof. The proper inclusion (19) is display (5.7) of Theorem 4.7, supplied at the ordinary primitive canonical seed by Theorem 4.8. By Theorem 5.8(7), q is injective if and only if $K = H_s$; hence (19) makes q a nontrivial (non-injective) covering, while Lemma 5.6 exhibits e as a topological embedding of the resolved pullback component. The peripheral class $[\gamma]$ supplied by Theorem 4.8 lies in $K \setminus H_s$, so its coset $[\gamma]H_s \in K/H_s$ is a non-base point of the based fiber (17). Finally, Theorem 5.8(4)–(5) identify the kernel pair of A with the kernel pair of q, so every point collision of A occurs inside a single q-fiber. □

Remark 5.10. This corollary is unconditional at the canonical primitive seed. Its transport to a named KL direct-decoration depth carries the same-seed alignment qualification of Proposition 4.11.

Corollary 5.11 (The canonical sheet defect is infinite)**.** *At every ordinary primitive canonical seed,*

$$[K : H_s] = \infty. \tag{20}$$

Consequently q is an infinite-sheeted cover, and for every $z \in P_K$,

$$A^{-1}(e(z)) = q^{-1}(z)$$

is infinite. Equivalently: every non-singleton fiber of the completed localized inclusion (that is, every fiber over the embedded resolved complement component $e(P_K)$) is infinite; fibers over the filled part are singletons and are not at issue. In particular, no finite enumeration of the raw localization sheets can be complete.

Proof. Use the excess component X^{exc} (the component denoted X in Theorem 4.8, written X^{exc} here to avoid the abstract symbols X_s, X_t of this section), the point $x \in X^{\mathrm{exc}}$, and the peripheral class $[\gamma]$ supplied by Theorem 4.8. Winding about x defines a homomorphism

$$w\colon \pi_1(\Omega_s, x_s) \longrightarrow \mathbb{Z}$$

(here $x_s = z_*$, the localization basepoint, under the instantiation of this section's scope note). Every source-local loop is supported in the chosen neighborhood of the central component, which misses x; hence

$$w(H_s) = 0.$$

The peripheral class satisfies

$$w([\gamma]) = 1, \qquad [\gamma] \in K.$$

Because K is a subgroup, $[\gamma]^n \in K$ for every $n \in \mathbb{Z}$. If two cosets were equal,

$$[\gamma]^m H_s = [\gamma]^n H_s,$$

then $[\gamma]^{m-n} \in H_s$. Applying w would give $m - n = 0$. Thus the cosets $[\gamma]^n H_s$, $n \in \mathbb{Z}$, are pairwise distinct, proving (20).

Theorem 5.8 identifies the fiber cardinality of q with this index and identifies the A-fibers with the q-fibers, which gives the remaining assertions. □

Remark 5.12. Here $K/H_s = a_*^{-1}(H_t)/H_s$ is a pointed set of left cosets; it is not asserted to be a quotient group unless normality of H_s is separately proved. Any carrier (e.g. a ribbon tree) that identifies these cosets to a point hides the analytic obstruction and is not proof-faithful. The minimal replacement state recorded by the underlying research notes is

$$\mathfrak{S}_{\mathrm{loc}} = \big(\mathrm{Comp}(F^{-1}B),\ \{\tau(X)\},\ H_s, H_t,\ H_s\backslash\pi_1(\Omega_s),\ a_*^{-1}(H_t)/H_s\big). \quad (9.1)$$

Theorem 5.13 (Unique embedded intermediate factor)**.** *Suppose $r_C\colon C \to X_s$ is a connected based intermediate cover and*

$$\widehat{X}_s \xrightarrow{q_C} C \xrightarrow{e_C} \widehat{X}_t \quad (21)$$

satisfies, with q_C and e_C based,

$$r_C q_C = p_s, \qquad p_t e_C = a r_C, \qquad A = e_C q_C, \quad (22)$$

and e_C is injective. Then the subgroup of $C \to X_s$ is exactly K, and there is a unique based isomorphism of covers

$$C \cong P_K \quad (23)$$

which identifies (21) *with the canonical factorization $\widehat{X}_s \xrightarrow{q} P_K \xrightarrow{e} \widehat{X}_t$.*

Proof. Let

$$L = (r_C)_*\pi_1(C) \le \pi_1(X_s).$$

Existence of q_C gives $H_s \le L$. Existence of e_C gives

$$a_*(L) \le H_t,$$

so $L \leq K$.

Apply the subgroup-reflection criterion (Theorem 3.4, applied to the covers with subgroups L and H_t) to the lift $e_C \colon C \to \widehat{X}_t$ of the physical embedding a. Since e_C is injective, that criterion gives

$$a_*^{-1}(H_t) = L.$$

The left side is K, hence $L = K$. The classification of connected based covers now gives the unique based isomorphism (23), and uniqueness of based lifts identifies both factor maps. □

Remark 5.14. Thus P_K is the unique connected intermediate cover on which the remaining leg is embedded; replacing K/H_s by a coarser finite label set without proving the corresponding factor injective would destroy this optimality.

5.1. **The sheet-transport groupoid.**

Proposition 5.15 (Exact sheet transport)**.** *The transport maps M_δ of Definition 5.3 satisfy:*

1. *M_δ is a bijection, with inverse $M_{\delta^{-1}}$;*
2. *M_δ depends only on the homotopy class of δ relative to its endpoints;*
3. *for composable paths,*

$$M_{\delta_1 * \delta_2} = M_{\delta_2} \circ M_{\delta_1}; \tag{24}$$

4. *the constant path induces the identity; and*
5. *at the based point these transports give the monodromy action on the pointed fiber K/H_s.*

Hence

$$\Pi_1(P_K) \longrightarrow \mathbf{Set}_{\mathrm{bij}}, \qquad z \longmapsto q^{-1}(z), \quad [\delta] \longmapsto M_\delta \tag{25}$$

is a functor. Its category of elements is the canonical sheet-transport groupoid $\mathcal{G}_q$. Its full object set is

$$\big\{(z, \widehat{u}) : z \in P_K,\ \widehat{u} \in q^{-1}(z)\big\},$$

canonically the point set of $\widehat{X}_s$. The pointed coset set K/H_s is only its based fiber (equivalently, the object set after restricting (25) *to the based action groupoid).*

Proof. Every assertion is forced by uniqueness and homotopy invariance of lifts. Lifting δ and then δ^{-1} returns to the initial point, which proves bijectivity, giving (1). A homotopy relative endpoints lifts after an initial sheet is fixed, so the terminal sheet is constant through the homotopy, giving (2). The lift of a concatenation is the concatenation of the two successive lifts; the first-traversed factor δ_1 acts first and therefore appears innermost in the function composition, proving (24) and (3). Constant paths lift to constant paths, giving (4). Restricting (25) to the based fundamental group gives the usual monodromy action on the cover fiber, identified with K/H_s by the inversion convention, giving (5). □

Remark 5.16. No regularity of the cover is used. If H_s is not normal in K, the based fiber K/H_s is not a quotient group, but the groupoid (25) still gives complete path transport. Under the inversion convention, continuation by a based loop g acts by the explicit formula (2), $k^{-1}H_s \mapsto g^{-1}k^{-1}H_s$, not by a silently assumed quotient-group multiplication.

Proposition 5.17 (Compact-segment lifting classification)**.** *Let $\gamma\colon [0,1] \to \widehat{X}_t$ have image in the embedded resolved component $e(P_K)$, and put $\delta = e^{-1}\gamma$. Then the source paths*

$$\widehat{\delta}\colon [0,1] \to \widehat{X}_s \quad \text{such that} \quad A\widehat{\delta} = \gamma \tag{26}$$

are exactly the q-lifts of δ. There is one and only one such path for each initial sheet in $q^{-1}(\delta(0))$, and its final sheet is given by M_δ.

The same statement applies separately on every compact parameter interval on which a target path lies in $e(P_K)$.

Proof. If $q\widehat{\delta} = \delta$, then

$$A\widehat{\delta} = eq\widehat{\delta} = e\delta = \gamma.$$

Conversely, if (26) holds, then

$$e(q\widehat{\delta}) = A\widehat{\delta} = \gamma = e\delta.$$

Injectivity of e (Lemma 5.6) gives $q\widehat{\delta} = \delta$. Covering path lifting supplies exactly one lift from each initial sheet, and Proposition 5.15 computes its endpoint. □

Remark 5.18. Proposition 5.17 is the precise compact-segment replacement for treating A as an embedding. It does not classify a target path which leaves and re-enters $e(P_K)$, glue choices over nested intervals, prove convergence to declared prime ends, or prove the KKL unique-proper-lift predicate.

Proposition 5.19 (Topological properness, with its exact boundary)**.** *Let $\delta\colon (0,1) \to P_K$ be topologically proper. Then every q-lift $\widehat{\delta}\colon (0,1) \to \widehat{X}_s$ is topologically proper.*

If, in addition, $a(X_s)$ is open in X_t, then $e(P_K)$ is open in $\widehat{X}_t$. Let $\gamma\colon (0,1) \to \widehat{X}_t$ be topologically proper and let J be a component of $\gamma^{-1}(e(P_K))$. After parametrizing J as an open interval,

$$\delta_J = e^{-1}\gamma|_J$$

is topologically proper, and so are all of its q-lifts.

Proof. For compact $C \subset \widehat{X}_s$, the set $q(C)$ is compact. Therefore

$$\widehat{\delta}^{-1}(C) \subset \delta^{-1}(q(C)).$$

The right side is compact in $(0,1)$ by properness of δ. Since $\widehat{X}_s$ is Hausdorff, C is closed, so $\widehat{\delta}^{-1}(C)$ is closed in $(0,1)$, hence closed in the right side. It is therefore compact. This proves properness of the lift.

If $a(X_s)$ is open, then $p_t^{-1}(a(X_s))$ is open. Lemma 5.6 identifies the selected component $e(P_K)$ with an open component of that preimage. For compact $C \subset P_K$,

$$\delta_J^{-1}(C) = J \cap \gamma^{-1}(e(C)).$$

The set

$$S = \gamma^{-1}(e(C))$$

is compact because $e(C)$ is compact and γ is proper. Moreover $S \subset \gamma^{-1}(e(P_K))$. The latter set is open in $(0,1)$, and its components are open intervals which are also closed relative to that open set. Hence $J \cap S$ is closed in S, and is compact. The definition $\delta_J = e^{-1}\gamma|_J$ identifies $J \cap S$ with $\delta_J^{-1}(C)$. Thus δ_J is proper. Apply the first paragraph to every q-lift. □

Remark 5.20. Topological properness means escape from compact subsets. It does not give convergence to a declared prime end, subordination to one marked ideal interval, or uniqueness among the infinitely many possible proper sheet lifts.

6. Exact enlargement tests

Root or marking enlargement changes the spaces and the subgroups. The following exact tests decide when the old sheet data survives; the accompanying warnings record why the algebraic tests cannot replace a geometric audit.

Theorem 6.1 (Based-coset no-merger/no-birth criterion)**.** *For the based coset map $\overline{\varphi}\colon K/H \to K^+/H^+$ of Definition 5.4:*

$$\overline{\varphi} \text{ is injective} \iff K \cap \varphi^{-1}(H^+) = H; \tag{27}$$

$$\overline{\varphi} \text{ is surjective} \iff K^+ = \varphi(K)H^+; \tag{28}$$

and therefore

$$\overline{\varphi} \text{ is bijective} \iff \begin{cases} K \cap \varphi^{-1}(H^+) = H, \\ K^+ = \varphi(K)H^+. \end{cases} \tag{29}$$

If (27) *fails, every*

$$k \in \big(K \cap \varphi^{-1}(H^+)\big) \setminus H \tag{30}$$

is an exact based-coset merger certificate*: the non-base coset kH maps to the base coset H^+.*

If (28) *fails, every*

$$k^+ \in K^+ \setminus \varphi(K)H^+ \tag{31}$$

is an exact based-coset birth certificate*: its coset has no old preimage.*

Proof. Suppose

$$\overline{\varphi}(k_1H) = \overline{\varphi}(k_2H).$$

By the definition (9), this is equivalent to

$$\varphi(k_2^{-1}k_1) \in H^+.$$

Since $k_2^{-1}k_1 \in K$, injectivity holds exactly when this implication forces $k_2^{-1}k_1 \in H$, which is precisely the equality $K \cap \varphi^{-1}(H^+) = H$ in (27). Taking $k_2 = 1$ gives the explicit certificate (30).

A coset k^+H^+ lies in the image of $\overline{\varphi}$ exactly when there is $k \in K$ with

$$k^+H^+ = \varphi(k)H^+,$$

equivalently

$$k^+ \in \varphi(K)H^+.$$

Every target coset has a preimage exactly when (28) holds. Its failure is witnessed by (31). Combining (27) and (28) proves (29). □

Corollary 6.2 (Subgroup diagram for the enlarged based fiber)**.** *Suppose an old and an enlarged localization give a commutative group diagram: homomorphisms $a_*\colon G_s \to G_t$, $a_*^+\colon G_s^+ \to G_t^+$, $\varphi_s\colon G_s \to G_s^+$, $\varphi_t\colon G_t \to G_t^+$ with*

$$a_*^+ \circ \varphi_s = \varphi_t \circ a_*, \tag{32}$$

and with

$$\varphi_s(H_s) \le H_s^+, \qquad \varphi_t(H_t) \le H_t^+.$$

Put

$$K = a_*^{-1}(H_t), \qquad K^+ = (a_*^+)^{-1}(H_t^+).$$

Then commutativity gives

$$\varphi_s(K) \le K^+,$$

so the old-to-new based-coset map $\overline{\varphi_s}\colon K/H_s \to K^+/H_s^+$ exists. It has no coset mergers and no coset births if and only if

$$K \cap \varphi_s^{-1}(H_s^+) = H_s, \qquad K^+ = \varphi_s(K)H_s^+. \tag{33}$$

Proof. If $k \in K$, then $a_*(k) \in H_t$. Hence, by (32),

$$a_*^+(\varphi_s(k)) = \varphi_t(a_*(k)) \in \varphi_t(H_t) \subseteq H_t^+.$$

Thus $\varphi_s(k) \in K^+$. Apply Theorem 6.1 with $\varphi = \varphi_s$, $H = H_s$, $H^+ = H_s^+$. □

Remark 6.3. The first condition in (33) is *reflection*: no old non-base sheet becomes new-base. The second is *exhaustion*: every new sheet is represented by an old sheet. Commutativity of (32) alone proves neither.

7. Kernel normal form and the exponential defect

Throughout this section we work at an ordinary primitive canonical seed with the shared local neighborhood and based representatives of Definition 4.2, and we write

$$a = i^p\colon \Omega_s \hookrightarrow \Omega_t, \qquad N_{\mathrm{fill}} := \ker\big[a_*\colon \pi_1(\Omega_s) \to \pi_1(\Omega_t)\big], \qquad K = a_*^{-1}(H_t).$$

We call a_* the *hole-filling homomorphism*: it is induced by the inclusion that fills back into the target every component of $F^{-1}B \setminus B$ removed from the source.

7.1. **The sheet fiber is the hole-filling kernel.**

Theorem 7.1 (Kernel normal form)**.** *At an ordinary primitive canonical seed, the restriction $a_*|_{H_s}\colon H_s \to H_t$ is an isomorphism. Consequently*

$$K = N_{\mathrm{fill}} \rtimes H_s,$$

and the map

$$N_{\mathrm{fill}} \longrightarrow K/H_s, \qquad n \longmapsto nH_s,$$

is a canonical pointed bijection.

Proof. By Lemma 4.1, $D_A \setminus A$ is an annulus; let c be its positively oriented core generator, so $\pi_1(D_A \setminus A) \cong \mathbb{Z}$ is generated by $[c]$. The image of $[c]$ in $\pi_1(\Omega_s)$, and likewise in $\pi_1(\Omega_t)$, has winding number one about every point of A; since the winding number about a fixed point of A is a homomorphism to $\mathbb{Z}$, the image of $[c]^n$ is nontrivial for every $n \neq 0$. Hence both induced maps

$$\pi_1(D_A \setminus A) \cong \mathbb{Z} \longrightarrow H_s, \qquad \pi_1(D_A \setminus A) \cong \mathbb{Z} \longrightarrow H_t$$

are isomorphisms onto their images. Because a restricts to the identity on the common physical neighborhood $D_A \setminus A$, the triangle formed by these two maps and $a_*|_{H_s}$ commutes, and $a_*|_{H_s}\colon H_s \to H_t$ is an isomorphism.

Let $k \in K$. Since $a_*(k) \in H_t$, there is a unique $h \in H_s$ with $a_*(h) = a_*(k)$; then $n := kh^{-1} \in \ker a_* = N_{\mathrm{fill}}$ and $k = nh$. Injectivity of $a_*|_{H_s}$ gives $N_{\mathrm{fill}} \cap H_s = \{1\}$, and N_{fill}, being the kernel of a_* on $\pi_1(\Omega_s)$, is normal in K. Hence $K = N_{\mathrm{fill}} \rtimes H_s$. Finally, $k = nh$ gives $kH_s = nH_s$, and if $nH_s = n'H_s$ with $n, n' \in N_{\mathrm{fill}}$ then $n^{-1}n' \in N_{\mathrm{fill}} \cap H_s = \{1\}$; so $n \mapsto nH_s$ is a bijection onto K/H_s, pointed because $1 \mapsto H_s$. □

Remark 7.2. The pointed coset set K/H_s of Theorem 5.8 is not in general a group. Theorem 7.1 shows that at the ordinary canonical seed, with the shared based representatives of Definition 4.2, it nevertheless carries a canonical group model: the kernel of the hole-filling homomorphism. In one sentence, *canonical localization forgets exactly the kernel of the source-to-target hole-filling map.* The pointed-coset discipline of Sections 5–6 remains the correct general-position language; the group model is a feature of the canonical seed with the shared neighborhood, not of the abstract setting.

7.2. **The exact excess count.**

Theorem 7.3 (Exact excess count)**.** *Let $f(z) = z^2 + c$ be primitively renormalizable of period $p > 1$ with ordinary return $F = f^p\colon U^p \to U^0$ and central continuum $A = K_p$. Then*

$$F^{-1}(A) = A \sqcup X_1 \sqcup \cdots \sqcup X_{2^p-2},$$

where each X_j is a full continuum disjoint from B, and $F|_{X_j}\colon X_j \to A$ is a homeomorphism.

Proof. Differentiating the composition,

$$F'(z) = \prod_{r=0}^{p-1} f'\big(f^r(z)\big) = 2^p \prod_{r=0}^{p-1} f^r(z),$$

so z is a critical point of F exactly when $f^r(z) = 0$ for some $0 \le r < p$, with critical value $F(z) = f^{p-r}\big(f^r(z)\big) = f^{p-r}(0)$. Since $0 \in A$ and $f^j(A) \subset K_j$, we get $f^{p-r}(0) \in K_{p-r}$. For $1 \le r < p$ the index $p-r$ lies in $\{1, \dots, p-1\}$, so by primitive disjointness the critical value lies in a member of $B \setminus A$ and in particular not in A. For $r = 0$ the critical point is $z = 0$ itself, with critical value $f^p(0) \in A$. Hence *the only critical point of F whose critical value lies in A is* 0.

Step 1: isolate A in the source. By Corollary 3.13 the preimage $F^{-1}(A)$ has finitely many components, each a full continuum. Choose a Jordan neighborhood $N_A \Subset U^p$ of A whose closure misses every component of $F^{-1}(A)$ other than A, with $\partial N_A \cap F^{-1}(A) = \varnothing$. Then $F(\partial N_A)$ is compact and disjoint from A.

Step 2: choose the target neighborhood. Choose a Jordan neighborhood $V \supset A$ with $\overline{V} \subset U^0$, small enough that $\overline{V}$ is disjoint from $B \setminus A$, from every noncentral critical value $f^{p-r}(0)$ $(1 \le r < p)$, and from $F(\partial N_A)$.

Step 3: the central component. Let V_0 be the component of $F^{-1}(V)$ containing A (hence containing 0). Since V_0 is connected, meets N_A, and satisfies $F(V_0) \subset V$, while $V \cap F(\partial N_A) = \varnothing$, the set V_0 cannot meet ∂N_A; hence $V_0 \Subset N_A$. Every critical point of F lying in $F^{-1}(V)$ has critical value in V, hence equals 0; so V_0 is the unique component containing a critical point, and $F|_{V_0} \colon V_0 \to V$ is proper of degree 2. Because N_A contains no component of $F^{-1}(A)$ other than A,

$$F^{-1}(A) \cap V_0 \subset F^{-1}(A) \cap N_A = A, \qquad \text{so} \qquad F^{-1}(A) \cap V_0 = A.$$

Step 4: the univalent components. Every other component V_j of $F^{-1}(V)$ contains no critical point, so it carries a proper unbranched covering of the simply connected domain V, hence a biholomorphism. The degrees over V sum to $\deg F = 2^p$, so there are exactly $2^p - 2$ biholomorphic components $V_1, \dots, V_{2^p-2}$.

Step 5: the excess continua. For $j \ge 1$ set $X_j = (F|_{V_j})^{-1}(A)$: a homeomorphic copy of A, compact, connected, and a component of $F^{-1}(A)$.

Step 6: fullness. Each X_j is a component of the p-fold pullback of the full continuum A, so it is full by Corollary 3.13.

Step 7: disjointness from B. This is the argument of Proposition 4.5: an intersection $X_j \cap K_l$ would map into $A \cap K_l = \varnothing$ for $l \neq 0$ by primitivity, while $X_j \cap A = \varnothing$ because distinct components of $F^{-1}(A)$ are disjoint. □

Corollary 7.4 (Exponential homological defect)**.** *By Corollary 3.13 the set $F^{-1}(B)$ has finitely many components, all full continua, among them B's own components and the excess continua X_j. Apply Corollary 3.14 to this* entire *finite disjoint family and retain the disks belonging to the X_j: this*

yields pairwise disjoint isolating Jordan disks $D_j \supset X_j$ with

$$D_j \cap \big(F^{-1}(B) \setminus X_j\big) = \varnothing,$$

so in particular $D_j \cap B = \varnothing$ and $\partial D_j \subset \Omega_s$. Choose based paths $\eta_j \subset \Omega_s$, peripheral loops $\gamma_j = \eta_j * \partial D_j * \eta_j^{-1}$, and points $x_j \in X_j$. Then each γ_j represents a class of $N_{\mathrm{fill}} = \ker a_*$, and the winding vector

$$W = \big(\mathrm{Ind}_{x_1}, \dots, \mathrm{Ind}_{x_{2^p-2}}\big) \colon N_{\mathrm{fill}} \longrightarrow \mathbb{Z}^{\,2^p-2}$$

is a surjective homomorphism with $W([\gamma_j]) = e_j$. Consequently

$$\operatorname{rank} H_1\big(N_{\mathrm{fill}}; \mathbb{Z}\big) \;\geq\; 2^p - 2,$$

and, via Theorem 7.1, the canonical sheet fiber K/H_s carries at least $2^p - 2$ independent peripheral winding coordinates.

Proof. Each x_j lies in $F^{-1}B$, so $x_j \notin \Omega_s$ and Ind_{x_j} is a well-defined homomorphism $\pi_1(\Omega_s) \to \mathbb{Z}$. Each D_j is disjoint from B, hence contained in Ω_t, and γ_j contracts across D_j there, so $a_*[\gamma_j] = 1$ and $[\gamma_j] \in N_{\mathrm{fill}}$. Pairwise disjointness of the disks gives $\mathrm{Ind}_{x_i}([\gamma_j]) = \delta_{ij}$, i.e. $W([\gamma_j]) = e_j$; hence W is surjective. A surjection of N_{fill} onto the free abelian group of rank $2^p - 2$ factors through $H_1(N_{\mathrm{fill}}; \mathbb{Z})$, giving the rank bound. □

Remark 7.5. Corollary 7.4 sharpens Corollary 5.11 from "the sheet defect is infinite" to "the sheet defect has at least $2^p - 2$ independent winding coordinates." Any finite bookkeeping of the canonical localization must therefore lose not one but exponentially many independent peripheral directions, at every ordinary primitive seed.

7.3. **The unicritical analogue.**

Lemma 7.6 (Degree-d pullback dichotomy)**.** *Let $g \colon U' \to U$ be a proper holomorphic map of degree d between Jordan disks whose only critical point ω has local degree d, and let $K \subset U$ be a full continuum. If $g(\omega) \notin K$, then $g^{-1}(K)$ has exactly d components, each a full continuum mapped homeomorphically onto K. If $g(\omega) \in K$, then $g^{-1}(K)$ is a single full continuum, mapped properly with degree d.*

Proof. Choose nested Jordan neighborhoods $W_1 \Supset W_2 \Supset \cdots$ of K with $\bigcap_k W_k = K$, chosen to avoid $g(\omega)$ when $g(\omega) \notin K$. *Unbranched case:* over each simply connected W_k the restriction of g is a proper unbranched covering, hence splits into d biholomorphic inverse branches; the nested intersections of the branch images are d disjoint full continua mapped homeomorphically onto K (fullness passes to nested intersections of Jordan domains). *Branched case:* since the local model of g at ω is $z \mapsto z^d$, the fiber $g^{-1}(g(\omega))$ is the single point ω; as every component of $g^{-1}(W_k)$ maps onto W_k by properness and $g(\omega) \in W_k$, every component contains ω, so $g^{-1}(W_k)$ is connected and maps with full degree d. By Riemann–Hurwitz with one ramification point of multiplicity $d-1$, its Euler characteristic is $d \cdot 1 - (d-1) = 1$, so each $g^{-1}(W_k)$ is a simply connected Jordan domain; the

nested intersection $g^{-1}(K) = \bigcap_k g^{-1}(W_k)$ is a single full continuum mapped properly with degree d. □

Theorem 7.7 (Unicritical analogue)**.** *Let $f(z) = z^d + c$ with $d \geq 2$ be primitively renormalizable of period $p > 1$, with ordinary return $F = f^p$ and central continuum A. Then*

$$F^{-1}(A) = A \sqcup X_1 \sqcup \cdots \sqcup X_{d^p - d},$$

with every X_j a full continuum disjoint from B and $F|_{X_j} \colon X_j \to A$ a homeomorphism, and the corresponding winding vector is a surjection $N_{\text{fill}} \twoheadrightarrow \mathbb{Z}^{d^p - d}$.

Proof. Now $F'(z) = d^p \prod_{r=0}^{p-1} \big(f^r(z)\big)^{d-1}$, so the critical points of F are again the solutions of $f^r(z) = 0$, $0 \leq r < p$, with critical values $f^{p-r}(0)$; exactly as in Theorem 7.3, only $z = 0$ has critical value in A, the central component of $F^{-1}(V)$ has degree d, and the degree budget leaves exactly $d^p - d$ unbranched degree-one components. The isolation of A by N_A, the choice of V avoiding $F(\partial N_A)$ and the noncentral critical values, the fullness of the X_j via the degree-d analogue of Corollary 3.13 (Lemma 7.6 replacing the quadratic dichotomy; the finite-full-pullback induction then applies verbatim), the disjointness from B, and the disk isolation from the full finite family of components of $F^{-1}(B)$ all repeat verbatim. We phrase this as the unicritical analogue of the forced-collision setting: the localization construction of Definition 2.4 is stated for general degree, but the term *canonical renormalization* is used by the source in the quadratic setting, and we attribute no degree-d canonical renormalization definition to it. □

7.4. **Future work.**

Remark 7.8 (Toward the full kernel). Let r be the number of source-only components of $F^{-1}(B) \setminus B$, so that $r \geq 2^p - 2$ by Theorem 7.3. A finite-connectivity argument with compatible peripheral bases should identify $\pi_1(\Omega_s) \cong F_{p+r}$ and $\pi_1(\Omega_t) \cong F_p$ (free groups of the indicated ranks) and identify a_* with the free retraction killing the r extra peripheral generators. This would make N_{fill} a free group of countably infinite rank and sharpen the rank bound of Corollary 7.4 to

$$\operatorname{rank} H_1\big(N_{\text{fill}}; \mathbb{Z}\big) = \aleph_0;$$

the winding vector W would then be the canonical finite quotient attached to the $2^p - 2$ excess preimages of A. We do not claim this here: the finite-connectivity and peripheral-basis lemma is not yet written.

8. Exact base change

Notation and standing conventions.

- $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ is the unit disk, $\operatorname{Aut}(\mathbb{D})$ its group of conformal automorphisms, and $d_{\mathcal{E}}$ the hyperbolic metric of a domain $\mathcal{E} \cong \mathbb{D}$. $C \Subset \mathcal{E}$ means that C has compact closure in $\mathcal{E}$.

- All spaces are Hausdorff. For a covering map $p\colon \widehat{X} \to X$, $\mathrm{Deck}(p)$ denotes its deck group. A covering is *universal* when its total space is simply connected; the deck group of a universal covering acts simply transitively on each fiber. A nonregular infinite covering may have trivial deck group, so the universal typing is part of the data, never a convenience.
- For a based covering $p\colon (\widehat{X}, \widehat{x}) \to (X, x)$, its subgroup is $H = p_*\pi_1(\widehat{X}, \widehat{x}) \leq \pi_1(X, x)$. For subgroups $H \leq K$, the *pointed coset set* $K/H = \{kH : k \in K\}$ is a pointed set with basepoint H; no group structure is asserted unless $H \trianglelefteq K$.
- $a\colon (X_s, x_s) \hookrightarrow (X_t, x_t)$ denotes the based topological embedding underlying the canonical localization, with based connected covers of subgroups H_s, H_t, based lift A, and $K := a_*^{-1}(H_t) \leq \pi_1(X_s, x_s)$, so that $H_s \leq K$. The *canonical sheet resolution* is the factorization $A = e \circ q$ through the resolved pullback component, with q a covering whose based fiber is the pointed coset set K/H_s and e a topological embedding; $[K : H_s]$ is the *sheet defect*. (These are supplied by the resolution cluster; see Theorem 5.8 and Corollary 5.11 there.)
- For maps $i\colon X \to Z$ and $f\colon Y \to Z$, the fiber product is $X \times_Z Y = \{(x, y) : i(x) = f(y)\}$; the *kernel pair* of $g\colon X \to Y$ is $X \times_Y X$. A commutative square is *cartesian* when it is a pullback square.
- $\mathbf{U}^r$ denotes the r-th stage of the exact Kahn equalizer tower on the complementary-cover locus, with the two stage maps $i_r, f_r\colon \mathbf{U}^{r+1} \to \mathbf{U}^r$ (localized inclusion and dynamics), so that $\mathbf{U}^{r+2} = \{(x, y) \in \mathbf{U}^{r+1} \times \mathbf{U}^{r+1} : i_r(x) = f_r(y)\}$ and $i_{r+1}(x, y) = y$.
- For a discrete subgroup $\Gamma \leq \mathrm{Aut}(\mathbb{D})$, $\Lambda(\Gamma) \subset \partial\mathbb{D}$ is its limit set; Γ is *elementary* when $\Lambda(\Gamma)$ contains at most two points. A hyperbolic element of $\mathrm{Aut}(\mathbb{D})$ has one repelling and one attracting fixed point on $\partial\mathbb{D}$.
- Overlines denote continuous extensions to conformal ideal closures, e.g. $\overline{\Phi}$, $\overline{L}_{\mathrm{KL}}$. The ideal closures of the Kahn–Lyubich pseudo-pieces used here are the Hausdorff conformal ideal closures of the source texts; on the resolved side the point-valued ideal boundary is formed after the Fuchsian limit set has been removed (Kahn I, §2.3).
- $\widehat{A}_s, \widehat{A}_t$ denote the source and target universal-cover tiles of an actual Kahn–Lyubich pseudo-piece (KL II, §3.2.1, p. 12); t_s is the positive source annular deck generator; $\omega_-, \omega_+ \in \partial\widehat{A}_s$ are its two fixed prime ends, identified by the source with the named infinitely wrapping λ/J vertices; $L_{\mathrm{KL}}\colon \widehat{A}_s \to \widehat{A}_t$ is one source-indexed actual KL inclusion tile map.
- $\mathcal{E}, \mathcal{R}, T, q, e, L, N, G, H, h, \zeta_\pm$ always refer to the canonical resolved system and its annular-tower typing, Definition 8.1.

Definition 8.1 (Canonical resolved system; annular-tower typing)**.** Let $\mathcal{E} \cong \mathbb{D}$, and let

$$\mathcal{E} \xrightarrow{q} \mathcal{R} \xrightarrow{e} T \tag{34}$$

be a holomorphic factorization such that:

(1) q is a universal covering;
(2) e is a holomorphic embedding;
(3) $L := e \circ q \colon \mathcal{E} \to T$ is nonconstant.

Write $N := \mathrm{Deck}(q)$. The system has *nontrivial kernel* when $N \neq \{1\}$.

We further impose the *annular-tower typing*: the factorization (34) arises from the degree-one annular tower of the canonical sheet resolution, so that one fixed based disk action contains

$$G = N \rtimes H, \qquad H = \langle h \rangle \cong \mathbb{Z}, \tag{35}$$

where G is the deck group of the composite universal cover of the base piece (hence discrete, torsion-free, and acting freely on $\mathcal{E} \cong \mathbb{D}$), and the wrapping generator h is hyperbolic, with repelling and attracting fixed points

$$\zeta_-, \zeta_+ \in \partial\mathbb{D}, \tag{36}$$

the two source-annular wrapping points.

Definition 8.2 (Actual KL tile system)**.** Let

$$L_{\mathrm{KL}} \colon \widehat{A}_s \longrightarrow \widehat{A}_t \tag{37}$$

be one source-indexed actual Kahn–Lyubich inclusion tile map. Both tiles are universal covers of their ambient annuli (KL II, §3.2.1, p. 12). Let t_s be the positive source annular deck generator and let

$$\omega_-, \omega_+ \in \partial\widehat{A}_s \tag{38}$$

be its two fixed prime ends; in the horizontal-strip uniformization with deck translation $z \mapsto z + 1$ these are the two ends of the strip (KL II, §5.5). The locked source identifies these ends with the named infinitely wrapping λ/J vertices (KL II, §3.2.4, p. 13) and supplies continuous values

$$\overline{L}_{\mathrm{KL}}(\omega_-), \qquad \overline{L}_{\mathrm{KL}}(\omega_+) \tag{39}$$

in the Hausdorff conformal ideal closure of the target pseudo-piece (KL II, §3.2.4, p. 13).

No injectivity of L_{KL} is assumed.

Definition 8.3 (Whole-tile end-compatible comparison)**.** A *whole-tile end-compatible comparison* from the canonical resolved system (34) to the actual KL tile system (37) is a pair

$$\Phi \colon \mathcal{E} \longrightarrow \widehat{A}_s, \qquad B \colon T \longrightarrow \widehat{A}_t \tag{40}$$

satisfying all of the following.

(1) **Based universal-cover identification.** Φ is a biholomorphism and is a based isomorphism of universal covers over the same oriented source annulus.

(2) **Deck-generator compatibility.**

$$\Phi h = t_s \Phi. \tag{41}$$

(3) **Target separation.** B is continuous and injective on $L(\mathcal{E})$. A whole-target-tile isomorphism or embedding is a special case.

(4) **Composite square.**

$$\begin{array}{ccc} \mathcal{E} & \xrightarrow{L=eq} & T \\ {\scriptstyle\Phi}\downarrow & & \downarrow{\scriptstyle B} \\ \widehat{A}_s & \xrightarrow{L_{\mathrm{KL}}} & \widehat{A}_t \end{array} \qquad L_{\mathrm{KL}}\,\Phi = B\,L. \tag{42}$$

(5) **Named-end compatibility.** In the conformal ideal closures,

$$\overline{\Phi}(\zeta_-) = \omega_-, \qquad \overline{\Phi}(\zeta_+) = \omega_+. \tag{43}$$

Remark 8.4 (Status of clause (5)). Clause (5) follows from clauses (1)–(2) *with the biholomorphic typing of clause (1)*: after disk uniformization, Φ is a disk automorphism, extends homeomorphically to the conformal ideal circles, and conjugates the two hyperbolic transformations h and t_s; it therefore sends their repelling and attracting fixed points to the corresponding fixed points. Clause (5) is listed separately because it is the exact endpoint datum used by the proof of Theorem 10.5 and must survive every proposed final-root enlargement. Algebraic generator conjugacy by an arbitrary *interior* homeomorphism would not suffice: on the strip $\mathbb{R} \times (0,1)$ the homeomorphism $u(x,y) = (x - 1/y,\, y)$ commutes with the deck translation $h(x,y) = (x+1, y)$, yet $(n, 1/(2n)) \to +\infty$ while $u(n, 1/(2n)) = (-n, 1/(2n)) \to -\infty$; so generator conjugacy alone does not supply wrapping-end compatibility.

Remark 8.5 (No boundary extension of q or e is required). Definition 8.3 does not require a boundary extension of q or e. It requires only the actual KL composite extension (39) already supplied by the source.

Proposition 8.6 (Base-change propagation through exact equalizers)**.** *Let*

$$X \xrightarrow{i} Z \xleftarrow{f} Y$$

be continuous maps, and suppose

$$i = e \circ q, \qquad X \xrightarrow{q} P \xrightarrow{e} Z,$$

where q is a covering and e is an embedding. Form

$$E = X \times_Z Y, \qquad R = P \times_Z Y,$$

and define

$$q' \colon E \to R, \quad q'(x,y) = (q(x), y); \qquad e' \colon R \to Y, \quad e'(p,y) = y; \qquad i' \colon E \to Y, \quad i'(x,y) = y. \tag{44}$$

Then:

(1) *q' is the pullback covering of q (along the projection $R \to P$); in particular the square*

$$\begin{array}{ccc} E & \xrightarrow{q'} & R \\ \downarrow & & \downarrow \\ X & \xrightarrow{q} & P \end{array}$$

(vertical maps the first-coordinate projections) is cartesian;
(2) *e' is a topological embedding;*
(3) *$i' = e' \circ q'$;*
(4) *i' and q' have the same kernel pair; and*
(5) *for every $(p, y) \in R$,*

$$(q')^{-1}(p, y) = \{(x, y) : q(x) = p\} \cong q^{-1}(p). \tag{45}$$

(6) *if moreover e is* open *— equivalently, if $i = e \circ q$ is a local homeomorphism, i.e. an immersion in the holomorphic category — then e' is an open embedding and i' is again a local homeomorphism.*

Hence infinite collision fibers remain infinite under the full base-change equalizer.

Proof. (1) The displayed square is a pullback by direct substitution in the definitions: a point of $X \times_P R$ is a pair $(x, (p, y))$ with $q(x) = p$ and $e(p) = f(y)$, which is exactly a pair (x, y) with $e(q(x)) = f(y)$, i.e. with $i(x) = f(y)$; thus $X \times_P R = X \times_Z Y = E$, with q' corresponding to the second projection. The pullback of a covering is a covering, so q' is a covering; it is surjective because for $(p, y) \in R$ any $x \in q^{-1}(p)$ satisfies $i(x) = e(q(x)) = e(p) = f(y)$, so $(x, y) \in E$ and $q'(x, y) = (p, y)$.

(2) If $e'(p_1, y_1) = e'(p_2, y_2)$, then $y_1 = y_2 = y$. Membership in R gives

$$e(p_1) = f(y) = e(p_2),$$

and injectivity of e gives $p_1 = p_2$. On the image $\{y \in Y : f(y) \in e(P)\}$ the inverse is

$$y \longmapsto \big(e^{-1}(f(y)),\, y\big),$$

which is continuous because e is an embedding (so e^{-1} is continuous on $e(P)$). Hence e' is a topological embedding.

(3) Formula (44) gives $e'(q'(x, y)) = e'(q(x), y) = y = i'(x, y)$.

(4) Since e' is injective, $i'(u) = i'(v) \iff e'(q'(u)) = e'(q'(v)) \iff q'(u) = q'(v)$; the kernel pairs coincide.

(5) Formula (45) is immediate from the definition of q'.

(6) Suppose e is open, so that $e(P) \subset Z$ is open; then $i = e \circ q$ is a composite of a covering and an open embedding and is therefore a local homeomorphism, and conversely a local homeomorphism i forces $e = i \circ$ (local inverse of q) to be open. By (2), e' is injective with image $\{y \in Y : f(y) \in e(P)\} = f^{-1}(e(P))$, which is open in Y by continuity of f; hence e' is an open embedding. By (3), $i' = e' \circ q'$ is then a composite of a covering and an open embedding, hence again a local homeomorphism. □

Remark 8.7 (Clause (6) is what re-supplies Kahn's pullback hypothesis). Clause (6) is not decoration. Kahn's identification of the set-theoretic equalizer with the pullback in the category of complex 1-manifolds is available only when one of the two legs is an *immersion* (Kahn I, Lemma 7.8, p. 40, arXiv:math/0609045v3); on p. 41 the source exhibits the failure explicitly, with common ramification of order $m > 1$ turning the set-theoretic equalizer into m disks glued at a point while the true pullback has m components. Clauses (1)–(5) alone yield only that e' is a *topological* embedding, which does not by itself make i' open; clause (6) supplies exactly the missing openness, so that the hypothesis of Lemma 7.8 is regenerated at the next stage. Kahn establishes the same propagation independently, through Lemma 7.7, p. 40 and Lemma 7.9, p. 40, and records at §7.3.2, p. 42 that "i_1 is an immersion and f_1 is proper of degree d"; clause (6) is the base-change form of the same fact, and we record it so that Corollary 8.8 does not silently borrow it.

Corollary 8.8 (Propagation through the exact Kahn equalizer tower)**.** *Kahn's recursion on the complementary-cover locus is exactly such a base change:*

$$\mathbf{U}^{r+2} = \{(x, y) \in \mathbf{U}^{r+1} \times \mathbf{U}^{r+1} : i_r(x) = f_r(y)\}, \qquad i_{r+1}(x, y) = y.$$

If $i_r = e_r q_r$ with q_r a covering onto the resolved intermediate P_r and e_r an embedding, then setting

$$P_{r+1} = P_r \times_{\mathbf{U}^r} \mathbf{U}^{r+1}, \qquad q_{r+1}(x, y) = (q_r(x), y), \qquad e_{r+1}(p, y) = y,$$

one has $i_{r+1} = e_{r+1} q_{r+1}$, with q_{r+1} the pullback covering of q_r, e_{r+1} a topological embedding, the kernel pairs of i_{r+1} and q_{r+1} equal, and, for every $(p, y) \in P_{r+1}$,

$$q_{r+1}^{-1}(p, y) \cong q_r^{-1}(p).$$

By induction on r, the resolved factorization and its infinite full fibers propagate through every full, unselected exact Kahn equalizer on the complementary-cover locus.

Proof. Apply Proposition 8.6 with $X = Y = \mathbf{U}^{r+1}$, $Z = \mathbf{U}^r$, $i = i_r$, $f = f_r$, $q = q_r$, $e = e_r$; then $E = \mathbf{U}^{r+2}$, $R = P_{r+1}$, and the displayed maps are exactly (44). Conclusions (1)–(5) of the proposition give the asserted covering, embedding, factorization, kernel-pair, and fiber statements at stage $r + 1$. Conclusion (6) applies as well, because i_r is an immersion at every stage of Kahn's tower (Kahn I, Lemma 7.7, p. 40 and Lemma 7.9, p. 40, with the tower statement at §7.3.2, p. 42); it gives that e_{r+1} is an open embedding and i_{r+1} again an immersion, which is exactly the hypothesis Kahn's Lemma 7.8, p. 40 needs for the set-theoretic equalizer at stage $r + 2$ to be the pullback in the category of complex 1-manifolds (see Remark 8.7). The base stage is $r = 0$: on the complementary-cover locus i_0 coincides with the lift $\widehat{a}$ of Section 5, whose factorization is Theorem 5.8 instantiated at the canonical seed (Corollary 5.9). Iterating from that stage propagates the

factorization, together with the cardinality of every fiber, to all subsequent stages. □

Remark 8.9. The propagation acts on the complementary-cover locus only: it does not extend the factorization across a filled or prime-end completion, does not select a decoration-side component, and does not discharge the same-seed alignment hypothesis of Proposition 4.11. Note also that the restriction to the complementary locus $A_r = \mathbf{U}^r \setminus K^r$ is forced rather than convenient: Kahn I, §7.3.2, p. 42 records that $i\colon K' \to K$ is a homeomorphism and hence so is $i_1\colon K'' \to K'$, so every i_r is injective on the filled set and no nontrivial covering factor could exist over K^r.

9. Transport to tiles and pseudo-pieces

This section transports the canonical collision to the Kahn–Lyubich direct-decoration stages. Every Kahn–Lyubich-facing statement of this section is conditional on the *strong stage-alignment hypothesis*, Definition 9.1 below, which sharpens the alignment quantifier of Proposition 4.11 into a formal commuting-square hypothesis. The equivalences below are the exact transport mechanism; their negative resolution at canonical seeds is inherited from Theorem 4.8 under that hypothesis.

Notation for the two towers. Throughout this section $\mathbf{U}^r$ (and $A_r = \mathbf{U}^r \setminus K^r$) denotes *this paper's* equalizer tower over the canonical seed, as fixed in Section 4, while $\mathbf{U}^r_{\mathrm{KL}}$ (and $A_r^{\mathrm{KL}} = \mathbf{U}^r_{\mathrm{KL}} \setminus K(f)$) denotes the tower of ψ-quadratic-like domains on which the Kahn–Lyubich pseudo-puzzle of [KL08, §3.2] is run. The two symbols are *never* interchanged: the whole content of Definition 9.1 is a comparison between them, and identifying them notationally would make the definition vacuous (Remark 9.2).

Definition 9.1 (Strong stage alignment)**.** Suppose the canonical seed carries Kahn–Lyubich direct-decoration data with decoration parameters (q, ν) in the sense of Definition 3.10, and put $m = \nu q + 1$, $n = (\nu - 1)q + 1$. Write $I_{m,n}\colon A_m \to A_n$ for the Kahn composite of the exact equalizer tower over the canonical seed (Theorem 9.6) and $\iota_{m,n}\colon A_m^{\mathrm{KL}} \to A_n^{\mathrm{KL}}$ for the Kahn–Lyubich ambient ψ-ql arrow at the same depths. The seed is *strongly aligned* at the depths m, n if there exist basepoint-preserving conformal identifications $\Theta_m\colon A_m \to A_m^{\mathrm{KL}}$ and $\Theta_n\colon A_n \to A_n^{\mathrm{KL}}$ making the square

$$\begin{array}{ccc} A_m & \xrightarrow{\;I_{m,n}\;} & A_n \\ {\scriptstyle\Theta_m}\downarrow & & \downarrow{\scriptstyle\Theta_n} \\ A_m^{\mathrm{KL}} & \xrightarrow[\;\iota_{m,n}\;]{} & A_n^{\mathrm{KL}} \end{array}$$

commute, where Θ_m and Θ_n preserve

(i) the basepoints,
(ii) the positive annular generators,
(iii) the component and tile labels,

(iv) the selected source and target pseudo-piece copies, and
(v) the central-overlap gluing data.

Under strong alignment the two rows of the square are identified; the symbol $I_{m,n}$ is then used for both maps, but only where the identification has been declared to be in force.

Remark 9.2 (What the alignment hypothesis really asks: identify two seeds, not two constructions). It should be said plainly what Definition 9.1 does and does not compare, because a reading of the two towers as literally the same objects would make the hypothesis trivial.

(a) The ambient arrow on the Kahn–Lyubich side is not a separate construction. KL II, §3.2.1 opens with "Let (i, f) be a ψ-quadratic-like map" and imposes no condition on where that map came from; and KL II, §3.1, p. 10 constructs $\mathbf{U}^n_{\mathrm{KL}} \to \mathbf{U}^{n-1}_{\mathrm{KL}}$ merely as the "restriction" of (i, f), deferring for the general theory to Kahn I. There the restriction *is* the categorical pullback, built as the set-theoretic equalizer with the leg assignment $i_1(x, y) = y$, $f_1(x, y) = x$ and iterated into a tower (Kahn I, §§7.3.1–7.3.2, pp. 38–43, arXiv:math/0609045v3). Hence $\iota_{m,n}$ is the $(m-n)$-fold composite of exactly the same equalizer legs that define $I_{m,n}$. There is no separate path-space, surgery, or decoration route at the ambient level: path spaces enter one level down, in the construction of the pseudo-piece $\mathbf{Y}$ itself.

(b) Therefore the only free parameter is the seed. Strong alignment is not a comparison of two *constructions* but an identification of the two *seeds* to which one and the same construction is applied. Two readings are available.

- *The intended reading, used in this paper*: the KL puzzle machinery of §§2.1–3 of [KL08] is run on the canonical seed itself. This is legitimate, since KL impose no provenance condition on (i, f). Under it the two towers are literally the same, Θ_m and Θ_n are the identity, the hypothesis of Definition 9.1 holds trivially, and item (4) of Theorem 9.6 is *unconditional*.
- *The other reading*, which remains open: the machinery is run on a different seed. This is not idle — at the one place where KL identify the renormalized ψ-ql map, in the proof of Corollary 5.14 (KL II, p. 24; Ann. Sci. ÉNS 41 (2008)), they take $(\mathbf{U}', K')$ to be a *glued pseudo-piece* $(\mathbf{E}^{\chi-1}, K(Rf))$ of the Modified Principal Nest, whereas Kahn's canonical seed (Kahn I, §7.4.3, p. 45) is the dynamical localization of f^p at K_p with the cycle removed: a different ambient disk and a different removed set.

(c) Alignment is all-or-nothing. Because Θ_m and Θ_n are required to be *conformal* isomorphisms of annuli, they exist only if the moduli agree at both depths simultaneously. If the seeds coincide the identifications are the identity and nothing is being assumed; if they do not, there is no reason for the moduli to agree, and no passage of either source supplies such Θ_m, Θ_n. The hypothesis is therefore certainly satisfiable and certainly not vacuous,

but it is a *seed-identification* hypothesis, and that is exactly the residual open item recorded in Section 13.

Remark 9.3 (Alignment across distinct seeds is not supplied by decoration data alone). Under the second reading of Remark 9.2, mere existence of Kahn–Lyubich direct-decoration data — the piece $P = Y^{(\nu-1)q+1}$ and the decorations $Q_L, Q_R \subset P$ at the stated depths (KL II, §2.1, p. 6 and Figure 2.1, p. 7) — does *not* by itself supply the identifications Θ_m, Θ_n: the decoration data live on the bold Kahn–Lyubich stage, while the composite $I_{m,n}$ lives on the equalizer tower of the canonical seed, and no passage of the sources identifies the two with the listed structure preserved. Proving that the Kahn and Kahn–Lyubich constructions themselves furnish such Θ_m, Θ_n for *distinct* seeds is precisely the open source-alignment proposition of Section 13.

Remark 9.4 (Scope: Q_L, Q_R and what the conditional conclusion touches). One honesty note about the depths m, n. In the published KL II the pieces Q_L, Q_R occur only in §2.1, p. 6 (where they are defined) and in Figure 2.1, p. 7; they never reappear in §3 (pseudo-puzzle) or §5, and KL never form a pseudo-piece from them. Where KL do need an inclusion-type arrow into $\mathbf{P}$ — §5.6, Lemma 5.11, pp. 22–23 — the pieces used are the Q^v of depth $(2\nu-1)q+1$ supplied by Lemma 2.1, p. 6. Consequently, even if the alignment hypothesis were fully discharged, the conditional conclusion below would land on an arrow $\mathbf{Q}_L \to \mathbf{P}$ that KL II never constructs and never uses, and would touch no step of KL II's proof. This is consistent with the scope declared in Section 1.4 — this paper claims no effect on any argument of the sources — but it should be said out loud rather than left to inference.

Lemma 9.5 (Annular degree of the canonical inclusion)**.** *Let* $(i_r, f_r)\colon (\mathbf{U}^{r+1}, K^{r+1}) \to (\mathbf{U}^r, K^r)$ *be any stage of Kahn's pullback tower over the canonical seed, and put* $A_r = \mathbf{U}^r \setminus K^r$*. Then*

$$\deg\big(i_r\colon A_{r+1} \to A_r\big) = +1$$

as a map of oriented annuli. Consequently every composite $I_{m,n} = i_n \circ \cdots \circ i_{m-1}\colon A_m \to A_n$ *has annular degree* $+1$*; and at any depths at which the seed is strongly aligned in the sense of Definition 9.1, so does the Kahn–Lyubich ambient arrow* $\iota_{m,n}$*.*

Proof. Three statements of the published sources are used, none of them new.

(i) Homeomorphism on filled sets. A ψ-polynomial-like map satisfies $i^{-1}(K) = f^{-1}(K) = K'$ (Kahn I, Definition 7.1(4), p. 36, arXiv:math/0609045v3), and $i\colon K' \to K$ is a homeomorphism; Kahn transports this along the tower, observing at §7.3.2, p. 42 that since $i\colon K' \to K$ is a homeomorphism so is $i_1\colon K'' \to K'$. Each stage of the tower is again a degree-d ψ-polynomial-like map (Kahn I, Lemma 7.11, p. 42), so this holds at every r.

(ii) Embedding on an inner collar of controlled modulus. For a ψ-polynomial-like map one may choose $U \supset K$ with $U' = f^{-1}U$ such that $i|_{U'}$ is an embedding and $\operatorname{mod}(U \setminus i(U')) \geq \mu > 0$ (Kahn I, Theorem 7.3, p. 37), the uniformized statement being that the immersion is an embedding on a round sub-annulus $A(1, \rho')$ adjacent to the inner end (Kahn I, Lemma 7.4(2), p. 37). Kahn–Lyubich import exactly this: "i is an embedding in a neighborhood of K'" (KL II, Lemma 3.1, p. 10; Ann. Sci. ÉNS **41** (2008)).

(iii) The conclusion is stated outright for the canonical renormalization. In the proof of Kahn I, Theorem 8.11, p. 51, the canonical-renormalization inclusion is recorded as having degree one on the inner ideal boundary.

Now the argument. Fix r and apply (ii) at the stage (i_r, f_r): there is an annulus $N \subset A_{r+1}$ adjacent to the inner end of A_{r+1}, i.e. separating K^{r+1} from the outer end, on which i_r is injective. Let $C \subset N$ be a Jordan curve separating K^{r+1} from the outer end of A_{r+1}; since A_{r+1} is an annulus, $[C]$ generates $\pi_1(A_{r+1}) \cong \mathbb{Z}$, and we orient C so that $[C]$ is the positive generator. By injectivity of i_r on N, the image $i_r(C)$ is again a Jordan curve, and it lies in A_r because $i_r^{-1}(K^r) = K^{r+1}$. By (i), i_r carries K^{r+1} homeomorphically onto K^r, and i_r is proper toward the inner end; hence $i_r(C)$ separates K^r from the outer end of A_r, so $[i_r(C)]$ is a generator of $\pi_1(A_r) \cong \mathbb{Z}$. A holomorphic map is orientation preserving, so the generator obtained is the positive one. Therefore $(i_r)_*$ carries the positive generator to the positive generator, which is precisely $\deg(i_r) = +1$; this is (iii) at the seed stage, and the argument above propagates it to every r.

Annular degree is multiplicative under composition, so $\deg I_{m,n} = +1$. Finally, clause (ii) of Definition 9.1 requires Θ_m, Θ_n to preserve the positive annular generators, so under strong alignment $\iota_{m,n} = \Theta_n \circ I_{m,n} \circ \Theta_m^{-1}$ has annular degree $+1$ as well. □

Theorem 9.6 (Exact propagation)**.** *Let $(i_r, f_r)\colon (\mathbf{U}^{r+1}, K^{r+1}) \to (\mathbf{U}^r, K^r)$, $r \geq 0$, be Kahn's pullback iteration of the canonical seed $(i_0, f_0) = (\widehat{i}, \widehat{f})$, realized by the exact equalizers (2.1), and put $A_r = \mathbf{U}^r \setminus K^r$. Then:*

(1) *For every $r \geq 0$ there are points $x_r^+ \neq x_r^- \in A_{r+1}$ with $i_r(x_r^+) = i_r(x_r^-) = v_r \in A_r$; the choices made are recorded by a complete inverse-sheet word $\omega = (\epsilon_0, \epsilon_1, \dots)$ appended to the certificate (6.3).*

(2) *For all $m > n \geq 0$ the composed inclusion*

$$I_{m,n} = i_n \circ i_{n+1} \circ \cdots \circ i_{m-1}\colon A_m \to A_n \tag{7.4}$$

is noninjective:

$$I_{m,n}(u) = I_{m,n}(v), \qquad u \neq v. \tag{7.5}$$

(3) *If the seed carries KL direct-decoration data with decoration parameters (q, ν), then the two integers $m = \nu q + 1$, $n = (\nu - 1)q + 1$ at which item (2) is read off are the Kahn–Lyubich depths of Q_L, Q_R and of P.*

(4) *If the seed is strongly aligned at those depths in the sense of Definition 9.1, the corresponding KL ambient ψ-ql arrow, its compatible tile lifts, and its glued pseudo-piece immersion are noninjective.*

Proof. Base case. The collision (6.1)–(6.2) of Theorem 4.8 lives in the complementary annulus of the localized seed: it gives

$$x_0^+ \neq x_0^- \in A_1, \qquad i_0(x_0^+) = i_0(x_0^-) = v_0 \in A_0. \tag{7.1}$$

(The two distinct source-cover points map to one target-cover point, and v_0 lies off the filled continuum because both source points do.)

Induction step. Suppose

$$x_r^+ \neq x_r^- \in A_{r+1}, \qquad i_r(x_r^+) = i_r(x_r^-) = v_r \in A_r. \tag{7.2}$$

We claim the restriction $f_r\colon A_{r+1} \to A_r$ is an unbranched surjective covering. It is unbranched over A_r because $K^{r+1} = f_r^{-1}(K^r)$ and, connectivity of the filled set being equivalent to all branch values of f_r lying in K^r (Kahn I, Definition 7.1 and the remark following it, p. 36), no branch value lies in A_r. It is *surjective* for a different and stronger reason: f_r is a proper branched covering of degree d onto the disk $\mathbf{U}^r$ (Kahn I, Definition 7.1(3), p. 36), a property inherited at every stage of the tower — the leg π_g of the pullback is proper of degree d whenever the other leg is an immersion (Kahn I, Lemma 7.9, p. 40), Kahn records at §7.3.2, p. 42 that "f_1 is proper of degree d", and Lemma 7.11, p. 42 states that the pullback is again a degree-d ψ-polynomial-like map. A proper map of positive degree onto $\mathbf{U}^r$ is onto, and $f_r^{-1}(K^r) = K^{r+1}$ (Kahn I, Definition 7.1(4), p. 36) forces $f_r(A_{r+1}) \supset A_r$; as A_r is connected (it is the complementary annulus of a connected full set in a disk), $f_r\colon A_{r+1} \to A_r$ is an unbranched covering onto A_r.

Choose $y_r \in A_{r+1}$ with $f_r(y_r) = v_r$. In the exact equalizer (2.1) put

$$x_{r+1}^{\pm} = (x_r^{\pm}, y_r).$$

These are genuine points of $\mathbf{U}^{r+2}$ because $i_r(x_r^{\pm}) = v_r = f_r(y_r)$; they are distinct because their first coordinates differ; and

$$x_{r+1}^+ \neq x_{r+1}^-, \qquad i_{r+1}(x_{r+1}^{\pm}) = y_r =: v_{r+1}. \tag{7.3}$$

Moreover $x_{r+1}^{\pm} \in A_{r+2}$, since $K^{r+2} = i_{r+1}^{-1}(K^{r+1})$ and $i_{r+1}(x_{r+1}^{\pm}) = y_r \notin K^{r+1}$. This proves item (1); the recorded choices of the y_r constitute the complete inverse-sheet word ω, so no unlabeled existence step is used.

Item (2). At $r = m-1$, (7.3) provides two distinct points $u = x_{m-1}^+$, $v = x_{m-1}^-$ of A_m with the same i_{m-1}-image. Applying the remaining factors $i_n \circ \cdots \circ i_{m-2}$ to that common image yields (7.5).

Item (3). Item (2) already gives noninjectivity for *all* $m > n \geq 0$, so item (3) does no more than fix two integers. For KL's direct-decoration pair $P = Y^{(\nu-1)q+1}$, $Q_L, Q_R \subset P$ of depth $\nu q+1$ (KL II, §2.1, p. 6, and Figure 2.1, p. 7), those integers are $m = \nu q + 1$ and $n = (\nu-1)q + 1$. What is being asserted is exactly this: *the integers at which (7.5) is read off are the Kahn–Lyubich depths.* The identification of a tower stage index over the period-p

canonical seed with a Yoccoz puzzle depth for the pre-renormalization map is not asserted here; it is part of the strong stage-alignment hypothesis and is used only in item (4). See also Remark 9.4 on what KL II does and does not build at those depths.

Item (4). Under the strong-alignment hypothesis of item (4), the identifications Θ_m, Θ_n of Definition 9.1 identify $I_{m,n}$ with the KL ambient ψ-ql arrow, whose annular degree is $+1$ by Lemma 9.5 — derived, not assumed. Let L be a compatible universal-cover tile lift, with deck generators T_s, T_t and degree-one equivariance $L \circ T_s = T_t \circ L$, and let $\widetilde{u}, \widetilde{v}$ be lifts of u, v in the source tile. Since $L(\widetilde{u})$ and $L(\widetilde{v})$ project to the equal points $I_{m,n}(u) = I_{m,n}(v)$, there is a unique integer k with $L(\widetilde{v}) = T_t^k L(\widetilde{u})$. Equivariance gives

$$L\big(T_s^{-k}\widetilde{v}\big) = T_t^{-k} L(\widetilde{v}) = L(\widetilde{u}), \tag{7.6}$$

and $T_s^{-k}\widetilde{v} \neq \widetilde{u}$ because these two points project to $v \neq u$; so (7.6) is a collision of L at two distinct source points. KL's construction embeds each source tile in its source pseudo-piece — the glued universal-cover copies $\widehat{A}_i$ are embedded, with pairwise disjoint images, by the proof of KL II, Lemma 3.3, p. 12 — so the identification (7.6) survives the source gluing: it is a same-tile collision of the full inclusion-induced pseudo-piece immersion, which is therefore noninjective. □

Remark 9.7 (Consumption of the KL-facing items). Items (1)–(2) are unconditional at the canonical seed, and item (3) requires only the KL direct-decoration data, which fix the depths m, n. Item (4) is the Kahn–Lyubich-facing part of the theorem, and it carries the strong stage-alignment hypothesis (Definition 9.1) explicitly in its statement: the alignment data it invokes are exactly the identifications Θ_m, Θ_n of that definition, and no statement of this paper uses item (4) without them. Under the intended reading of Remark 9.2 — KL's puzzle machinery run over the canonical seed itself — those identifications are the identity and item (4) is unconditional; it is only the distinct-seed reading that leaves it conditional.

Theorem 9.8 (Tile/base injectivity equivalence)**.** *Let (I, L) be a degree-one annular lift datum (Definition 3.3). Then, after choosing compatible positive deck generators,*

$$L \circ T_s = T_t \circ L,$$

and

$$L \text{ is injective} \quad \Longleftrightarrow \quad I \text{ is injective.}$$

Proof. Equivariance. Both $L \circ T_s$ and $T_t \circ L$ are lifts of $I \circ \pi_s$ through π_t, and their relative deck displacement is the integer $I_*(1) = 1$; by uniqueness of lifts this gives $L \circ T_s = T_t \circ L$.

Base injectivity implies lift injectivity. Assume I is injective and suppose

$$L(x) = L(y).$$

Projecting by π_t and using $\pi_t \circ L = I \circ \pi_s$ gives $I(\pi_s(x)) = I(\pi_s(y))$, hence $\pi_s(x) = \pi_s(y)$. Universal-cover fibers are deck orbits, so there is a unique $k \in \mathbb{Z}$ with $y = T_s^k x$. Equivariance gives

$$L(y) = L(T_s^k x) = T_t^k L(x).$$

Combined with $L(x) = L(y)$, the deck transformation T_t^k fixes $L(x)$. A deck transformation of a covering acts freely; hence $k = 0$, and then $x = y$. Thus L is injective.

Lift injectivity implies base injectivity. Assume L is injective and suppose, toward a contradiction, that

$$I(u) = I(v), \qquad u \neq v.$$

Choose lifts $x, y \in \widetilde{A}_s$ with $\pi_s(x) = u$, $\pi_s(y) = v$. By the lifting relation, $L(x)$ and $L(y)$ project to the same point of A_t, so there is a unique $k \in \mathbb{Z}$ with

$$L(y) = T_t^k L(x).$$

Equivariance gives

$$L(T_s^{-k} y) = T_t^{-k} L(y) = L(x),$$

while

$$\pi_s(T_s^{-k} y) = v \neq u = \pi_s(x),$$

so the two source points are distinct. This contradicts injectivity of L. Hence I is injective, which proves the equivalence. □

Proposition 9.9 (Properness criterion)**.** *Let $I : A_s \to A_t$ be a proper holomorphic immersion between connected annuli. If its oriented annular degree is one, then I is a biholomorphism. In particular, every universal-cover lift of I is injective.*

Proof. A holomorphic immersion between Riemann surfaces is a local biholomorphism; therefore $I(A_s)$ is open in A_t. A proper continuous map from a locally compact Hausdorff space to a Hausdorff space is closed; hence $I(A_s)$ is also closed in A_t. The target annulus is connected and the image is nonempty, so

$$I(A_s) = A_t.$$

A proper local homeomorphism is a covering map. For completeness: each target point has a discrete fiber by local injectivity and a compact fiber by properness, hence a finite fiber. Choose pairwise disjoint neighborhoods of the finitely many preimages on which I is invertible; after the target neighborhood is shrunk, properness excludes additional preimages, so the target point has an evenly covered neighborhood.

The number N of sheets of a connected covering of annuli is the index of the induced subgroup

$$I_* \pi_1(A_s) \subset \pi_1(A_t).$$

For an orientation-preserving holomorphic annulus covering, I_* is multiplication by N. The assumed degree is one, so $N = 1$, and I is bijective. A

bijective local biholomorphism has a holomorphic inverse, so I is a biholomorphism. The final assertion follows from Theorem 9.8. □

Remark 9.10 (Notation for the gluing criterion). In Proposition 9.11 the symbols are those of the Kahn–Lyubich gluing data at the aligned direct-decoration stage (KL II, §3.2.1 and Lemma 3.3, pp. 11–12): Q is the central overlap piece of the glued pseudo-piece model, Q_i its intersection with the i-th source tile, $P_{j(i)}$ the target component carrying the image of the i-th tile, $i \mapsto j(i)$ the component map between source and target tile labels, e_i the embedding of the central overlap Q_i into the i-th source tile, $\widetilde{e}_{j(i)}$ the corresponding target-central embedding, and L_i the induced universal-cover tile lift. These are alignment data of the same-seed decoration stage (Proposition 4.11); the tile index i is a label and is unrelated to the inclusion arrows i_r.

Proposition 9.11 (Tile/glued pseudo-piece equivalence)**.** *Fix a KL direct-decoration pair for which the component map $i \mapsto j(i)$ between source and target tile labels is bijective and every target component $P_{j(i)}$ satisfies*

$$Q \cap P_{j(i)} = Q_i.$$

Write e_i for the embedding of the central overlap Q_i into the i-th source tile, $\widetilde{e}_{j(i)}$ for the corresponding target-central embedding, and L_i for the induced universal-cover tile lift, so that the overlap commutation $L_i \circ e_i = \widetilde{e}_{j(i)}$ holds on Q_i. Then

every tile lift L_i is injective $\iff$ *the glued pseudo-piece immersion is injective on its interior.*

Proof. Assume every tile lift is injective, and suppose a tile point x collides in the target quotient with a central point $z \in Q$. By bijectivity of the component map, the collision-relevant target overlap of the i-th tile is restricted to $P_{j(i)}$, and $Q \cap P_{j(i)} = Q_i$; hence $z \in Q_i$. Overlap commutation gives

$$L_i(e_i(z)) = \widetilde{e}_{j(i)}(z) = L_i(x),$$

and injectivity of L_i gives $x = e_i(z)$. Thus the collision was already one of the prescribed source gluing identifications, and the glued map is injective on its interior. Collisions between points of different tiles are excluded by bijectivity of the component map.

Conversely, injectivity of the glued map restricts to injectivity of each tile lift, because each universal-cover tile embeds in its source pseudo-piece before the prescribed central overlap identifications are imposed. □

Corollary 9.12 (Equivalence chain and its negative resolution)**.** *For the fixed direct-decoration pair, the following are equivalent:*

1. *the actual bold base-annulus immersion $I_{m,n} : \mathbf{U}^m \setminus K(f) \to \mathbf{U}^n \setminus K(f)$ is globally univalent;*
2. *every corresponding universal-cover tile lift is injective;*
3. *the glued pseudo-piece immersion is injective on its interior.*

If, additionally, this direct-decoration pair is strongly aligned with an ordinary primitive canonical seed in the sense of Definition 9.1, then all three statements are false.

Proof. The equivalence of (1) and (2) is Theorem 9.8, applied to the degree-one Kahn–Lyubich ambient annulus immersion; the equivalence of (2) and (3) is Proposition 9.11. The negative resolution at aligned canonical seeds is the forced-collision theorem, whose base collision the two equivalences transport to every full universal-cover tile and then to the glued pseudo-piece immersion. □

10. The whole-tile obstruction

The canonical resolution of Section 5 is an infinite covering onto an embedded intermediate factor. It is natural to ask whether that resolved object can be *identified* with the boundary-extended universal-cover tile of the Kahn–Lyubich construction. This section proves that it cannot, and the typing needed to say so is derived, not assumed: the annular resolution normal form (Theorem 10.1) *derives* the entire typed comparison data of Definition 8.3 — universal-cover typing, split deck group, hyperbolic wrapping generator — from any covering/embedding factorization of a degree-one immersion of finite-modulus annuli. The no-go then reads: *any end-compatible whole-tile comparison of a nontrivial connected annular resolution is impossible.* The only surviving hypothesis is nontriviality of the kernel, which at an ordinary primitive canonical seed is supplied at aligned stages by the infinite defect (Corollary 5.11) together with exact base change (Corollary 8.8). The obstruction concerns the comparison square only; it is not an injectivity theorem for the Kahn–Lyubich immersion, and Proposition 10.9 records exactly which weaker interfaces remain admissible.

Theorem 10.1 (Annular resolution normal form)**.** *Let $I\colon A_s \to A_t$ be an oriented degree-one holomorphic immersion of finite-modulus annuli admitting a factorization*

$$A_s \xrightarrow{q_0} P \xhookrightarrow{e_0} A_t,$$

with q_0 a connected surjective holomorphic *covering and e_0 a* holomorphic *embedding (so that the deck actions below are by conformal automorphisms). Let $\widetilde{A}_t \to A_t$ be the universal cover and form the based pullbacks*

$$E = A_s \times_{A_t} \widetilde{A}_t, \qquad R = P \times_{A_t} \widetilde{A}_t.$$

Then:

1. *E is the universal cover of A_s: the pullback cover corresponds to $\ker I_*$, and $I_*\colon \pi_1(A_s) \to \pi_1(A_t)$ is an isomorphism (degree one), so $\ker I_*$ is trivial; E is connected because I_* is surjective.*
2. *$R \to P$ is connected and corresponds to the subgroup $N := \ker(e_{0*}) \leq \pi_1(P)$; connectedness holds because e_{0*} is surjective, since $e_{0*}q_{0*} = I_*$ is.*

(3) *the induced map* $q\colon E \to R$ *is the universal cover of* R*, with deck group canonically isomorphic to* N*.*
(4) *with* $G = \pi_1(P)$ *and* $H = q_{0*}\pi_1(A_s)$*, the identity* $e_{0*}q_{0*} = I_* = \mathrm{id}_{\mathbb{Z}}$ *exhibits* q_{0*} *as a splitting of the extension* $1 \to N \to G \xrightarrow{e_{0*}} \mathbb{Z} \to 1$*; hence*

$$G = N \rtimes H, \qquad H \cong \mathbb{Z}.$$

(5) *the generator of* H *acts on* $E \cong \mathbb{D}$ *as a hyperbolic deck transformation, because* A_s *has finite modulus (the deck generator of the universal cover of a finite-modulus annulus is hyperbolic; only a punctured disk or plane would give a parabolic or trivial action).*

In particular, every datum of the typed comparison (Definition 8.3): the universal-cover typing of q*, the splitting* $G = N \rtimes H$*, and the hyperbolicity of the wrapping generator, is* derived *from the annular factorization; none of it is an independent hypothesis.*

Proof. (1) The pullback of the universal cover $\widetilde{A}_t \to A_t$ along I is the cover of A_s corresponding to $I_*^{-1}(1) = \ker I_*$. By the oriented degree-one hypothesis, $I_*\colon \pi_1(A_s) \to \pi_1(A_t)$ is the identity after the chosen generator identifications (no properness of I is assumed or needed); hence $\ker I_* = 1$ and the corresponding cover is universal. The pullback is connected because $\pi_1(A_t) = I_*\pi_1(A_s)$.

(2) Identically, $R \to P$ corresponds to $e_{0*}^{-1}(1) = \ker e_{0*} = N$, and is connected because e_{0*} is surjective: $e_{0*}q_{0*} = I_*$ is onto $\pi_1(A_t)$.

(3) The universal property of the pullback induces $q\colon E \to R$ over q_0, a covering map. E is simply connected by (1), so $E \to R$ is the universal cover of R, and its deck group is $\pi_1(R) \cong N$.

(4) Applying π_1 to the factorization gives $e_{0*}q_{0*} = I_*$, an isomorphism onto $\pi_1(A_t) \cong \mathbb{Z}$. Thus q_{0*} is injective, $H = q_{0*}\pi_1(A_s) \cong \mathbb{Z}$, and $e_{0*}|_H$ is an isomorphism onto $\mathbb{Z}$; so H is a complement of $N = \ker e_{0*}$ in G, i.e. $G = N \rtimes H$.

(5) Under (1), the H-action on E is the deck action of $\pi_1(A_s)$ on the universal cover of A_s. For a finite-modulus annulus this deck generator is a hyperbolic element of $\mathrm{Aut}(\mathbb{D})$; the parabolic case occurs only for the punctured disk (infinite modulus) and the trivial case only for the plane, both excluded. □

Remark 10.2 (The typing is a theorem, not an assumption). Theorem 10.1 replaces the typed hypotheses of Definition 8.3 by a derivation: whenever the canonical resolution restricts, at an aligned stage, to a covering/embedding factorization of a degree-one immersion of finite-modulus annuli, the entire annular-tower structure — universal q, split deck group $G = N \rtimes H$, hyperbolic wrapping generator — is automatic. The whole-tile obstruction (Theorem 10.5) therefore reads: *any end-compatible whole-tile comparison of a nontrivial connected annular resolution is impossible.* The only remaining hypothesis is nontriviality of N, which at an ordinary primitive canonical

seed is supplied, at aligned stages, by Corollary 5.11 together with the base-change propagation of Corollary 8.8.

Lemma 10.3 (Normal-kernel wrapping incidence)**.** *Let the canonical resolved system carry the annular-tower typing of Definition 8.1. If $N \neq \{1\}$, then*

$$\zeta_-, \zeta_+ \in \Lambda(N). \tag{46}$$

Proof. Let $\pi\colon G \to \mathbb{Z}$ be the projection determined by the splitting (35), normalized by $\pi(h) = 1$; its kernel is N. As a deck group acting freely on $\mathcal{E} \cong \mathbb{D}$, G is discrete and torsion-free.

Suppose first that G were elementary. A torsion-free elementary discrete subgroup of $\mathrm{Aut}(\mathbb{D})$ is cyclic; since G contains the hyperbolic element h, write $G = \langle g \rangle$ and $h = g^m$. Then

$$1 = \pi(h) = m\,\pi(g)$$

in $\mathbb{Z}$, forcing $m = \pm 1$. Hence $G = \langle h \rangle = H$ and $N = \ker \pi = \{1\}$, contrary to hypothesis. Therefore G is non-elementary.

The subgroup N is nontrivial and normal in G. Its limit set $\Lambda(N)$ is nonempty, closed, and G-invariant: normality gives

$$g\,\Lambda(N) = \Lambda(gNg^{-1}) = \Lambda(N) \qquad \text{for every } g \in G.$$

If $\Lambda(N)$ were finite, its one- or two-point set would be preserved by G, making G elementary. Thus $\Lambda(N)$ is infinite. Choose

$$\alpha \in \Lambda(N) \setminus \{\zeta_-, \zeta_+\}.$$

Because h normalizes N, every $h^k\alpha$ belongs to $\Lambda(N)$. Hyperbolic dynamics of h on $\partial\mathbb{D}$ gives

$$h^k\alpha \to \zeta_+ \quad (k \to +\infty), \qquad h^k\alpha \to \zeta_- \quad (k \to -\infty).$$

Closedness of $\Lambda(N)$ proves (46). □

Remark 10.4 (Standard Fuchsian facts). The Fuchsian facts used — classification of torsion-free elementary Fuchsian groups containing a hyperbolic element, nonemptiness and invariance of the limit set of a nontrivial normal subgroup, and convergence of hyperbolic iterates to their fixed endpoints — are standard; see [Bea83] or [Kat92]. In particular a nontrivial torsion-free discrete subgroup of $\mathrm{Aut}(\mathbb{D})$ contains a hyperbolic or parabolic element, whose fixed-point set on the circle is nonempty, so the limit set of a nontrivial normal subgroup is nonempty.

Theorem 10.5 (No end-compatible whole-tile square with nontrivial kernel)**.** *Let* (34) *be a canonical resolved system with the annular-tower typing of Definition 8.1, and let* (37) *be an actual KL tile system as in Definition 8.2. If a whole-tile end-compatible comparison (Φ, B) in the sense of Definition 8.3 exists, then*

$$N = \{1\}. \tag{47}$$

Equivalently, if $N \neq \{1\}$, no pair (Φ, B) satisfies clauses (1)–(5) of Definition 8.3.

Proof. Assume $N \neq \{1\}$. By Lemma 10.3,

$$\zeta_+ \in \Lambda(N).$$

Choose $z_0 \in \mathcal{E}$. By the definition of the limit set, there are distinct $\gamma_n \in N$ such that

$$\gamma_n z_0 \longrightarrow \zeta_+. \tag{48}$$

For arbitrary $z \in \mathcal{E}$, every γ_n is a hyperbolic isometry, so

$$d_{\mathcal{E}}(\gamma_n z, \gamma_n z_0) = d_{\mathcal{E}}(z, z_0). \tag{49}$$

Bounded hyperbolic distance and (48) imply

$$\gamma_n z \longrightarrow \zeta_+ \qquad \text{for every } z \in \mathcal{E}, \tag{50}$$

since hyperbolic disks of a fixed radius centered at points converging to a point of $\partial\mathbb{D}$ have Euclidean diameters tending to 0.

Because $N = \mathrm{Deck}(q)$,

$$L(\gamma_n z) = e\, q(\gamma_n z) = e\, q(z) = L(z). \tag{51}$$

Apply the composite square (42):

$$L_{\mathrm{KL}}\big(\Phi(\gamma_n z)\big) = B\, L(\gamma_n z) = B\, L(z). \tag{52}$$

By clause (1) of Definition 8.3, after disk uniformization Φ is a disk automorphism and extends to a homeomorphism $\overline{\Phi}$ of the conformal ideal circles; the named-end compatibility (43) together with (50) gives

$$\Phi(\gamma_n z) \longrightarrow \omega_+. \tag{53}$$

The source-locked continuity of L_{KL} at ω_+ (Definition 8.2, (39)) now gives

$$\overline{L}_{\mathrm{KL}}(\omega_+) = \lim_{n\to\infty} L_{\mathrm{KL}}\big(\Phi(\gamma_n z)\big) = B\, L(z). \tag{54}$$

Equation (54) holds for *every* $z \in \mathcal{E}$. The map L is nonconstant and B is injective on $L(\mathcal{E})$, so $B\,L$ is nonconstant. It cannot equal the one fixed value $\overline{L}_{\mathrm{KL}}(\omega_+)$ for every z. This contradiction proves (47).

The same proof works verbatim at ζ_-. One wrapping vertex is sufficient; the second is an independent symmetric certificate. □

Corollary 10.6 (No aligned annular realization of the canonical infinite resolution is a whole KL tile)**.** *Assume that, at an aligned stage, the propagated canonical factorization is realized as a holomorphic covering/embedding factorization of a degree-one immersion of finite-modulus annuli satisfying the hypotheses of Theorem 10.1. For the ordinary primitive canonical resolution the sheet defect is infinite:*

$$[K : H_s] = \infty.$$

Therefore no whole-tile end-compatible comparison satisfying the annular-tower typing of Definition 8.1 identifies its covering/embedding factorization

with an actual Kahn–Lyubich tile map. No statement is made about comparisons outside that typing, and no construction of a typed system from the canonical resolution is claimed here.

Proof. The statement applies once the canonical factorization is realized in the annular setting of Theorem 10.1; that realization is part of the hypothesis. The canonical winding theorem (Corollary 5.11, supplied by the resolution cluster) gives an infinite fiber for the canonical covering q before tilewise base change, and the exact fiber bijection of Proposition 8.6(5), iterated through the tower by Corollary 8.8, preserves it. Hence in any proposed realization of (34) satisfying the annular-tower typing the covering q has infinite fibers. Since q is by typing a *universal* covering, $N = \mathrm{Deck}(q)$ acts simply transitively on each fiber, so N is infinite; in particular $N \neq \{1\}$. Theorem 10.5 now denies the existence of the comparison. □

Corollary 10.7 (Exact optional input that fails)**.** *The ambient same-seed identity of the Kahn iterate tower does not imply the whole-tile comparison of Definition 8.3. If the canonical infinite factorization and an actual KL tile are both present, the optional identification*

$$\mathcal{J}_{\mathrm{whole}}: \qquad L_{\mathrm{KL}}\,\Phi = B\,(e\,q) \quad \textit{subject to clauses (1)–(5) of Definition 8.3} \tag{55}$$

is false.

Proof. By Corollary 10.6 the canonical infinite factorization has $N \neq \{1\}$, and by Theorem 10.5 clauses (1)–(5) are then jointly unsatisfiable. Hence (55) fails. □

Proposition 10.8 (Exact obstruction certificate)**.** *Every putative whole-tile end-compatible comparison with $N \neq \{1\}$ carries the finite certificate*

$$\mathfrak{C}_{\mathrm{whole}} = \begin{pmatrix} N \neq 1, \quad \zeta_\pm \in \Lambda(N), \quad \Phi h = t_s \Phi, \quad \overline{\Phi}(\zeta_\pm) = \omega_\pm, \\ L_{\mathrm{KL}}\Phi = B\,e\,q, \quad B|_{L(\mathcal{E})} \textit{ injective}, \quad \overline{L}_{\mathrm{KL}}(\omega_\pm) \textit{ defined} \end{pmatrix}, \tag{56}$$

and the complete collision itinerary is any deck sequence

$$\gamma_n z \longrightarrow \zeta_+, \qquad q(\gamma_n z) = q(z), \qquad L_{\mathrm{KL}}\big(\Phi(\gamma_n z)\big) = B\,L(z), \tag{57}$$

of collision type

$$\chi_{\mathrm{corr}} = \textit{nontrivial deck-limit orbit transported to a source-locked continuous wrapping end.} \tag{58}$$

No numerical experiment, local-connectivity assumption, or parameter shrinking enters the certificate.

Proof. The entries of (56) are precisely the hypotheses consumed by the proof of Theorem 10.5: $N \neq 1$ and Lemma 10.3 supply $\zeta_\pm \in \Lambda(N)$; clauses (2), (5), (4), and (3) of Definition 8.3 supply $\Phi h = t_s \Phi$, $\overline{\Phi}(\zeta_\pm) = \omega_\pm$, $L_{\mathrm{KL}}\Phi = B\,e\,q$, and injectivity of B on $L(\mathcal{E})$; and Definition 8.2 supplies the continuous values $\overline{L}_{\mathrm{KL}}(\omega_\pm)$. The itinerary (57) is exactly the sequence constructed in (48), (51), and (52), and its transport to the wrapping end in (53)–(54) is the

collision (58). All entries are finitely presented algebraic, group-theoretic, or source-locked boundary data; no numerics, no local connectivity, and no parameter shrinking are used anywhere in the proof. □

Proposition 10.9 (What remains admissible)**.** *Theorem 10.5 forbids only the conjunction of the five clauses of Definition 8.3. Each of the following sharply different interfaces is not contradicted by it.*

(1) ***Proper interior or collar restriction.*** *Let $C \Subset \mathcal{E}$, or let C be a proved end-collar on which no two points in one N-orbit are retained. A comparison $\Phi_C \colon C \to \widehat{A}_s$ is not contradicted by Theorem 10.5. To use this repair one must prove: (a) every proof-used Wave path is contained in the retained domain or charged to a uniformly bounded defect window; (b) the restriction is an embedding with complete access markings; (c) all omitted sheets and limit-set ends are entered in the defect ledger; and (d) the restriction is stable under final-root enlargement.*

(2) ***Non-covering immersion or relation-valued comparison.*** *The actual KL map can remain a noninjective immersion. A relation $\mathcal{C} \subset \mathcal{E} \times \widehat{A}_s$ may retain canonical sheets and actual KL accesses without conjugating the deck action. It must record every fiber, target footprint, and proper-path branch; quotienting the relation to one coarse label is forbidden unless every future observable is constant on its fibers.*

(3) ***Retiming.*** *At a retimed stage with $N = \{1\}$, the obstruction vanishes; the one-sheet branch transports the source conformal ideal closure through q. The existence of a universal bounded retiming rule is open. Moreover, retiming* within the unchanged exact Kahn equalizer tower *cannot make the canonical factorization one-sheeted: by Corollary 8.8, the infinite q-fiber is preserved by every finite base change.*

(4) ***Localization revision.*** *Synchronizing the source and target obstacle packets can change H_s, H_t, $K = a_*^{-1}(H_t)$, and K/H_s, and may restore subgroup reflection. Provided the physical embedding lifts between the revised covers, a revised localization is one-sheeted exactly when it restores the equality $a_*^{-1}(H_t) = H_s$. Every component, access, and root-enlargement clause must then be reproved for the revised packet.*

None of the positive obligations listed above follows from the no-go.

Proof. (1) The contradiction in Theorem 10.5 is produced by a deck-limit sequence $\gamma_n z \to \zeta_+$. Such sequences need not remain in C, and the restricted map Φ_C need not satisfy the N-invariance (51) used in the proof; the argument therefore does not run on C. Any collar no-go would have to prove invariance or recurrence of the deck-limit sequence inside that collar separately.

(2) A relation span supplies neither the single-valued composite square (clause (4)) nor the deck-generator conjugation (clause (2)), so no step of the proof of Theorem 10.5 applies to it. This branch is not vacuous, by the following positive control. Define

$$F(z) = \exp\bigl(4(z-1)\bigr), \qquad z \in \mathbb{D}.$$

Then $|F(z)| = \exp(4(\operatorname{Re} z - 1)) < 1$, $F'(z) = 4F(z) \neq 0$, and F extends continuously to $\overline{\mathbb{D}}$. It is noninjective because

$$F(\pi i/4) = F(-\pi i/4) = -e^{-4}.$$

It is not a covering onto its image: if it were, the simply connected domain would make it the universal cover, whose deck group acts transitively on every fiber. A deck transformation δ carrying $\pi i/4$ to $-\pi i/4$ would satisfy $\exp(4(\delta(z)-1)) = \exp(4(z-1))$, so the continuous function $4(\delta(z)-z)/(2\pi i)$ would be integer-valued on the connected disk, hence constant; its value at $\pi i/4$ is -1, so $\delta(z) = z - \pi i/2$, which is not an automorphism of $\mathbb{D}$. Thus noninjectivity together with continuous boundary extension is compatible exactly when the noninjectivity is not a global covering-deck invariance.

(3) If $N = \{1\}$, there is no nontrivial deck element, hence no sequence (48), and the deck-limit obstruction is absent; a one-sheeted factorization can be transported to the actual KL tile whenever the based map-identification and target embedding are independently supplied. For the preservation claim: any stage reached from the canonical seed by finitely many full exact Kahn equalizer base changes has, by Corollary 8.8, fibers in bijection with the seed fiber, which is infinite; hence N remains infinite at every such stage, and a trivial-kernel stage cannot be reached by retiming inside the unchanged tower.

(4) If the revised subgroup data satisfies $a_*^{-1}(H_t) = H_s$, then by the injectivity clause of the canonical sheet-resolution theorem (Theorem 5.8: q is injective if and only if $K = H_s$) the revised resolution is one-sheeted, so $N = \{1\}$ and Theorem 10.5 imposes no constraint; conversely, if the reflection equality fails, the revised q is again a nontrivial covering. The proviso that the physical embedding lifts between the revised covers is part of the hypothesis and is not established here. □

11. The ordinary canonical seed versus a general bold stage

Every unconditional theorem in this paper lives at an *ordinary primitive canonical seed*: an ordinary quadratic-like restriction of a quadratic polynomial, primitively renormalizable of period $p > 1$, with the physical nested inclusion $i^p \colon U^p \hookrightarrow U^0$ and the canonical localization of [Kah23] applied to (i^p, f^p) at $A = K_p$ with $B = \mathcal{K}_p$ removed. At such a seed the unlocalized inclusion is a genuine embedding, the localization subgroups are computed from the actual Jordan neighborhoods of Section 3, and the entire chain excess component → subgroup defect → collision → resolution → infinitude is unconditional.

A general bold Kahn–Lyubich stage is a different object, and the distinction must be kept everywhere. At a general bold ψ-ql stage, the checked sources supply exactly: an *immersion* on the bold ambient domains (not a physical nested inclusion), degree one at the inner annular end, injectivity on a sufficiently thin inner collar, and end-properness at the little-Julia-set ends. They supply neither global base-annulus univalence nor full properness, and the subgroup-reflection criterion of Section 3 applies to a bold stage only *after* one has realized the bold inclusion as the lift of a specified physical embedding between specified covers — a realization the sources do not provide and this paper does not claim. Consequently:

(1) at ordinary canonical seeds, noninjectivity is forced unconditionally (Theorem 4.8);
(2) at aligned direct-decoration stages of the same seed, it is forced conditionally on the strong stage-alignment hypothesis (Definition 9.1; Theorem 9.6, Proposition 4.11);
(3) at a general bold stage, this paper asserts *nothing*: a base collision there would require an independent argument, and the abstract counterexamples of Appendix B show only that the stated axioms cannot rule one out.

12. What this changes in localization arguments

The results classify as follows:

- *new dynamical theorem*: the forced canonical collision (Theorem 4.8);
- *new quantitative refinement*: the kernel normal form and the exponential winding defect (Theorem 7.1, Corollary 7.4);
- *standard topology specialized canonically*: the intermediate-cover factorization, coset transport, and exact base change;
- *conditional Kahn–Lyubich application*: the tile and pseudo-piece propagation;
- *conditional no-go*: the whole-tile comparison, with the typing now derived (Theorem 10.1);
- *supporting abstract counterexamples*: Appendix B.

We record the operational consequences, in decreasing order of generality.

1. The canonical localized object is a fibered pair. At an ordinary primitive canonical seed the correct canonical object is not an embedded pseudo-piece but the pair $(e(P_K), q)$: the embedded intermediate cover *together with* the covering q whose based fiber is the pointed coset set K/H_s, with its canonical path-lifting transport (Proposition 5.15). Forgetting q forgets infinitely many sheets (Corollary 5.11); any construction consuming the canonical localization must either carry the fibration or prove a future-faithful finite quotient preserving every proof-used path, access, and observable.

2. Embedding preservation may not be assumed. Any argument that treats Kahn localization as embedding-preserving is invalid at every ordinary primitive canonical seed — not occasionally, but always (Theorem 4.8); at a general bold stage no such statement is made (Section 11). The sources' own phrasing (immersion, never embedding) is thus sharp, and proofs that quote localization must be audited for silent embedding uses. The subgroup-reflection criterion (Theorem 3.4) is the exact test: injectivity of a localized lift is equivalent to the reflection equality $a_*^{-1}(H_t) = H_s$, which fails canonically.

3. Finite sheet bookkeeping is impossible without a quotient theorem. Because every non-singleton fiber (equivalently, every fiber over the embedded resolved complement component) is infinite, no finite raw sheet list, tile inventory, or bounded-multiplicity bookkeeping of the canonical localization is complete. Enlargement steps must pass the exact coset tests of Section 6 *and* a geometric birth audit; the algebraic tests are necessary conditions only (Remark 6.3): commutativity of the enlargement square proves neither reflection nor exhaustion.

4. The resolved cover is not a tile. The infinite canonical resolution cannot be identified, by any comparison satisfying the annular-tower typing of Definition 8.1, with a whole boundary-extended Kahn–Lyubich tile (Theorem 10.5, Corollary 10.6). Interfaces between the canonical resolution and tile combinatorics must therefore be one of: a proper interior or collar restriction, a non-covering immersed or relation-valued comparison, a retiming to a stage with trivial kernel (impossible inside the unchanged equalizer tower), or a revised localization restoring subgroup reflection (Proposition 10.9). None of these four is constructed here; they are the admissible shapes, not results.

5. What is untouched. The collision lives on the inclusion arrow only. Nothing here affects the dynamical covering arrow, the embedded inner collar, the homeomorphism on filled sets, the pullback-tower construction squares (which are genuine fiber products and preserve an embedding already present), or any statement of the sources within their stated scope. In particular this paper neither proves nor undermines any a priori bounds statement; it corrects the interface through which such statements may consume localization.

13. Exact identifications and verification status

The interface between this paper's theorems and the sources rests on six exact identifications. Three of them are now proved in the body of the paper: the local-pair property of Jordan neighborhoods (Lemma 4.1), the shared-neighborhood based subgroup representatives (Definition 4.2), and injectivity of the filled completion on the complement cover (Lemma 4.3). The other four have now been checked against the *published* sources —

Kahn I in the numbering of arXiv:math/0609045v3 (28 December 2023, 63 pp.) and KL II in the numbering of Ann. Sci. Éc. Norm. Supér. (4) **41** (2008) 57–84 — and are *confirmed*. We record the outcome, with the confirming loci, before restating the checklist in full.

(V1) *H_s, H_t versus Kahn's covering subgroups: confirmed.* Kahn's localization cover of $S \setminus A$ is the cover corresponding to the fundamental group of a local neighborhood $U \setminus A$ (Kahn I, §7.4.1, Theorem 7.12 and its proof, p. 44), with the dynamical instance at §7.4.2, p. 45. Kahn specifies the cover by an unbased subgroup; the based representatives used here are fixed once and for all in Definition 4.2, and the collision certificate is invariant under change of representative because the winding homomorphism kills every conjugate of H_s. Kahn's requirement on a local neighborhood is only that every component of its complement of A be an annulus; our specialization to Jordan neighborhoods is our own and is justified by Lemma 4.1. The subgroups are *not* those of the different, satellite-compatible construction of Kahn I, §7.4.4, p. 46.

(V2) *Excess-component position: confirmed.* The nesting $U^r = f^{-r}(U^0)$ with each $f\colon U^{r+1} \to U^r$ quadratic-like is Kahn I, §6, p. 31, so $F = f^p$ is proper of degree 2^p; the canonical choice $A = K_p$, $B = \mathcal{K}_p$ for (i^p, f^p) with central local degree two is Kahn I, §7.4.3, p. 45; and the removal asymmetry is literally Kahn's $T = (\mathbf{U} \setminus B) \cup A$ against $T' = (\mathbf{U}' \setminus f^{-1}B) \cup A'$ at Kahn I, §7.4.2, p. 45. Since $2^p - 2 > 0$ for $p > 1$, an excess component exists, and primitive disjointness gives $X \cap B = \varnothing$ (Theorem 7.3). The construction meant is the one of §7.4.3, not the variants of §7.4.4 (Definition 3.8).

(V3) *Collision surviving every equalizer: confirmed.* Kahn's equalizer coordinates $\mathbf{U}'' = \{(x, y) : i(x) = f(y)\}$ with $i_1(x, y) = y$, $f_1(x, y) = x$, the identity $K'' = i_1^{-1}(K') = f_1^{-1}(K')$, the location of all critical points of f_1 in K'', and the commuting tower $i_n \circ f_{n+1} = f_n \circ i_{n+1}$ with its iterates are all at Kahn I, §7.3.2, pp. 42–43; that each stage is again a degree-d ψ-polynomial-like map is Lemma 7.11, p. 42. The leg assignment is exactly as used in the induction of Theorem 9.6; had the legs been reversed the induction would fail. Surjectivity of f_r on the complementary annuli comes from Definition 7.1(3), p. 36, Lemma 7.9, p. 40 and the properness statement at §7.3.2, p. 42, as now argued in the proof rather than asserted. The immersion hypothesis that Lemma 7.8, p. 40 requires for the set-theoretic equalizer to be the pullback in the category of complex 1-manifolds is regenerated at every stage (Proposition 8.6(6), Remark 8.7).

(V4) *Direct-decoration depth arithmetic and the tile transport: confirmed.* $P = Y^{(\nu-1)q+1}$ and its two univalent f^q-pullbacks Q_L, Q_R of depth $q\nu + 1$ are KL II, §2.1, p. 6 and Figure 2.1, p. 7, so $m = \nu q + 1$ and $n = (\nu - 1)q + 1$ are the correct integers. The tile objects

are universal covers of exactly the annuli A_m: KL II, §3.2.1, p. 12 forms the pseudo-piece by gluing copies of the universal cover of $\mathbf{U}^m \setminus K(f)$ to Y. Each glued copy embeds with images pairwise disjoint (KL II, Lemma 3.3, p. 12), which is what the transport step uses; the ambient arrow is the immersion of KL II, Lemma 3.4(ii), p. 13, not the dynamical arrow.

The one remaining open item is the seed-identification reading of the strong stage-alignment hypothesis, and it is now stated sharply in Remark 9.2: the Kahn–Lyubich ambient arrow at depths $m > n$ is the $(m-n)$-fold composite of exactly Kahn's equalizer legs (KL II, §3.1, p. 10 referring to Kahn I, §§7.3.1–7.3.2, pp. 38–43), so the hypothesis does not compare two constructions but identifies two seeds. Run over the canonical seed itself — the reading intended here, and legitimate because KL II, §3.2.1 imposes no provenance condition on (i, f) — the identifications are the identity and Theorem 9.6(4) is unconditional. Run over a different seed — for instance the renormalized pair identified at KL II, Corollary 5.14, p. 24 with a glued pseudo-piece $\mathbf{E}^{\chi-1}$ of the Modified Principal Nest — the identifications would have to be conformal isomorphisms of annuli at both depths simultaneously, and no source supplies them. That is the open item, and the statements depending on it remain conditional.

The checks above were performed against the published texts, so the identifications are no longer unperformed. We nonetheless invite specialist confirmation, particularly of (V1) and (V4), and of the scope note of Remark 9.4 recording that KL II never forms a pseudo-piece from Q_L or Q_R. The checklist below is retained in full, with published loci, as the record of what was checked.

Remark 13.1 (the exact identifications, with their published verification loci)*.* The forced-collision theorem and the sheet-resolution factorization are stated on objects that this paper *identifies* with objects of the Kahn/KL sources. Six identifications carry the entire weight of the word "canonical". Three are now proved in the body (Lemma 4.1, Definition 4.2, Lemma 4.3); the remaining four are listed here with the published locus at which each was checked and the outcome of that check.

(1) H_s/H_t **versus Kahn's covering subgroups.** *Status: confirmed. Paper-side objects:* the based subgroups $H_s = \mathrm{im}[\pi_1(D_A \setminus A, z_*) \to \pi_1(\Omega_s, z_*)]$ and the target analogue (Definition 2.5). *Published locus:* Kahn I, §7.4.1, Theorem 7.12 and its proof, p. 44 — the localization cover of $S \setminus A$ is the cover corresponding to the fundamental group of $U \setminus A$ for U a local neighborhood of A; the general (possibly disconnected) case is on the same page; the dynamical instance is §7.4.2, p. 45. *Checked:* (i) Kahn specifies the cover by an *unbased* subgroup (a conjugacy class); the based H_s, H_t are representatives fixed in Definition 4.2, and the collision certificate is invariant under change of representative because the winding homomorphism kills

every conjugate of H_s; (ii) independence of the choice of local neighborhood follows from the deformation-retraction property of local pairs (Kahn I, §7.4.1, p. 44); (iii) these subgroups are *not* the explicitly described "loops that never cross γ_p" subgroup of Kahn's *other* (satellite-compatible) renormalization, Kahn I, §7.4.4, p. 46, which is a different construction with a different removal set. Note also that Kahn requires of a local neighborhood only that every component of its complement of A be an annulus; the Jordan choice made in this paper is our own specialization, justified by Lemma 4.1.

(2) **Excess-component position in the canonical seed.** *Status: confirmed. Paper-side objects:* the component $X \neq A$ of $F^{-1}A$ with $X \cap B = \varnothing$, and the removal asymmetry $X \subset F^{-1}B$, $X \cap B = \varnothing$ (Proposition 4.5, Theorem 7.3). *Published locus:* nesting and properness of the seed, Kahn I, §6, p. 31 ($U^r = f^{-r}(U^0)$, each $f\colon U^{r+1} \to U^r$ quadratic-like, so $F = f^p$ is proper of degree 2^p); the little filled Julia sets K_p, $K_p(j) = f^j(K_p)$, $\mathcal{K}_p$, and primitivity as pairwise disjointness, Kahn I, §1.1; the canonical choice $A = K_p$, $B = \mathcal{K}_p$ for (i^p, f^p) with central local degree two, Kahn I, §7.4.3, p. 45; and the removal asymmetry, which is literally Kahn's $T = (\mathbf{U} \setminus B) \cup A$ against $T' = (\mathbf{U}' \setminus f^{-1}B) \cup A'$, Kahn I, §7.4.2, p. 45. *Checked:* the central component has local degree two, so $2^p - 2 > 0$ forces an excess component for $p > 1$; the invariance $F(K_p(j)) = K_p(j)$ plus primitive disjointness gives $X \cap B = \varnothing$; and the source-side removal $X \subset F^{-1}B$ with target-side non-removal is exactly the T'/T mismatch. The construction meant throughout is the one of §7.4.3, not either variant of §7.4.4, p. 46 (Definition 3.8).

(3) **Collision surviving every equalizer.** *Status: confirmed. Paper-side step:* inductive propagation of the collision pair through the pullback tower in equalizer coordinates (Theorem 9.6). *Published locus:* Kahn I, §7.3.2, pp. 42–43: the equalizer coordinates $\mathbf{U}'' = \{(x, y) :\ i(x) = f(y)\}$, $i_1(x, y) = y$, $f_1(x, y) = x$; the iff-chain giving $K'' = i_1^{-1}(K') = f_1^{-1}(K')$; all critical points of f_1 in K''; the properness statement "f_1 is proper of degree d"; and the commuting tower $i_n \circ f_{n+1} = f_n \circ i_{n+1}$ with its iterates i^n. That each stage is again a degree-d ψ-polynomial-like map is Lemma 7.11, p. 42. *Checked:* f_n restricted to the complements of the filled sets is an unbranched covering (branch values in K^n by Definition 7.1 and the remark after it, p. 36, and $f_n^{-1}(K^n) = K^{n+1}$ by Definition 7.1(4)); it is *onto* each complementary annulus by properness of positive degree (Definition 7.1(3), p. 36; Lemma 7.9, p. 40; §7.3.2, p. 42) together with connectedness of A_n; the chosen pairs $(x_n^{\pm}, y_n)$ satisfy the equalizer membership condition $i_n(x_n^{\pm}) = f_n(y_n)$; the leg assignment is Kahn's, and reversing it would break the induction; and the propagated points remain in the complementary annuli at every depth. The immersion hypothesis needed for Lemma 7.8, p. 40 (without

which, as the source's own example on p. 41 shows, the set-theoretic equalizer need not be the pullback in the category of complex 1-manifolds) is regenerated at each stage by Proposition 8.6(6) and independently by Kahn I, Lemma 7.7, p. 40 and Lemma 7.9, p. 40.

(4) **Direct-decoration depths and tile transport.** *Status: confirmed* (the transport itself remains conditional on strong stage alignment; see below). *Paper-side step:* noninjectivity of the ambient annulus maps $I_{m,n}$, of their universal-cover tile lifts, and of the glued pseudo-piece immersions at the depths $m = \nu q + 1$, $n = (\nu - 1)q + 1$ (Theorem 9.6). *Published locus:* KL II, §2.1, p. 6 ($P = Y^{(\nu-1)q+1}$; the two univalent f^q-pullbacks $Q_L, Q_R \subset P$ of depth $q\nu+1$) and Figure 2.1, p. 7; KL II, §3.2.1, pp. 11–12 (pseudo-piece as path classes, and equivalently as Y with copies of the universal cover of $\mathbf{U}^m \setminus K(f)$ glued on); Lemma 3.3, p. 12 (the glued copies embed, with pairwise disjoint images); Lemma 3.4, pp. 12–13 (the inclusion-induced arrow is an immersion, with component index map "surjective but not necessarily injective"; the dynamical arrow is a branched covering). *Checked:* the paper's ambient map at depths (m, n) is the *inclusion-induced* arrow of KL II, not the dynamical arrow (which is an isomorphism on univalent pieces and is not contested); the depth arithmetic matches §2.1 exactly; the tile objects are universal covers of exactly the annuli A_m; the collision transported from the canonical seed lands in the correct glued copy, using the disjoint-image property of Lemma 3.3, p. 12 — not Lemma 3.4, which supplies only immersions; and the realization quantifier is kept separate. Two scope facts belong here: KL II never forms a pseudo-piece from Q_L or Q_R (Remark 9.4), and the annular degree $+1$ used in the transport is *derived* in Lemma 9.5 from Kahn I, Theorem 7.3, p. 37, Lemma 7.4(2), p. 37 and KL II, Lemma 3.1, p. 10, and is stated outright for the canonical renormalization in the proof of Kahn I, Theorem 8.11, p. 51.

The residual open item is not on this list: it is the seed-identification reading of the strong stage-alignment hypothesis, discussed in Remark 9.2.

Source verification (documentary; not a mathematical proof). Every locus above was checked against the published texts: Kahn I in arXiv:math/0609045v3 (28 December 2023, 63 pp.), whose sectioning is that used throughout this paper, and KL II in Ann. Sci. Éc. Norm. Supér. (4) **41** (2008) 57–84 and the identically sectioned arXiv:math/0609046v2 (12 February 2007). The SHA-256–locked local TEX archives from which the paraphrases of Section 2 were originally prepared correspond to exactly those two versions; their line-number concordance is recorded in Appendix D. The authoritative statements of the forced-collision theorem and the reflection criterion are Theorems 4.8 and 3.4 as proved in Sections 3–5 of this paper. This checklist adds no new mathematical claim. □

Scope of machine verification. The accompanying validation script for the forced-collision theorem machine-checks: the source-lock strings and hashes of every cited passage; the arithmetic $2^p - 2 > 0$ for $2 \le p \le 64$; and the dependency routing of the accompanying research ledger (the collision and resolution theorems recorded as proved with no open hypotheses; all global program statements recorded as open). The exact rational certificates of Appendix B (two Sturm chains, nine Rouché–Taylor disks, one exact polynomial identity and factorization) are machine-produced and reverified by an exact integer-arithmetic validator. *No topological or analytic step of Sections 3–10 is machine-verified.*

Appendix A. Source disclosures and the annulus immersion lemma

This appendix records Kahn's annulus immersion lemma, with a faithful paraphrase of its proof, and the four places where the locked sources demote the quadratic-like embedding arrow to an immersion. The citation conventions of Appendix D apply throughout.

Lemma A.1 (annulus immersion lemma; Kahn I, Lemma 7.4, p. 37)**.** *Let $d > 1$ be an integer, let $r > 1$, and set $r' = r^{1/d}$. Let $I\colon \mathbb{A}(1, r') \to \mathbb{A}(1, r)$ be a holomorphic immersion that extends continuously to a degree-one map $\mathbb{T} \to \mathbb{T}$. For $\rho \in (1, r)$ set $\rho' = \rho^{1/d}$. Then:*

1. *$I(\mathbb{A}(1, \rho')) \subset \mathbb{A}(1, s)$, where $s = s(\rho, r)$ and $\rho - s$ is uniformly positive on every compact subset of $\{(r, \rho) :\ 1 < \rho < r \le \infty\}$;*
2. *$I|_{\mathbb{A}(1,\rho')}$ is an embedding for some $\rho = \rho(r)$ that is uniformly greater than 1 whenever r is bounded away from 1.*

Combining (1) and (2) (Kahn's remark following Lemma 7.4, p. 38): for r bounded away from 1 there are uniform $1 < s < \rho$ such that $I|_{\mathbb{A}(1,\rho')}$ is an embedding and $I(\mathbb{A}(1, \rho')) \subset \mathbb{A}(1, s)$.

Remark A.2 (scope of the assertion). The lemma asserts an embedding only on the *inner collar* $\mathbb{A}(1, \rho')$ adjacent to the unit circle, together with a uniform containment of its image. Kahn neither asserts nor uses injectivity of I on its whole domain $\mathbb{A}(1, r')$; the entire point of the restriction theorem below is to *restrict* to a collar on which the immersion becomes an embedding. No statement in the locked source upgrades this to a global embedding of any localized inclusion.

Proof (faithful paraphrase of Kahn I, proof of Lemma 7.4, pp. 37–38). (1) Double $\mathbb{A}(1, r')$ and $\mathbb{A}(1, r)$ across $\mathbb{T}$ to $\mathbb{A}(1/r', r')$ and $\mathbb{A}(1/r, r)$ and extend I by Schwarz reflection. Because $\rho' = \rho^{1/d}$ and $r' = r^{1/d}$, the Poincaré distance from $\mathbb{T}$ to $\mathbb{T}_{\rho'}$ in $\mathbb{A}(1, r')$ equals the Poincaré distance from $\mathbb{T}$ to $\mathbb{T}_\rho$ in $\mathbb{A}(1, r)$. For (ρ, r) in a compact subset of $\{1 < \rho < r < \infty\}$ the reflected map contracts the Poincaré metric uniformly, so $I(\mathbb{T}_{\rho'})$ lies a definite Poincaré distance closer to $\mathbb{T}$ than $\mathbb{T}_\rho$; this yields $s < \rho$ with $\rho - s$

uniformly positive. If $r = \infty$ then I is the identity and the estimate is trivial. A sequence/compactness argument interpolates between the two cases on compact subsets of $\{1 < \rho < r \leq \infty\}$.

(2) For r bounded away from 1, the image $I(\mathbb{T}_{\sqrt{r'}})$ is bounded away from $\mathbb{T}$; choose a uniform $\rho > 1$ so that $\Lambda := \mathbb{A}(1, \rho)$ is disjoint from $I(\mathbb{T}_{\sqrt{r'}})$. Let Λ' be the component of $I^{-1}(\Lambda)$ adjacent to $\mathbb{T}$. Then $I \colon \mathbb{T} \cup \Lambda' \to \mathbb{T} \cup \Lambda$ is a covering map, of degree one because it has degree one on $\mathbb{T}$; hence $I|_{\Lambda'}$ is an embedding. Poincaré contraction on the doubled annulus $\mathbb{A}(1/r', r')$ places the outer boundary of Λ' outside $\mathbb{A}(1, \rho')$, so $\mathbb{A}(1, \rho') \subset \Lambda'$, and $I|_{\mathbb{A}(1,\rho')}$ is an embedding. □

Where the sources demote the embedding to an immersion.

Remark A.3 (where the quadratic-like embedding becomes only an immersion). The sources disclose, at four separate places, that the inclusion-type arrow of the quadratic-like theory degenerates to an immersion in the ψ-quadratic-like theory; none of these places asserts a global embedding of the localized inclusion.

(1) **Kahn I, §6.1, pp. 31–32.** In the quadratic-like sections the map $i \colon U^{n+1} \setminus \mathcal{K} \to U^n \setminus \mathcal{K}$ is literally the inclusion; footnote 15 on p. 31 announces that in the ψ-quadratic-like sections (§§7–8) this embedding will be replaced by an immersion, "so it is good to have notation for it." The same passage notes that the triangle formed by the covering f, the inclusion $U^{n+1} \setminus \widetilde{\mathcal{K}} \subset U^{n+1} \setminus \mathcal{K}$, and i does *not* commute; passage along that hole-filling leg is controlled quantitatively via Property D (Kahn I, §2.7).

(2) **Kahn I, §1.4.** ψ-quadratic-like maps are introduced as the generalization in which the inclusion $\mathbf{U} \to \mathbf{V}$ is replaced by an immersion i whose properties imply only that the multivalued $f \circ i^{-1}$ has a single-valued quadratic-like *restriction*.

(3) **Kahn I, §8.1.** At each stage of the ψ-quadratic-like tower there are three maps: a covering $f \colon \mathbf{U}^{n+1} \setminus f^{-1}\mathcal{K} \to \mathbf{U}^n \setminus \mathcal{K}$; an inclusion $\mathbf{U}^{n+1} \setminus f^{-1}\mathcal{K} \hookrightarrow \mathbf{U}^{n+1} \setminus \mathcal{K}$; and an immersion $i \colon \mathbf{U}^{n+1} \setminus \mathcal{K} \to \mathbf{U}^n \setminus \mathcal{K}$. The source states that the one significant change from the quadratic-like setting is that the third map, previously an embedding, is now an immersion. The subsequent proofs use, about i, only (a) properness on the ends corresponding to $\mathcal{K}$ and (b) that the iterated immersion is homotopic to a homeomorphism; point-injectivity of i is neither claimed nor used.

(4) **KL II, §3.2.2, Lemma 3.4, pp. 12–13.** The naturality lemma keeps two arrows strictly separate: the dynamical arrow $\mathbf{f} \colon \mathbf{Y} \to \mathbf{Z}$ induced by $f \colon Y \to Z$ is a branched covering of the same degree k (an isomorphism of glued universal-cover copies when $k = 1$; Lemma 3.4(i), p. 13), while the inclusion arrow $\mathbf{i} \colon \mathbf{Y} \to \mathbf{Z}$ induced by $Y \subset Z$ is an immersion built from component-wise immersions

of universal covers, with the component assignment $j(i)$ described as "surjective but not necessarily injective" (proof of Lemma 3.4(ii), p. 13; p. 68 of the journal pagination). This is the closest published anticipation of Theorem 4.8, and it is discussed as such in Section 1.3, where the three ways in which the present theorem differs from it are set out. Ideal-boundary behavior is likewise not embedding-like: the boundary extension of $\mathbf{i}$ exists, but KL state that $\mathbf{i}(\partial\mathbf{Y})$ is not contained in $\partial\mathbf{Z}$ unless $\mathbf{Z} = \mathbf{Y}$ (KL II, §3.2.4, p. 13) — a stronger assertion than the "need not" used elsewhere in this paper — and the extremal-length arguments of KL II use maximal initial lifts that may terminate on the O-boundary (KL II, §5.5).

Source verification (documentary; not a mathematical proof). Each item paraphrases the cited passage of the published sources: items (1)–(3) against Kahn I, arXiv:math/0609045v3 (28 December 2023); item (4) against KL II, Ann. Sci. Éc. Norm. Supér. (4) **41** (2008) 57–84, and the identically sectioned arXiv:math/0609046v2. No further claim is made: in particular none of the cited passages asserts, and no other passage of either source asserts, that the localized inclusion $\widehat{i}$ of dynamical localization is globally injective. The five-word phrase quoted in item (4) is the only verbatim material retained from either source. □

Appendix B. The two abstract counterexamples

The theorems of this appendix calibrate the rest of the paper. They show that the hypotheses the sources actually provide — degree-one boundary behavior with an immersion for the annulus lemma; the complete abstract ψ-quadratic-like axioms — do not imply global injectivity. Both are abstract: neither is a counterexample to any actual dynamically generated Kahn or Kahn–Lyubich map, and both are fully compatible with the embedded inner collar and the local embedding near the filled set that the sources prove. Their role is to close every shortcut: after these examples, any injectivity assertion for an actual localized inclusion must use its dynamical construction history, and Sections 4–5 show that for the canonical primitive stages the answer is negative for dynamical reasons.

Notation and certificate conventions. Throughout, $\mathbb{A}(1,\rho) = \{z \in \mathbb{C} : 1 < |z| < \rho\}$ is the open round annulus, $\mathbb{T} = \{|z| = 1\}$, and $\mathbb{D}_R = \{z : |z| < R\}$. For $|a| < 1$ the Blaschke factor is

$$B_a(z) = \frac{z - a}{1 - \overline{a}\, z}.$$

For a Gaussian rational w we use the exact rational modulus bounds

$$L(w) = \max\big(|\Re w|, |\Im w|\big), \qquad U(w) = |\Re w| + |\Im w|, \qquad L(w) \le |w| \le U(w). \tag{N.1}$$

Sturm certificate. For $p \in \mathbb{Z}[t]$, the exact Euclidean Sturm chain is the signed remainder sequence of (p, p'); if $V(\pm\infty)$ denote the numbers of sign

changes in the leading-coefficient sign lists at $\mp$infinity, Sturm's theorem gives the number of distinct real roots $N_{\mathbb{R}}(p) = V(-\infty) - V(+\infty)$. A certificate consists of the chain's degree list and the two sign lists; all entries are exact integers/rationals. *Rouché–Taylor certificate.* For a polynomial P, a Gaussian-rational center c, a rational radius ρ, and an integer multiplier $m \geq 1$, the certificate is the exact rational inequality

$$L\big(P'(c)\big)\,\rho \;>\; m\Big[\,U\big(P(c)\big) + \sum_{k\geq 2} U\Big(\tfrac{P^{(k)}(c)}{k!}\Big)\rho^k\Big]. \tag{N.2}$$

By (N.1), (N.2) implies $|P'(c)\zeta| > \big|P(c) + \sum_{k\geq 2} \frac{P^{(k)}(c)}{k!}\zeta^k\big|$ on $|\zeta| = \rho$, so Rouché's theorem (comparing $P(c+\zeta)$ with $P'(c)\zeta$) places exactly one zero of P in $\{|z-c| < \rho\}$. No floating point enters: centers, coefficients, ρ, and both sides of (N.2) are rational, and the accompanying validator records the exact cross-multiplied fractions. For circle maps, deg denotes topological degree and $\mathrm{wind}(\gamma, w)$ the winding number of a loop γ about w.

Definition B.1 (FJ-set)**.** An *FJ-set* is a compact, connected, full plane continuum containing more than one point (full: its complement is connected). This is Kahn's definition (Kahn I, §1.5).

Definition B.2 (ψ-polynomial-like map, after Kahn)**.** A *ψ-polynomial-like map of degree d* is a pair

$$(i, f) : (\mathbf{U}', K') \longrightarrow (\mathbf{U}, K)$$

such that: $\mathbf{U}'$ and $\mathbf{U}$ are disks; $i : \mathbf{U}' \to \mathbf{U}$ is a holomorphic immersion; $f : \mathbf{U}' \to \mathbf{U}$ is a proper holomorphic branched cover of degree d; $K \subset \mathbf{U}$ is an FJ-set; and the common inverse

$$K' = i^{-1}(K) = f^{-1}(K)$$

is connected. (Paraphrased from Kahn I, Definition 7.1, p. 36, arXiv:math/0609045v3; properness of f is clause (3) there. Kahn–Lyubich restate the same disk, branched-cover, full-continuum, and common-inverse clauses in KL II, §3.1, p. 9.) For $d = 2$ we say ψ-quadratic-like.

Remark B.3 (Plane-case exceptionality). When the target $\mathbf{U}$ is the whole plane, removability forces i to be an isomorphism (Kahn I, remark following Definition 7.1, p. 36). The counterexample of Theorem B.13 uses bounded hyperbolic disks and therefore does not fall into that exceptional case.

Definition B.4 (The Blaschke quotient R)**.** Let

$$r' = \frac{1287}{1000}, \qquad d = 16, \qquad r = (r')^{16},$$

and set

$$\mathcal{A} = \Big\{\tfrac{1+6i}{10},\ \tfrac{-3-i}{10},\ \tfrac{-7+2i}{10}\Big\}, \qquad \mathcal{B} = \Big\{\tfrac{-6+2i}{10},\ \tfrac{4i}{10}\Big\},$$

$$R(z) = \frac{\prod_{a\in\mathcal{A}} B_a(z)}{\prod_{b\in\mathcal{B}} B_b(z)}. \tag{A.1}$$

Clearing denominators, $R = N/D_0$ where, in descending powers of z,

$$N : 2000 + 4000i,\ 19600 + 2200i,\ 28460 - 23880i,\ 4072 - 29046i, \\ -6630 - 10210i,\ -1950 - 1525i, \tag{A.2}$$

$$D_0 : 1525 + 1950i,\ 10210 + 6630i,\ 29046 - 4072i,\ 23880 - 28460i, \\ -2200 - 19600i,\ -4000 - 2000i. \tag{A.3}$$

Every coefficient is a Gaussian integer, so all subsequent sign and Rouché checks are finite rational calculations.

B.1. **Failure of injectivity under Kahn's annulus hypotheses.**

Lemma B.5. *The map R of Definition B.4 has no zero and no pole on the closed annulus $1 \le |z| \le r'$. In particular $u(z) = \log|R(z)|$ is harmonic on an open neighborhood of $\overline{\mathbb{A}(1, r')}$.*

Proof. The zeros and poles of a Blaschke factor and its reciprocal occur among $a,\ b,\ 1/\overline{a},\ 1/\overline{b}$. The points $a \in \mathcal{A}$, $b \in \mathcal{B}$ lie in the unit disk, with

$$\max_{a\in\mathcal{A}} |a| = \frac{\sqrt{53}}{10}, \qquad \max_{b\in\mathcal{B}} |b| = \frac{\sqrt{40}}{10}.$$

Direct squaring gives

$$\left(\tfrac{1287}{1000}\right)^2 \tfrac{53}{100} < 1, \qquad \left(\tfrac{1287}{1000}\right)^2 \tfrac{40}{100} < 1,$$

so every reciprocal point $1/\overline{a}$, $1/\overline{b}$ lies outside $|z| = r'$. Hence R has neither zero nor pole on $1 \le |z| \le r'$, and $u = \log|R|$ is harmonic on a neighborhood of the closed annulus. □

Lemma B.6. *$|R| \equiv 1$ on $\mathbb{T}$, and the restriction $R|_{\mathbb{T}} : \mathbb{T} \to \mathbb{T}$ is an orientation-preserving homeomorphism of degree one; indeed $\frac{d}{d\theta} \arg R(e^{i\theta}) > 0$ for every θ.*

Proof. On $\mathbb{T}$ every Blaschke factor has modulus one, so $|R| \equiv 1$ there. The topological degree is the factor count $\deg(R|_{\mathbb{T}}) = |\mathcal{A}| - |\mathcal{B}| = 3 - 2 = 1$. For the pointwise orientation statement: if $z = e^{i\theta}$ then

$$\frac{d}{d\theta} \arg B_a(e^{i\theta}) = \frac{1 - |a|^2}{|e^{i\theta} - a|^2},$$

hence

$$\frac{d}{d\theta} \arg R(e^{i\theta}) = \sum_{a\in\mathcal{A}} \frac{1 - |a|^2}{|e^{i\theta} - a|^2} - \sum_{b\in\mathcal{B}} \frac{1 - |b|^2}{|e^{i\theta} - b|^2}. \tag{A.4}$$

Substitute $t = \tan(\theta/2)$ and multiply by the product of the positive Poisson denominators; the sign of (A.4) is then the sign of the integer polynomial

$$p(t) = 619271t^{10} - 5100608t^9 + 35876739t^8 - 169721152t^7 \\ + 491196326t^6 - 885663808t^5 + 1065558054t^4 \\ - 957958592t^3 + 669681731t^2 - 236915328t + 60062535. \tag{A.5}$$

Certificate S1. The exact Euclidean Sturm chain of p has degree list $10, 9, 8, 7, 6, 5, 4, 3, 2, 1, 0$, with sign lists

$$+--++++--+--\ \text{at}\ -\infty, \qquad ++--+--++++-\ \text{at}\ +\infty,$$

each with five sign changes; Sturm's theorem gives $N_{\mathbb{R}}(p) = 5 - 5 = 0$. Since $p(0) = 60062535 > 0$ and $\mathrm{lc}(p) = 619271 > 0$, we conclude $p(t) > 0$ for all real t, and the point $t = \infty$ (i.e. $\theta = \pi$) is positive as well. Thus (A.4) is strictly positive everywhere, so $R|_{\mathbb{T}}$ is a local orientation-preserving homeomorphism; together with degree one it is an orientation-preserving homeomorphism $\mathbb{T} \to \mathbb{T}$. □

Lemma B.7. *$R(\mathbb{A}(1, r')) \subset \mathbb{A}(1, r)$; more precisely $|R(z)| > 1$ for $1 < |z| < r'$, $|R(z)| > 1$ on $|z| = r'$, and $|R(z)| < 48 < (r')^{16} = r$ for $1 < |z| < r'$. Moreover $r' = r^{1/16}$ identically, the exact modulus relation of Kahn's annulus lemma.*

Proof. Lower bound. Parameterize the outer source circle by

$$z(t) = r' \, \frac{1 - t^2 + 2it}{1 + t^2}, \qquad t \in \mathbb{R}.$$

Exact substitution into $R = N/D_0$ gives

$$|N(z(t))|^2 - |D_0(z(t))|^2 = \frac{656369\, q(t)}{32000000000000000000000000\,(1 + t^2)^4}, \tag{A.6}$$

where

$$\begin{aligned} q(t) = {} & 1175116031365640042695178891 9\, t^8 \\ & - 1375798396389252774929713008 00\, t^7 \\ & + 9589017992342884495518311284 76\, t^6 \\ & - 4106906924234287705167320942400\, t^5 \\ & + 1077937218965500653409381977191 4\, t^4 \\ & - 1668032877573975295283172798240 0\, t^3 \\ & + 1404219926227446396377435331407 6\, t^2 \\ & - 5586354003042199421157378340800\, t \\ & + 1394955241714223750805412881719. \end{aligned} \tag{A.7}$$

The denominator in (A.6) is positive. *Certificate S2.* The exact Sturm chain of q has degree list $8, 7, 6, 5, 4, 3, 2, 1, 0$ and sign lists

$$+--++--++\ \text{at}\ -\infty, \qquad ++--++--+\ \text{at}\ +\infty,$$

each with four sign changes, so q has no real root. Since $q(0) = 1394955241714223750805412881719 > 0$ with positive leading coefficient, $q(t) > 0$ on the real line and at $t = \infty$. Hence $|R(z)| > 1$ on $|z| = r'$. By Lemma B.5, $u = \log|R|$ is harmonic on a neighborhood of the closed annulus; $u = 0$ on $\mathbb{T}$ (Lemma B.6) and $u > 0$ on $|z| = r'$. The minimum principle gives $u \geq 0$ on the closed annulus and the

strong minimum principle gives $|R(z)| > 1$ for $1 < |z| < r'$. *Upper bound.* For $s = |a| < 1$ and $\rho > 1$ with $\rho s < 1$, a direct calculation on $|z| = \rho$ gives

$$\max_{|z|=\rho} |B_a(z)| = \frac{\rho - s}{1 - \rho s}, \tag{A.8}$$

and the right side is increasing in both ρ and s; the maximum modulus principle extends the bound over $1 \le |z| \le \rho$. Also $|B_b(z)| > 1$ for $|z| > 1$, so each reciprocal factor $1/B_b$ has modulus less than one there. Using the exact rational bounds

$$\frac{\sqrt{37}}{10} < \frac{61}{100}, \qquad \frac{\sqrt{10}}{10} < \frac{32}{100}, \qquad \frac{\sqrt{53}}{10} < \frac{73}{100}$$

for the moduli of the three numerator zeros, (A.1) and (A.8) give, for $1 < |z| < r'$,

$$|R(z)| < \prod_{s \in \{61/100,\, 32/100,\, 73/100\}} \frac{r' - s}{1 - r's} = \frac{5697579109375}{119480253283} < 48 < (r')^{16} = r,$$

where the last two inequalities are integer cross-multiplications. Combining the two bounds proves $R(\mathbb{A}(1, r')) \subset \mathbb{A}(1, r)$, and $r = (r')^{16}$ by definition, so $r' = r^{1/16}$. □

Lemma B.8. *$R'(z) \neq 0$ for $1 < |z| < r'$; hence R is a holomorphic immersion on $\mathbb{A}(1, r')$.*

Proof. Up to a nonzero constant multiple, the numerator of R' is

$$\begin{aligned} P(z) = {} & (-1268000 + 501000i)z^8 + (-1247000 + 7116800i)z^7 \\ & + (17024420 + 10563170i)z^6 + (33992992 - 21977616i)z^5 \\ & - 52469243i\, z^4 + (-33992992 - 21977616i)z^3 \\ & + (-17024420 + 10563170i)z^2 + (1247000 + 7116800i)z \\ & + (1268000 + 501000i). \end{aligned} \tag{A.9}$$

All eight zeros of P are isolated by exact Rouché–Taylor certificates (N.2) with common radius $\rho = 10^{-4}$ at the following Gaussian-rational centers (every displayed decimal terminates and is exact):

j	c_j	m_j
1	$-1.469179 + 0.035737i$	126
2	$-1.187253 + 1.185352i$	362
3	$-0.818448 + 1.003716i$	118
4	$-0.680250 + 0.016547i$	451
5	$-0.487955 + 0.598410i$	122
6	$-0.421815 + 0.421140i$	192
7	$0.407227 + 0.223067i$	182
8	$1.888874 + 1.034668i$	144

Certificates R1–R8. For every j, exact substitution into (A.9) verifies the rational inequality

$$L\big(P'(c_j)\big)\rho > m_j\Big[U\big(P(c_j)\big) + \sum_{k=2}^{8} U\Big(\tfrac{P^{(k)}(c_j)}{k!}\Big)\rho^k\Big]. \tag{A.10}$$

No floating-point estimate occurs in (A.10): all centers, coefficients, ρ, and both sides are rational, and the validator records the exact cross-multiplied fractions. By (N.1), (A.10) gives, on $|\zeta| = \rho$,

$$|P'(c_j)\zeta| > \Big|P(c_j) + \sum_{k=2}^{8} \tfrac{P^{(k)}(c_j)}{k!}\zeta^k\Big|,$$

so Rouché's theorem, comparing $P(c_j+\zeta)$ with $P'(c_j)\zeta$, shows that each disk $\Delta_j = \{|z - c_j| < \rho\}$ contains exactly one zero of P. Exact squared-distance comparisons show the eight disks are pairwise disjoint; since $\deg P = 8$, they exhaust all zeros of P. Classification by exact squaring of rational quantities: for $j = 4,5,6,7$, $|c_j| + \rho < 1$; for $j = 1,2,3,8$, $|c_j| - \rho > 1.2901 > r'$. Therefore P has no zero in $1 < |z| < r'$. The denominator of R' has no zero there by Lemma B.5, so $R'(z) \neq 0$ on the source annulus. □

Lemma B.9. *Let $z_1 = -1 + \frac{4}{5}i$. Then $1 < |z_1| < r'$, and there exists $z_2 \in \mathbb{A}(1, r')$ with $z_2 \neq z_1$ and $R(z_2) = R(z_1)$. Hence R is not injective on $\mathbb{A}(1, r')$.*

Proof. First, $|z_1|^2 = \frac{41}{25}$, so $1 < |z_1| < r'$. Exact substitution gives

$$w = R(z_1) = -\frac{559602300}{1858025273} + \frac{1886709225}{1858025273}\,i. \tag{A.11}$$

The polynomial $N - wD_0$ has the exact factor $z - z_1$. Removing that factor and multiplying by a nonzero constant, the remaining quartic is

$$\begin{aligned} Q(z) = {} & (1649705408450 + 1129218801775i)z^4 \\ & + (8374862949240 - 2102550942545i)z^3 \\ & + (5597363347032 - 11487498744901i)z^2 \\ & + (-2960676892680 - 7024430543585i)z \\ & - 1525645358450 - 501687535025i. \end{aligned} \tag{A.12}$$

Certificate R9. Set $c = -0.570412 + 1.153470i$ (exact terminating decimals) and $\rho_1 = 10^{-5}$. The exact rational Taylor inequality

$$L\big(Q'(c)\big)\rho_1 > 11\Big[U\big(Q(c)\big) + \sum_{k=2}^{4} U\Big(\tfrac{Q^{(k)}(c)}{k!}\Big)\rho_1^k\Big] \tag{A.13}$$

holds, so by Rouché's theorem Q has a zero z_2 in $|z - c| < \rho_1$. The exact center modulus is $|c|^2 = \frac{413965722661}{250000000000}$, and squaring rational endpoints gives

$$1 < |c| - \rho_1 < |z_2| < |c| + \rho_1 < 1.287 = r'.$$

The z_2-disk is disjoint from z_1, and D_0 has no zero in the source annulus (Lemma B.5), so z_2 is a genuine solution of $R(z) = w$. Therefore $z_2 \neq z_1$ and $R(z_2) = w = R(z_1)$, proving noninjectivity. □

Theorem B.10. *Let r', $d = 16$, $r = (r')^{16}$, and R be as in Definition B.4. Then*

$$R : \mathbb{A}(1, r') \longrightarrow \mathbb{A}(1, r)$$

is a holomorphic immersion, extends analytically across $\mathbb{T}$, and restricts to an orientation-preserving degree-one homeomorphism $\mathbb{T} \to \mathbb{T}$, with the exact modulus relation $r' = r^{1/d}$ of Kahn's annulus lemma; but R is not injective. Consequently, the strengthening "a holomorphic immersion $I : \mathbb{A}(1, r^{1/d}) \to \mathbb{A}(1, r)$ that extends to the inner circle with degree one must be globally injective" is false, even when the inner-circle map is an orientation-preserving analytic homeomorphism.

Proof. By Lemma B.5, R is holomorphic and pole-free on a neighborhood of the closed annulus $1 \leq |z| \leq r'$; in particular it extends analytically across $\mathbb{T}$. By Lemma B.7, R maps $\mathbb{A}(1, r')$ into $\mathbb{A}(1, r)$ with the exact relation $r' = r^{1/16}$. By Lemma B.6, the boundary restriction is an orientation-preserving degree-one homeomorphism of $\mathbb{T}$. By Lemma B.8, R is an immersion on the source annulus. By Lemma B.9, two distinct source points have the same image. □

Remark B.11. Theorem B.10 closes one possible shortcut:

inner-circle degree one + analytic boundary extension + holomorphic immersion + Kahn's modulus relation	$\not\Longrightarrow$ global injectivity.

Therefore the direct-decoration embedding gate cannot be solved by strengthening Kahn's inner-collar lemma through only its stated annular hypotheses. The actual Kahn–Lyubich question remains narrower: one must either (1) prove global univalence from additional dynamically generated path-space/fiber-product structure; (2) produce an actual KL base-annulus collision together with its lifted deck certificate; or (3) replace the full tile by a collar-truncated, sheet-sensitive carrier program.

Remark B.12 (Status of the trichotomy). The three-way trichotomy of Remark B.11 is transcribed from the underlying research notes, which record it as the honest residual state of the direct-decoration gate after the no-go; none of the three branches is proved there or here. The notes are provenance for this remark, not authority: the mathematical content asserted in this paper is exactly Theorem B.10.

B.2. **The abstract ψ-quadratic-like counterexample.**

Theorem B.13. *There is a degree-two ψ-polynomial-like map $(i, f) : (\mathbf{U}', K') \to (\mathbf{U}, K)$ in the exact sense of Definition B.2 such that:*

(1) $\mathbf{U}'$ *and* $\mathbf{U}$ *are disks;*
(2) $K \Subset \mathbf{U}$ *is an FJ-set;*
(3) $f : \mathbf{U}' \to \mathbf{U}$ *is a proper holomorphic branched cover of degree two;*
(4) $i : \mathbf{U}' \to \mathbf{U}$ *is a holomorphic immersion;*
(5) *the common inverse* $K' = i^{-1}(K) = f^{-1}(K)$ *is connected;*
(6) $i|_{K'} : K' \to K$ *is a homeomorphism;*
(7) *the complementary-annulus map* $i : \mathbf{U}' \setminus K' \to \mathbf{U} \setminus K$ *induces* $+1$ *on fundamental groups; but*
(8) i *is not injective, even on* $\mathbf{U}' \setminus K'$.

Consequently, neither the proper degree-two companion map nor the common connected full-continuum pullback axiom upgrades the immersion in an abstract ψ-quadratic-like pair to a global embedding.

Proof. Construction. Put

$$R = 75, \qquad P = 100, \qquad \beta = \frac{2\pi}{P} = \frac{\pi}{50}, \qquad \Delta = e^{2\beta} - 1, \tag{B.1}$$

$$\mathbf{U}' = \mathbb{D}_{75}, \qquad \mathbf{U} = \mathbb{D}_{5625}, \qquad f(z) = z^2, \qquad K = [0,1], \qquad K' = [-1,1], \tag{B.2}$$

$$i(z) = \frac{e^{\beta(z+1)} - 1}{\Delta}. \tag{B.3}$$

Step 1: disk and FJ-set clauses. $\mathbb{D}_{75}$ and $\mathbb{D}_{5625}$ are simply connected plane domains, hence disks. The intervals $K = [0,1]$ and $K' = [-1,1]$ are compact, connected, and nondegenerate; a closed segment does not separate the plane, so each is full. Thus K (and also K') is an FJ-set, and $K \Subset \mathbb{D}_{5625}$, $K' \Subset \mathbb{D}_{75}$. No local-connectivity assertion about a Julia set is used: these continua are explicit intervals. *Step 2: the proper degree-two branch and its exact inverse.* The map $f(z) = z^2 : \mathbb{D}_{75} \to \mathbb{D}_{5625}$ is onto of degree two counted with multiplicity; its only critical point is 0, of local degree two, with the single branch value $0 \in K$. It is proper: if $C \Subset \mathbb{D}_{5625}$, choose $r_0 < 5625$ with $C \subset \overline{\mathbb{D}}_{r_0}$; then $f^{-1}(C) \subset \overline{\mathbb{D}}_{\sqrt{r_0}} \Subset \mathbb{D}_{75}$, and $f^{-1}(C)$ is closed, hence compact. Finally $z^2 \in [0,1] \iff z \in [-1,1]$: writing $z = \rho e^{i\theta}$, the condition $z^2 \geq 0$ forces $2\theta \equiv 0 \pmod{2\pi}$, so z is real, and $z^2 \leq 1$ gives $|z| \leq 1$. Hence $f^{-1}(K) = K'$. *Step 3: i is an immersion into* $\mathbf{U}$. Differentiating (B.3),

$$i'(z) = \frac{\beta e^{\beta(z+1)}}{\Delta}, \tag{B.4}$$

with $\beta > 0$, $\Delta > 0$, and a nonvanishing exponential; so $i' \neq 0$ throughout $\mathbf{U}'$. For containment without numerical approximation: for $|z| < 75$, $\Re(z+1) \leq |z| + 1 < 76$, hence

$$|i(z)| \leq \frac{e^{\beta\Re(z+1)} + 1}{e^{2\beta} - 1} < \frac{e^{76\beta} + 1}{e^{2\beta} - 1}. \tag{B.5}$$

Using only the elementary bounds $3 < \pi < \frac{22}{7}$, $e < 3$, and $e^x > 1 + x$ for $x > 0$ — where $e < 3$ follows from $e = 2 + \sum_{n\geq 2} \frac{1}{n!} < 2 + \sum_{n\geq 2} 2^{-(n-1)} = 3$, the

bound $\pi < \frac{22}{7}$ from the positive-integral identity $\frac{22}{7} - \pi = \int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,dx > 0$, and $3 < \pi$ from the inscribed regular hexagon — we get

$$76\beta = \frac{38\pi}{25} < \frac{38}{25}\cdot\frac{22}{7} = \frac{836}{175} < 5, \qquad e^{2\beta} - 1 > 2\beta = \frac{\pi}{25} > \frac{3}{25}.$$

Substituting in (B.5),

$$|i(z)| < \frac{3^5+1}{3/25} = \frac{6100}{3} < 2034 < 5625,$$

so $i(\mathbf{U}') \subset \mathbf{U}$. *Step 4: exact common-inverse equality and the homeomorphism on K'.* For $t \in [0,1]$,

$$i(z) = t \iff e^{\beta(z+1)} = 1 + \Delta t,$$

and the right side is a positive real in $[1, e^{2\beta}]$. Enumerating all complex logarithms,

$$z = -1 + \frac{\log(1+\Delta t)}{\beta} + 100ik, \qquad k \in \mathbb{Z}. \tag{B.6}$$

As t runs through $[0,1]$, $0 \le \log(1+\Delta t) \le 2\beta$, so the real part in (B.6) runs through $[-1,1]$. If $k \neq 0$ then $|\Im z| \ge 100 > 75$ and $z \notin \mathbf{U}'$. Hence only $k = 0$ survives, and $i^{-1}(K) = [-1,1] = K'$. Combined with Step 2, $i^{-1}(K) = f^{-1}(K) = K'$, which is connected. On K', (B.4) is positive real, and $i(-1) = 0$, $i(1) = (e^{2\beta}-1)/\Delta = 1$; thus $i|_{K'}$ is strictly increasing and a homeomorphism $K' \to K$, as Kahn's general theorem predicts. *Step 5: collision criterion.* Equality of two i-values has the exact form

$$i(z) = i(w) \iff e^{\beta(z-w)} = 1 \iff z - w = 100ik \quad (k \in \mathbb{Z}). \tag{B.7}$$

Step 6: the complementary-annulus degree is $+1$. Let $\gamma(\theta) = 2e^{i\theta}$, $0 \le \theta \le 2\pi$, a positive generator of $\pi_1(\mathbf{U}' \setminus K')$; since $i^{-1}(K) = K'$, its image lies in $\mathbf{U} \setminus K$. Compute the winding about $1/2 \in K$: the solutions of $i(z) = 1/2$ are $z = x_0 + 100ik$ with

$$x_0 = -1 + \frac{1}{\beta}\log\Big(1 + \frac{\Delta}{2}\Big),$$

and $1 < 1 + \frac{\Delta}{2} < e^{2\beta}$ gives $-1 < x_0 < 1$. Thus exactly one solution lies in $\mathbb{D}_2$, namely the $k = 0$ one, and it is simple by (B.4). The argument principle gives $\operatorname{wind}(i \circ \gamma, 1/2) = 1$. Hence the induced homomorphism on the two annular fundamental groups sends positive generator to positive generator. Degree $+1$ still does not imply injectivity. *Step 7: exact remote self-collision.* Take $z_+ = 50i$, $z_- = -50i$. They are distinct, satisfy $|z_\pm| = 50 < 75$, and are not in $[-1,1]$. Since $\beta(z_+ - z_-) = \frac{\pi}{50}(100i) = 2\pi i$, criterion (B.7) gives $i(z_+) = i(z_-)$. The common value is explicit:

$$i(z_\pm) = \frac{-e^{\beta} - 1}{\Delta} < 0;$$

by Step 3 it lies in $\mathbf{U}$, and since $K = [0,1]$ it lies in $\mathbf{U} \setminus K$. Thus $i : \mathbf{U}' \setminus K' \to \mathbf{U} \setminus K$ has a genuine self-collision, proving clause (8) and the theorem. $\square$

Proposition B.14. *The pair (i, f) of Theorem B.13 does not contradict Kahn's local-embedding theorem: i is injective on every set of imaginary diameter less than* 100*, and in particular on a sufficiently small neighborhood of $K' = [-1, 1]$. Hence the local polynomial-like restriction and the embedded inner annular collar of Kahn's theorem remain valid for this pair; what fails is precisely the unsupported passage from the local collar to the entire complementary annulus.*

Proof. By the exact collision criterion (B.7), every nontrivial collision $i(z) = i(w)$, $z \neq w$, has $z - w = 100ik$ with $k \neq 0$, hence imaginary displacement at least 100. Consequently i is injective on every set of imaginary diameter less than 100, including a sufficiently small neighborhood of $[-1, 1]$. Kahn proves exactly a polynomial-like restriction: an embedding on a neighborhood selected near K', obtained by local injectivity near K' followed by truncation to an inner annular collar (Kahn I, Theorem 7.3, p. 37, with proof on pp. 38–39); no global embedding of $\mathbf{U}'$ is asserted there, and Kahn–Lyubich import only the local embedding near K' (KL II, Lemma 3.1, p. 10). The present pair satisfies those local conclusions and refutes only the global upgrade. □

Remark B.15. The abstract whole-annulus upgrade is closed negatively:

$$\boxed{\text{full abstract } \psi\text{-ql axioms } \not\Longrightarrow \text{ global injectivity of } i.}$$

The remaining direct-decoration gate is narrower and more geometric: prove global univalence for the *actual* Kahn–Lyubich path-space/localization immersion, or replace the full pseudo-piece by the embedded inner collars and record every path that exits them in a bounded defect ledger. No global MLC obligation is promoted.

Remark B.16 (Audit points for the abstract counterexample). The following audit points delimit exactly what Theorem B.13 does and does not establish. They are transcribed from the underlying research notes, which are provenance for this remark, not authority.

- Not merely an immersion-only example: the pair includes the proper degree-two companion $f(z) = z^2$, not an unrelated noninjective disk immersion.
- Exact common inverse, not coarse image agreement: Step 4 enumerates every logarithmic sheet; the period $P = 100$ and source radius 75 exclude all nonzero sheets from the preimage of $[0, 1]$; the collision occurs halfway between adjacent real-positive sheets and does not create a hidden second component of K'.
- The collision is outside the protected continuum: both source points lie outside K' and their common value is negative, hence outside K — a collision of the actual complementary-annulus map, not one absorbed by the filled set.
- The annular degree remains one, established by the argument principle rather than inferred from a label, so the example survives the degree-one condition used in the tile/base reduction.

- No hidden critical point: i' is everywhere nonzero; the self-collision is caused by exponential periodicity, not branching of i.
- Properness is assigned to the correct arrow: f is proper; i is not proper on the whole complementary annulus, and the ψ-ql definition does not require it to be — reassigning the properness of f to i would conflate the two arrows.
- Abstract pair versus actual KL path-space stage: the theorem refutes a deduction from the complete abstract axioms; a theorem about the actual KL inclusion immersion must use its nested puzzle/path-space construction and prove one of: global base-annulus injectivity; full properness; complete path and homotopy lifting; or an equivalent no-return-to-an-old-sheet statement. Absent such a theorem, only Kahn's certified inner collar may be treated as embedded.

Appendix C. Certificate conventions and data

The certificate conventions (exact Sturm chains over $\mathbb{Z}[t]$; Rouché–Taylor disk certificates over Gaussian rationals with the L/U modulus bounds) are stated in Appendix B. The full certificate data — the Sturm chain degree and sign lists for certificates S1–S2, the nine Rouché–Taylor centers, radii, and cross-multiplied rational inequalities R1–R9, and the exact factorization $N - wD_0 = (z - z_1) \cdot (\text{quartic})$ — are reproduced in the ancillary file `certificates.json` together with the validating script `validate_certificates.py`; they are exact rational identities and can be re-checked by integer arithmetic alone.

Appendix D. Source lock and provenance concordance

Every citation in the body of this paper is given in published form: "Kahn I, §X / Theorem N, p. P" refers to [Kah23] in the numbering of arXiv:math/0609045v3 (28 December 2023, 63 pp.), the current and only public full version, for which the arXiv abstract page records no journal venue; "KL II, Lemma N, p. P" refers to [KL08] in the numbering of Ann. Sci. Éc. Norm. Supér. (4) **41** (2008) 57–84 and of the identically sectioned preprint arXiv:math/0609046v2 (12 February 2007, 29 pp.). Where a page reference is to the journal pagination of KL II rather than to the preprint it is flagged in place.

This appendix records only the *provenance* of those paraphrases. They were prepared against the following SHA-256–locked local TEX archives, whose contents correspond to exactly the two versions named above (the kl06 files are dated 28 December 2023, the kl08 files 12 February 2007):

- kl06: the TEX source archive of [Kah23] (files `intro.tex`, `pullback.tex`, `psiql2.tex`, `wad.tex`, `conclusions.tex`);
- kl08: the TEX source archive of [KL08] (files `psiql.tex`, `kingdom.tex`, `moduli.tex`).

The archive SHA-256 hashes, together with a file-to-section concordance, are printed in the ancillary file `SOURCELOCK.txt` distributed with this manuscript; the certificate data and validating script are the ancillary files `certificates.json` and `validate_certificates.py`. The concordance most used here is `psiql2.tex` = Kahn I §7 (with §7.1 at line 15, §7.2 at line 54, §7.3.1 at line 188, §7.3.2 at line 350, §7.4.1 at line 434, §7.4.2 at line 510, §7.4.3 at line 549, §7.4.4 at line 593, §7.4.5 at line 612), `pullback.tex` = Kahn I §6, `conclusions.tex` = Kahn I §8, `intro.tex` = Kahn I §1, `wad.tex` = Kahn I §2; and, on the Kahn–Lyubich side, `kingdom.tex` = KL II §2, `psiql.tex` = KL II §3, `moduli.tex` = KL II §5.

Four typographical corrections to `kl06` are adopted silently in the paraphrases of Section 2 and collected here: $\pi_{T,A}$ for $\pi_{S,A}$ twice in the localization construction (§7.4.1, p. 44); an undefined B' read as B in the hypothesis list for dynamical localization (§7.4.2, p. 45); and K'' read as K' in the proof of the polynomial-like restriction theorem (§7.2, p. 38). In the locked archive these occur at `psiql2.tex` lines 461, 491, 524 and 156 respectively.

A citation hazard worth flagging. KL II (2006–2008) cites Kahn I as [K] in the numbering of its v1/v2, whose sectioning differs from that of the 2023 v3 used throughout this paper. A reader following a cross-reference out of KL II into Kahn I must re-locate it in v3. All Kahn-side citations here are given in the v3 numbering.

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